Unit 13 Overview: Shapes on a Coordinate Plane

Unit Narrative

The content of this unit revisits concepts that were introduced in previous units. Students apply these concepts while using coordinate geometry to determine points on a directed line segment, slopes of lines, and length of line segments. The spiraling of this content was purposely done so that students can review a majority of the content before summative assessments. Many of the lessons utilize Explorations and Collaborations so that students have spaced recall of the content, thus helping the student to solidify the properties, postulates, and theorems used in the unit. Provide students the opportunity to find the necessary information without your support. By having to locate the information on their own, students will have a better chance of remembering the information.

Unit 13 begins with support re-teaching parallel and perpendicular lines. Students learned parallel and perpendicular lines in Algebra 1, but may have forgotten some of the information. Students then apply previously learned definitions, properties, or theorems of circles using coordinate geometry. Section 1 concludes by building a student’s understanding of using coordinate geometry to determine a midpoint to partitioning line segments, concluding with using weighted averages.

In section 2, students use coordinate geometry to classify circles, triangles, or quadrilaterals. Students use properties of these figures to justify their classification. Students then apply coordinate geometry to solve problems involving circles, triangles, or quadrilaterals. Section 2 concludes with a culminating activity, where students are given coordinates of vertices and asked to identify the figure, justifying their choice.

In section 3, students continue to utilize coordinate geometry to determine if shapes are congruent or similar. Students revisit transformations to verify the congruence and similarity. The section concludes with finding area and perimeter of composite figures, using coordinate geometry to determine lengths. The last lesson involves finding area and perimeter of composite figures using coordinate geometry, and then using that information to solve real-world problems.

Unit At-A-Glance

Section	Benchmarks Addressed	Lesson	Title
Section 1: Lines and Circles	**MA.912.GR.3.1** **MA.912.GR.3.2** **MA.912.GR.3.3** **MA.912.GR.3.4** **MA.912.GR.7.2**	Lesson 1	Parallel and Perpendicular Lines
		Lesson 2	Circles on the Coordinate Plane
		Lesson 3	Determining Points on a Circle
		Lesson 4	Partitioning Line Segments
		Lesson 5	Weighted Averages
Section 2: Classifying Triangles and Quadrilaterals	**MA.912.GR.3.2** **MA.912.GR.3.3**	Lesson 6	Classifying Triangles
		Lesson 7	Centroids of Triangles
		Lesson 8	Classifying Quadrilaterals – Part 1
		Lesson 9	Classifying Quadrilaterals – Part 2
		Lesson 10	Classifying Quadrilaterals – Part 3
		Lesson 11	What Is The Figure?
Section 3: Problem-Solving with Polygons	**MA.912.GR.3.2** **MA.912.GR.3.3** **MA.912.GR.3.4**	Lesson 12	Congruence and Similarity
		Lesson 13	Finding Perimeter and Area on the Coordinate Plane
		Lesson 14	Using Coordinate Geometry to Solve Real-World Problems

Differentiation

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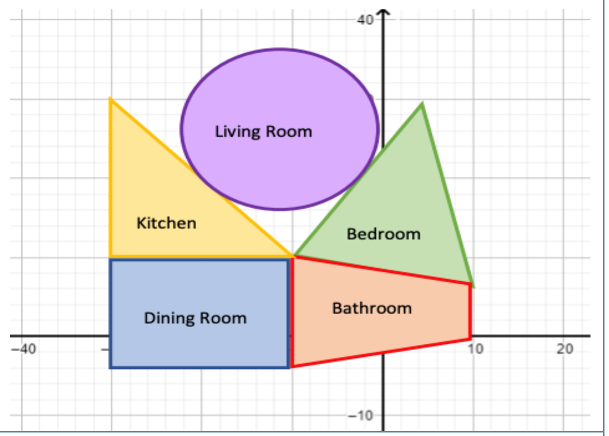
<table>
<tr><td>Engagement
The "WHY" of learning</td><td>

- Home Renovation Project:
  - Step 1: Design the floor plan of your current or dream home. Each room must be either a circle, triangle, or quadrilateral.
  - Step 2: Determine the dimensions of each room.
  - Step 3: Use what you know of proportional relationships and distance to scale a coordinate plane to fit the dimensions of your home. The line y = 0 represents sea level, and the line x = 0 represents the midpoint of your front door.
  - Step 4: Draw your home according to the scaled dimensions. Include all vertices of each room as a coordinate pair. Additionally, determine the area of each polygon, and compare those dimensions to the square footage of your home to verify your answers.



</td></tr>
<tr><td>Representation
The "WHAT" of learning</td><td>

- Transform your classroom into a life-size coordinate plane for this unit. Use blue tape, tacks and string, or other creative methods for constructing the floor or a classroom wall into an entirely new and unique learning surface. Rather than the coordinate plane living on paper, coordinate geometry can come to life – students can *walk or trace* a figure to experience the slope of its sides or the length of a side, or students could compare and contrast types of quadrilaterals on the wall, as though a touch screen came to life on their wall.

</td></tr>
<tr><td>Action & Expression
The "HOW" of learning</td><td>

- Students can apply a variety of methods for comparing and contrasting types of quadrilaterals. Consider starting with an engaging digital activity like "Polygraph: Basic Quadrilaterals" from Desmos. With Polygraph, Desmos provides tools for developing informal language into formal vocabulary, because words should result from a need to describe our world – this is where they gain their power.

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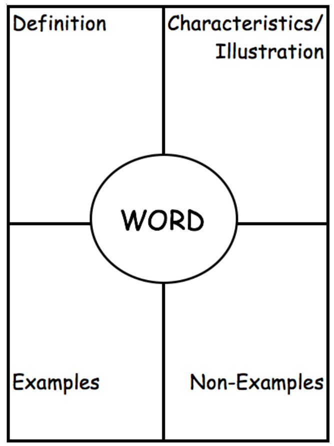
ELL Information



Students will be engaging with many shapes that are similar, yet uniquely different. This unit provides an opportunity to review some important vocabulary. Have students create a Frayer Model for the words in this topic:

- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. For this section, students will be introduced to many words that set the foundation for functions.
- Provide the definition, but have students work in pairs to illustrate the word (when possible, or else list characteristics), list examples, and list non-examples.
- For examples and non-examples, encourage students to consider this a work in progress, and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support" below:

- Area
- Composite figure
- Perimeter

Additional Differentiated Support

Scaffolding Resources

GeoGebra offers a variety of digital interactive tools that provide additional scaffolding as students make sense of information:

- Coordinate Geometry Exercise: Creating Parallelograms
- Distance in the Coordinate Plane (Parts 1 and 2, with hints embedded in each activity)
- Midpoints and Coordinates (Parts 1 and 2, with hints embedded in each activity)

Students engaging with digital manipulatives have the ability to co-construct their learning experiences. Consider offering GeoGebra and other digital manipulatives as scaffolding for students to opt-in for use during Guided Practice, Try It, or Your Turn! activities.

On-Ramp

Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Classifying Quadrilaterals
- Parallel and Perpendicular Lines
- Similar Triangles
- Coordinate Points in Real-World Context
- Perimeter and Area on the Coordinate Plane

Section Overview

Section 1: Lines and Circles (Lessons 1-5)

Benchmarks Addressed	Learning Targets	
MA.912.GR.3.1 Determine the weighted average of two or more points on a line. • *Clarification 1: Instruction includes using a number line and determining how changing the weights moves the weighted average of points on the number line.* **MA.912.GR.3.2** Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals. • *Clarification 1: Instruction includes using the distance or midpoint formulas and knowledge of slope to classify or justify definitions, properties and theorems.* **MA.912.GR.3.3** Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals. • *Clarification 1: Problems involving lines include the coordinates of a point on a line segment including the midpoint.* • *Clarification 2: Problems involving circles include determining points on a given circle and finding tangent lines.* • *Clarification 3: Problems involving triangles include median and centroid.* • *Clarification 4: Problems involving quadrilaterals include using parallel and perpendicular slope criteria.* **MA.912.GR.3.4** Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons. **MA.912.GR.7.2** Given a mathematical or real-world context, derive and create the equation of a circle using key features. • *Clarification 1: Instruction includes using the Pythagorean Theorem and completing the square.* • *Clarification 2: Within the Geometry course, key features are limited to the radius, diameter, and the center.*	Lesson 1	• I can use coordinate geometry to identify parallel and perpendicular lines.
	Lesson 2	• I can use coordinate geometry to determine parts of a circle in a mathematical context. • I can draw a circle using given definitions, properties, and theorems in a mathematical context.
	Lesson 3	• I can determine whether a point is on the circumference of a circle, in a circle, or outside a circle. • I can find the center of a circle given two points. • I can write the equation of a circle given the endpoints of a diameter.
	Lesson 4	• I can use coordinate geometry to partition a line segment.
	Lesson 5	• I can determine the weighted average of two or more points on a line. • I can explain how changing the weight moves the weighted average of points.

<table>
<tr><th colspan="3">Section 2: Classifying Triangles and Quadrilaterals (Lessons 6-11)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="6">MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals.
• Clarification 1: Instruction includes using the distance or midpoint formulas and knowledge of slope to classify or justify definitions, properties and theorems.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.
• Clarification 1: Problems involving lines include the coordinates of a point on a line segment including the midpoint.
• Clarification 2: Problems involving circles include determining points on a given circle and finding tangent lines.
• Clarification 3: Problems involving triangles include median and centroid.
• Clarification 4: Problems involving quadrilaterals include using parallel and perpendicular slope criteria.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.</td><td>Lesson 6</td><td>• I can use coordinate geometry to classify triangles in a mathematical or real-world context.</td></tr>
<tr><td>Lesson 7</td><td>• I can use coordinate geometry to identify properties of triangles.</td></tr>
<tr><td>Lesson 8</td><td>• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
• I can draw a quadrilateral on a coordinate plane using given definitions, properties, or theorems.</td></tr>
<tr><td>Lesson 9</td><td>• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
• I can use coordinate geometry to solve geometric problems involving quadrilaterals in a mathematical or real-world context</td></tr>
<tr><td>Lesson 10</td><td>• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.</td></tr>
<tr><td>Lesson 11</td><td>• I can use coordinate geometry to justify definitions, properties, and theorems of triangles and quadrilaterals.</td></tr>
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<tr><th colspan="3">Section 3: Problem-Solving with Polygons (Lessons 12-14)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="3">MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals.
Clarification 1: Instruction includes using the distance or midpoint formulas and knowledge of slope to classify or justify definitions, properties and theorems.
MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles and quadrilaterals.
Clarification 1: Problems involving lines include the coordinates of a point on a line segment including the midpoint.
Clarification 2: Problems involving circles include determining points on a given circle and finding tangent lines.
Clarification 3: Problems involving triangles include median and centroid.
Clarification 4: Problems involving quadrilaterals include using parallel and perpendicular slope criteria.
MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.</td>
<td>Lesson 12</td><td>I can use coordinate geometry to justify congruence and similarity in polygons.</td></tr>
<tr><td>Lesson 13</td><td>I can find the perimeter or area of polygons and composite figures drawn on the coordinate plane.</td></tr>
<tr><td>Lesson 14</td><td>I can use coordinate geometry to classify triangles and quadrilaterals.
I can use coordinate geometry to solve real-world problems on the coordinate plane involving perimeter or area of polygons.
I can use coordinate geometry to solve real-world geometric problems involving circles, triangles, and quadrilaterals.</td></tr>
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Vertical Alignment

Algebra 1 Unit 4 Key Features of Linear Functions	→	**Geometry Unit 13 Coordinate Geometry**

Prior Course(s)	Course Level
MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.	**MA.912.GR.3.1** Determine the weighted average of two or more points on a line. **MA.912.GR.3.2** Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. **MA.912.GR.3.3** Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals. **MA.912.GR.3.4** Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons. **MA.912.GR.7.2** Given a mathematical or real-world context, derive and create the equation of a circle using key features.

13.1 Parallel and Perpendicular Lines

Lesson Introduction

Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.	Learning Target	• I can use coordinate geometry to classify parallel and perpendicular lines.

	Building On	Working Toward
Benchmark Coherence	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.	N/A

Essential Questions	• What characteristics do slopes of parallel lines have? • What characteristics do slopes of perpendicular lines have? • How can a linear equation be converted from standard form to slope-intercept form? • How can the slope be identified in slope-intercept form?
Lesson Narrative	The lesson guides students to discover that parallel lines have the same slope, and that the slopes of perpendicular lines are opposite reciprocals. Students work with graphs and equations to identify the slopes. The lesson also requires students to convert from the standard form of a linear equation to the slope-intercept form of an equation.

Component Summary	13.1.1 Warm-Up (~5 Min)	Artificial Intelligence and Algorithms *(SE p. 281)*	13.1.3 Guided Instruction (15 Min)	Formal Introduction of Parallel and Perpendicular Lines *(SE p. 284)*
	13.1.2 Collaboration (15 Min)	Deriving the Relationships for Parallel and Perpendicular Lines *(SE p. 282)*	13.1.4 Guided Practice (10 Min)	Practice on Identifying Parallel and Perpendicular Lines *(SE p. 286)*
	13.1.5 Wrap-Up (~5 Min)	Students summarize their learning *(SE p. 287)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Parallel • Perpendicular Additional Terms: • Slope	• Career Profile Video • (optional) Graphing Utility	• 13.1.1 Warm-Up contains a video to accompany the question prompts. • If choosing to make 13.1.2 question 13 into a Desmos Card Sort, create the Card Sort.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may make computational errors when converting from standard form to slope-intercept form. • Students may identify lines as perpendicular if the slopes are opposite, instead of opposite reciprocals.	• Consider using a graphing utility, such as Desmos or GeoGebra, with this lesson.
	Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
	Think-Pair-Share There are multiple opportunities highlighted throughout the lesson to Think-Pair-Share. Students work individually and then collaborate. Give students the time to think individually before sharing with a partner or group in the multiple exercises during the span of the lesson.	**MA.K12.MTR.5.1: Use Patterns and Structure** Lesson 13.1 requires students to apply patterns and structures to derive the relationships between parallel and perpendicular lines. Throughout the practice exercises in the lesson, students should look for similarities of characteristics of parallel and perpendicular lines.
Differentiate Instruction	Lesson 13.1 lends itself to differentiate to a visual approach. Consider either using an online graphing utility with the class, or having students use a graphing calculator on personal devices. The Collaboration involves a large amount of graphing and writing equations. Desmos can also highlight the relationship between standard form of a linear equation and the slope-intercept form of a linear equation.	

Lesson Notes

13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Component Overview: *Students read a short narrative and watch a video about artificial intelligence and algorithms. Then, they are asked to respond to three questions.*

13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Artificial intelligence enables a machine to simulate human behavior and decisions. Machine learning is when a machine uses artificial intelligence to learn from past data without a programmer having to re-program the machine.

Many of the women in the video used the word "algorithm."

1. Describe one task you would like a machine to be programmed to do.
   Answers will vary. Drive my car.

   Both artificial intelligence and machine learning rely on mistakes to help improve the programming.

2. Describe a mistake you have made that helped you improve.
   Answers will vary. Making mistakes helps me to see how a problem works correctly and then I can understand and remember the procedures for the problem.

Consider asking the following questions to guide student thinking and responses:

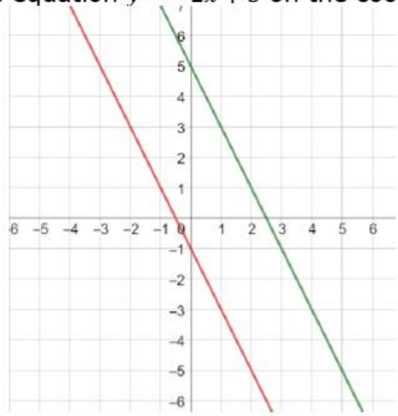
- *What are some forms of artificial intelligence that are used in everyday life?*
- *How are algorithms used in the study of mathematics?*

13.1.2 Collaboration: Parallel and Perpendicular Lines

Component Overview: *This Collaboration has students apply multiple skills for discovering the characteristics of parallel lines, perpendicular lines, and their slopes. Students will be required to graph by hand and convert from the standard form of a linear equation to the slope-intercept form of an equation.*

13.1.2 Collaboration: Parallel and Perpendicular Lines

1. Graph the equation $y = -2x + 5$ on the coordinate plane.



2. Rewrite the equation $6x + 3y = -3$ in slope-intercept form.
   $y = -2x - 1$
3. Graph the equation $6x + 3y = -3$ using slope-intercept form.
   See above
4. What similarities do the two equations have?
   They have the same slope.
5. What differences do the two equations have?
   They have different y-intercepts.

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify.

Consider having students draw slope triangles or count the slope on the graph to provide a visual approach.

Monitor student work when converting from standard form to slope-intercept form. Students may forget to subtract $6x$ from each side, and may forget to divide each term by 3 to isolate y.

6. Complete the statement.

   The equations $y = -2x + 5$ and $6x + 3y = -3$ resulted in

   - parallel
   - perpendicular
   - vertical

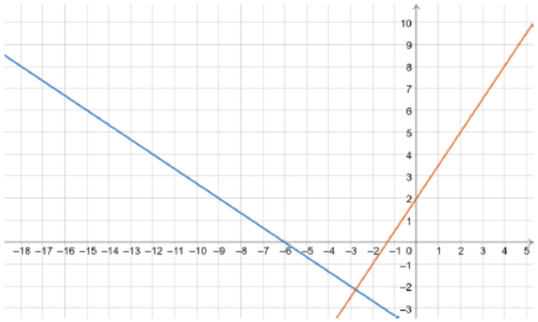
   lines.

   The slopes of the equations are

   - equal.
   - undefined.
   - opposite reciprocals.

7. Rewrite the equation $4y - 6x = 8$ in slope-intercept form.
   $y = \frac{3}{2}x + 2$

8. Graph the equation $4y - 6x = 8$ using slope-intercept form.



9. Graph the equation $y = -\frac{2}{3}x - 4$.
   See above.

10. What do you notice about the slopes in the two equations?
    They are the opposite reciprocal of each other.

For a kinesthetic approach to questions 8-9, consider having students work in partners. Have one student graph the equation on question 8, and have another student graph the equation on question 9.

Think-Pair-Share
Provide students with the opportunity to share with a partner what the students notice about the graph and the slopes.

13.1.2 Collaboration: Parallel and Perpendicular Lines (continued)

11. Complete the statement.

Equation $4y - 6x = 8$ and equation $y = -\frac{2}{3}x - 4$ form

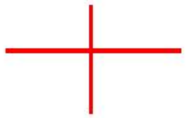
- o parallel
- o perpendicular
- o vertical

lines; therefore, the slopes of the equations are

- o equal.
- o undefined.
- o opposite reciprocals.

Think-Pair-Share
Have students complete questions 11 and 12 as a Think-Pair-Share. Give students a few seconds to ponder the two questions, and then have students share their responses.

12. Complete the table.

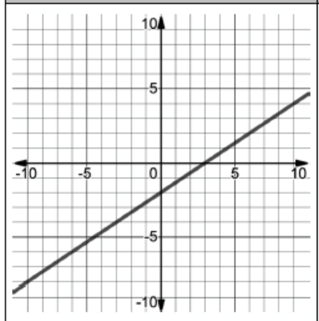
Type	Sketch	Slope
Parallel lines		*Equal*
Perpendicular lines		*Opposite Reciprocals*

13.1.2 Collaboration: Parallel and Perpendicular Lines (continued)

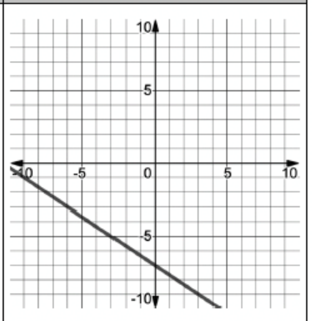
13. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel	*A & C*	*E & G*
Perpendicular	*B & D*	*F & H*

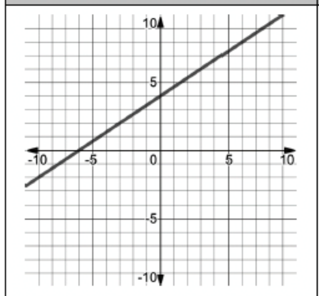
Line A



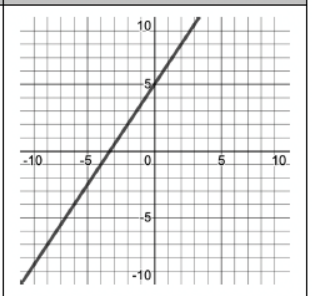
Line B



Line C



Line D



Equation E	Equation F
$y+2=\frac{2}{3}(x+9)$	$2x+3y=-21$
Equation G	**Equation H**
$-2x+3y=-6$	$y=\frac{3}{2}x+5$

For a challenge, consider having students write the equations of each line by only giving them the graphs.

Consider making question 13 into a Desmos Card Sort for a digital extension.

For Lines A-D, ask students about the strategies they will choose to sort the lines.

For Equations E-H, ask students to share their strategies for matching the graphs to the equations.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines

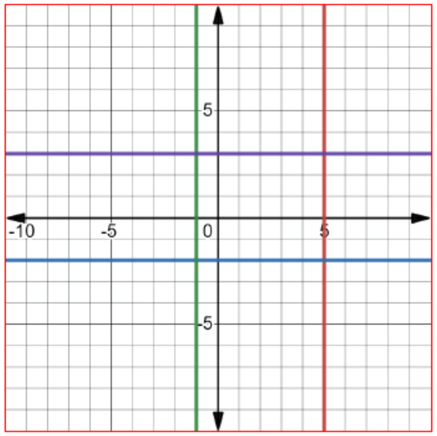
Component Overview: *Students work through two questions on parallel and perpendicular lines.*

13.1.3 Guided Instruction: Parallel and Perpendicular Lines

Lines are **parallel** if the slopes of the lines are the same.

Lines are **perpendicular** if the slopes are opposites and reciprocals.

1. Four lines are shown on the coordinate plane.



a. What do you notice about the four lines?
Answers will vary. The purple and blue lines are parallel. The green and red lines are parallel. The purple line is perpendicular to the green and red lines, the blue line is perpendicular to the green and red lines.

This section of the lesson provides students the opportunity to refine and apply what was learned in the Guided Discovery. Question 1 gives students a graph and asks students to apply properties with a rectangle. On question 2, students are given a graph and must find whether two lines are parallel. Allow students the opportunity to discuss what they notice about the four lines in question 1.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines (continued)

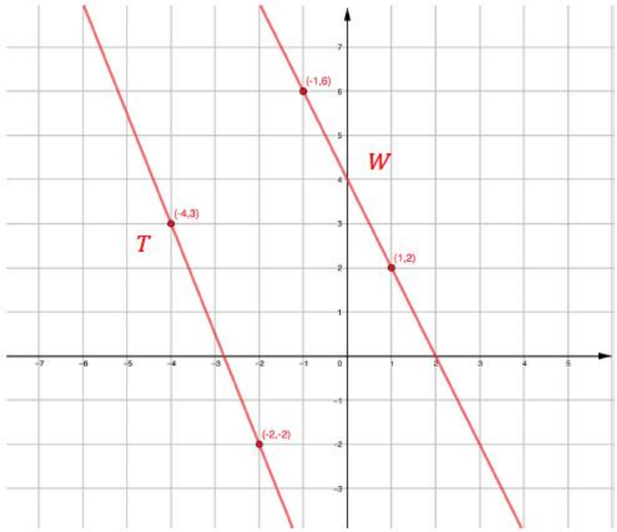
b. In the graph of the four lines, the four lines create a rectangle. Explain how each of the properties of a rectangle could be shown.

Property	Explanation
Opposite sides are parallel.	*The purple and blue lines are opposite each other and parallel because they have the same slope. The green and red lines are opposite each other and parallel because they have the same slope.*
All angles are right angles.	*The purple and blue lines are perpendicular (opposite reciprocal slopes) to the red line forming eight right angles. The purple and blue lines are perpendicular (opposite reciprocal slopes) to the green line forming eight right angles.*

Consider making question 1b a Think-Pair-Share.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines (continued)

2. Line W has points $(-1,6)$ and $(1,2)$ on it. Line T has points $(-4,3)$ and $(-2,-2)$ on it.



   a. Graph line W and line T.

   b. Determine if line W and line T are parallel lines, perpendicular lines, or neither. Justify your answer with math.

      The slope of $W = \frac{2-6}{1+1} = -2$

      The slope of $T = \frac{-2-3}{-2+4} = \frac{-5}{2}$

      The lines are neither parallel nor perpendicular because their slopes are neither the same nor opposite reciprocals.

Consider having students draw slope triangles on the graph.

For a challenge, have students create and graph a line that is parallel to either line W or line T. For an additional extension, have students graph a line that is perpendicular to either line W or line T.

3. Line N is written as $y = \frac{1}{3}x - 8$. Line V is written as $6x + 2y = 4$.

   Determine if line N and line V are parallel lines, perpendicular lines, or neither. Justify your answer with math.

   $y = \frac{1}{3}x - 8$

   $6x + 2y = 4$

   $2y = -6x + 4$

   $y = -3x + 2$

   The lines are perpendicular because their slopes are opposite reciprocals.

Monitor students' work on question 3, as there are multiple errors that can be made when converting from standard form to slope-intercept form.

13.1.4 Guided Practice: Parallel and Perpendicular Lines

Component Overview: *Students work with a partner or group on two separate tasks of determining parallel and perpendicular lines.*

13.1.4 Guided Practice: Parallel and Perpendicular Lines

1. Given that line M is $3x - y = 24$, determine if the following lines are parallel to line M.

   Complete the table.

Equation of Line	Work	Is the line parallel to line M?
$y = 3x + 8$	$y = 3x - 24$, *slope* $= 3$ $y = 3x + 8$, *slope* $= 3$	○ Yes ○ No
$y = -3x - 1$	$y = 3x - 24$, *slope* $= 3$ $y = -3x - 1$, *slope* $= -3$	○ Yes ○ No
$3y = x - 4$	$y = 3x - 24$, *slope* $= 3$ $y = \frac{1}{3}x - \frac{4}{3}$, *slope* $= \frac{1}{3}$	○ Yes ○ No
$3x - y = 3$	$y = 3x - 24$, *slope* $= 3$ $y = 3x - 3$, *slope* $= 3$	○ Yes ○ No

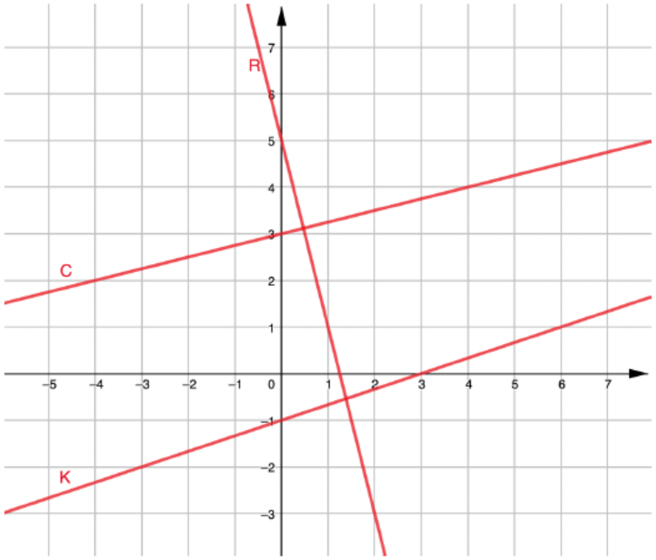
Consider asking the following question to students as they begin 13.1.4.

After converting $x - y = 24$ from standard form to slope-intercept form, what strategies can you use to determine whether the lines are parallel to line M?

Look for students who may have a common error of forgetting to divide by -1 after subtracting x from each side when converting $x - y = 24$ from standard form to slope-intercept form.

For students who finish early, challenge them to create a linear function that would be parallel and perpendicular to $x - y = 24$

2. Line R is written as $4x + y = 5$, line K is written as $12y = 4x - 12$, and line C is written as $4y - 12 = x$.



a. Graph lines R, K, and C.
Images drawn on graph.

b. Determine which two lines are perpendicular lines. Justify your answer with math.

R; $y = -4x + 5$, *slope* $= -4$

K; $y = \frac{1}{3}x - 1$, *slope* $= \frac{1}{3}$

C; $y = \frac{1}{4}x + 3$, *slope* $= \frac{1}{4}$

Lines R and C are perpendicular to each other because their slopes are opposite reciprocals.

Consider asking students the following question before beginning question 2:
Which lines do you think will be perpendicular to one another? Explain your reasoning.

Consider having students do a Think-Pair-Share for questions 2a and 2b. Have students work silently, then pair with another classmate to share.

13.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

13.1.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

   Answers will vary.

13.2 Circles on the Coordinate Plane

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals. MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.	Learning Targets	• I can use coordinate geometry to determine parts of a circle in a mathematical context. • I can draw a circle using given definitions, properties, or theorems in a mathematical context. • I can use coordinate geometry to solve mathematical problems involving perimeter and area of polygons

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How do you find the lengths of segments of circles on the coordinate plane? • How can you use coordinate geometry to determine parts of a circle? • How do you draw a circle on the coordinate plane given definitions, properties, or theorems in a mathematical context?
Lesson Narrative	Lesson 13.2 requires students to apply theorems relating to segments in circles to find the lengths of segments on the coordinate plane. This lesson is an extension of the study of circles in Unit 10. The lesson is also an extension of Lesson 13.1, as students are required to find slopes and write equations of lines in point-slope form and slope-intercept form.

Component Summary				
	13.2.1 Warm-Up (~5 min)	Bell Work *(SE p. 288)*	13.2.3 Try It (20 min)	Practice on finding segments in circles on the coordinate plane *(SE p. 291)*
	13.2.2 Exploration (20 min)	Discovering the lengths of segments and parts of circles on the coordinate plane *(SE p. 289)*	13.2.4 Wrap-Up (~5 min)	Students summarize their learning *(SE p. 293)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • N/A <u>Additional Terms:</u> • Chord • Secant • Slope • Tangent	• (Optional) Graphing Utility	• If planning on using the 13.2.3 Try It as a group activity, prepare the activities for the groups.

	Common Misconceptions	Things to Consider
Teacher Tips	• Student may struggle with finding coordinates of points when the coordinates are not given. • Students may not substitute coordinates correctly into the formula for point-slope form. • Students may struggle with recalling prior knowledge on secant and tangent lines to help them find the lengths of segments.	• Consider allowing students to use a graphing utility to enhance Lesson 13.2. • Consider reviewing the distance formula and the Pythagorean Theorem, if needed.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect Lesson 13.2 gives an opportunity for students to compare the content of the lesson to previous standards on circles and equations of lines. Students will need to apply knowledge and theorems from previous lessons to complete the tasks in 13.2, and to examine various methods for solving problems.	**MA.K12.MTR.4.1: Discussion and Reflection** Students should be engaging in discussions throughout the lesson and should be able to communicate their strategies and methods for solving problems. Students should also be asking questions to their peers and justifying their answers. Lesson 13.2 provides an opportunity for students to engage in discussions in mathematics.
Differentiate Instruction	There are several parts to this lesson that can be differentiated. Consider structuring the Try It Section into a Station Rotation and differentiate by ability level. Question 1 would be a beginner level. Question 2 would be the advanced level, and question 3 would be the intermediate level. Consider assigning students into groups of 2-3 and have two to three groups per question for potentially up to 9 groups working at the same time. Give students a determinate amount of time to complete the problem at their station. Reform groups of 3, where each person in the new group has completed a separate problem and students become the teachers of their problem.	

Lesson Notes:

13.2.1 Warm-Up: Bell Work

Component Overview: *Students solve a question to determine which equation of a line is perpendicular to a given equation. This question relates to lesson 13.1.*

13.2.1 Warm-Up: Bell Work

1. Which equation represents a line that is perpendicular to the line $y = -\frac{1}{3}x + 3$?

   A. $3x + y = 4$
   B. $x - 3y = 24$
   C. $y - 3x = -4$
   D. $x + 3y = 3$

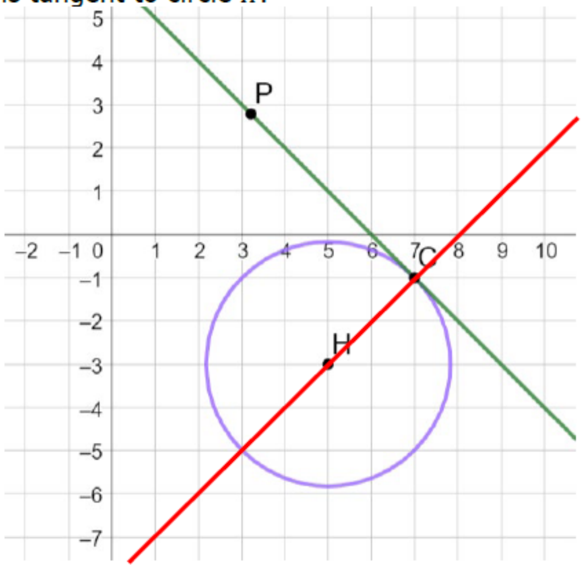
Use student understanding to inform your instruction, and to determine where during this and further lessons, remediation may be needed. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

13.2.2 Exploration: Circles on the Coordinate Plane

Component Overview: *This Collaboration guides students through finding the lengths of segments and parts of circles on the coordinate plane. Students apply skills from Unit 10, as well as Lesson 13.1. Students will be expected to write equations of lines, find lengths and distances of segments of circles, and find areas and perimeters of triangles.*

13.2.2 Exploration: Circles on the Coordinate Plane

1. Circle H is shown, where point C is on the circle and line PC is tangent to circle H.



a. Use a straightedge to draw a line that goes through points C and H.
See above.

b. Discuss with your partner how the slope of tangent line PC and the slope of the radius CH are related.
The slopes of PC and CH are opposite reciprocals.

c. What property of circles explains this relationship?
If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.

d. Write the equation of line PC.
$y = -x + 6$

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify.

Students may need a refresher on properties of tangent lines. Use the following questions to refresh student memory:

- *How do you find the slopes of CH and PC?*
- *What do you notice about the slopes?*
- *What type of angle is formed when a tangent line meets a radius?*

e. Write the equation of line CH.
$y = x - 8$

f. Line PC and line CH intersect at point $C\ (7, -1)$. Use the equations of lines PC and CH to confirm this point.
$-1 = (-7) + 6$
$-1 = 7 - 8$

2. Circle M has a center at $(-2,1)$ and a tangent line with the equation $y = 2x - 5$.

a. Write the equation of the line perpendicular to the tangent line.
$1 = \frac{-1}{2}(-2) + b; \quad b = 0$
$y = \frac{-1}{2}x$

b. Find the point where the tangent line touches circle M. Call this point R.
$\frac{-1}{2}x = 2x - 5; \quad x = 2;\ y = -1$

c. What is the radius of circle M?
$\sqrt{(2+2)^2 + (-1-1)^2} = \sqrt{20} = 2\sqrt{5}$

d. Write the equation of circle M.
$(x+2)^2 + (y-1)^2 = 20$

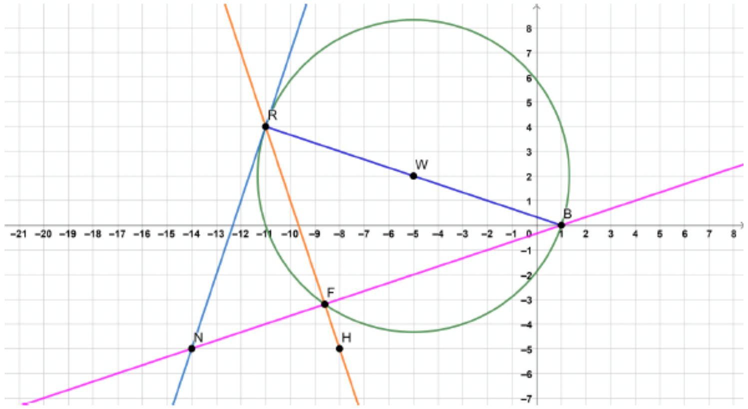
Consider using a graphing utility for questions 1d-1f, and 2a-2d.

Consider reviewing with students how to write the equation of a line when given the point and the slope. Use MLR 7, and have students Compare and Connect from previous content to the content of the current lesson.

How can you find the length of the radius of circle M?

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

3. Circle W is shown, with secant lines RH and NB, tangent line NR, and chord RB.



a. Complete the table.

Line Segment	Length
$\overline{NB}$	$d = \sqrt{(-5 - 0)^2 + (-14 - 1)^2} = 5\sqrt{10}$
$\overline{NR}$	$d = \sqrt{(-5 - 4)^2 + (-14 + 11)^2} = 3\sqrt{10}$

b. Determine the length of $\overline{NF}$.

$(NR)^2 = (NF)(NB)$

$\left(3\sqrt{10}\right)^2 = (NF)\left(5\sqrt{10}\right)$

$NF = \frac{18}{\sqrt{10}}$

To help students get started with the table, ask the following questions:

- How do you find the length of NB?
- How do you find the length of NR?
- What strategies can you use to find the lengths of NB and NR?

Common Misconception

Students may struggle with finding NF because point F is not a whole number, and cannot easily be identified on the coordinate plane. Have students find the equations of lines RH and NB. Set the equations equal to find point F. Then, use the distance formula to find NF.

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

c. Complete the table.

Line	Slope	Equation in Point-Slope Form
$\overleftrightarrow{NR}$	3	$y+5=3(x+14)$
$\overleftrightarrow{RH}$	-3	$y-4=-3(x+11)$
$\overleftrightarrow{NB}$	$\frac{1}{3}$	$y+5=\frac{1}{3}(x+14)$

d. Rewrite the equation of $\overleftrightarrow{RH}$ in slope-intercept form.
$y=-3x-29$

e. Rewrite the equation of $\overleftrightarrow{NB}$ in slope-intercept form.
$y=\frac{1}{3}x-\frac{1}{3}$

f. Use the slope-intercept forms of $\overleftrightarrow{RH}$ and $\overleftrightarrow{NB}$ to find point F.
$-3x-29=\frac{1}{3}x-\frac{1}{3}$; $(-8.6,-3.2)$

g. Work with your partner to determine if ΔBRN is isosceles. Show your work as justification.
It is not isosceles.
$BR=\sqrt{(-11-1)^2+(4-0)^2}=4\sqrt{10}$
$RN=\sqrt{(-5-4)^2+(-14+11)^2}=3\sqrt{10}$
$NB=\sqrt{(-5-0)^2+(-14-1)^2}=5\sqrt{10}$

Consider giving students the formula for Point-Slope Form to assist with question 3c.

Common Misconception
Monitor students' work when substituting coordinates into point-slope form. Continue to monitor students when rewriting the equations of lines RH and NB in slope-intercept form.

Ask the following questions to guide students on question G:

- *What mathematical strategy would be used to determine whether triangle BRN is isosceles?*
- *What is the defining characteristic of an isosceles triangle?*

13.2.2 Exploration: Circles on the Coordinate Plane (continued)

h. What could be used to justify that ΔRFB is a right triangle?

The slopes can be used to justify that the triangle is a right triangle. If two lines have opposite reciprocal slopes then they are perpendicular and thus intersect at a right angle, creating a right triangle. The slopes of $\overline{RF}$ $(m=-3)$ *and* $\overline{FB}$ $\left(m=\frac{1}{3}\right)$ *are perpendicular.*

i. What is the area of ΔRFB?

$$RF=\sqrt{(-11+8.6)^2+(4+3.2)^2}=\sqrt{57.6}$$

$$NB-NF=FB;\ 5\sqrt{10}-\frac{18}{\sqrt{10}}=\frac{32}{\sqrt{10}}$$

$Area=\frac{1}{2}(RF)(FB);\quad Area=38.4$ *square units*

j. What is the perimeter of ΔRFN?

Perimeter $=RF+FN+NR$

$RF+FN+NR=\sqrt{57.6}+\frac{18}{\sqrt{10}}+3\sqrt{10}\approx 22.768$ *units*

If necessary, remind students of the formulas for the area and the perimeter of a triangle.

13.2.3 Try It: Circles on the Coordinate Plane

Component Overview: *Students will work on various practice problems relating to circles on the coordinate plane. Question 1 has students find a tangent point and the equation of a circle. On question 2, students find distances of segments of circles, then find slopes. Question 3 requires students to find distances, and write the equation of a circle.*

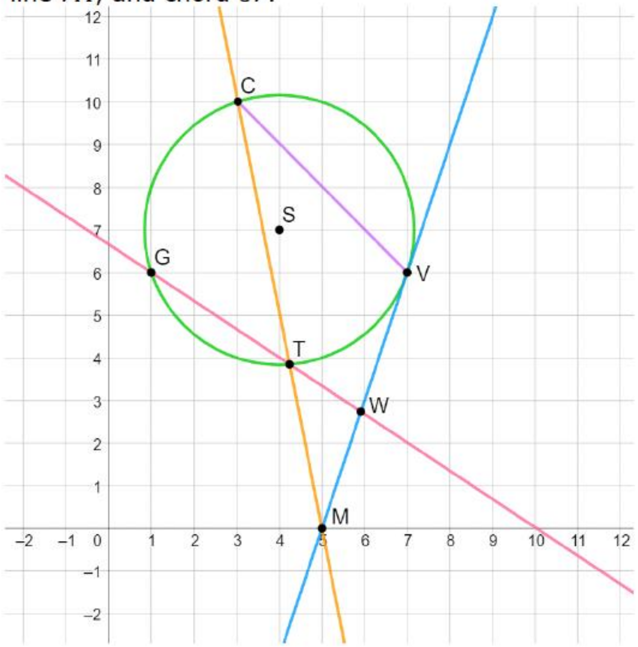
13.2.3 Try It: Circles on the Coordinate Plane

1. Circle G has a center at $(9, -2)$ and a tangent line with the equation $3x + 2y = 36$.

   a. At what point does the tangent line touch circle G?
   $\frac{-3}{2}x + 18 = \frac{2}{3}x - 8;\ (12, 0)$

   b. Write the equation of circle G.
   $(x - 9)^2 + (y + 2)^2 = 13$

2. Circle S is shown with secant lines CM and GW, tangent line VM, and chord CV.



Consider having students use a graphing utility to construct question 1 for a visual approach. Providing the visual approach can also help refresh students' memories with circle properties.

For a differentiation idea for 13.2.3, see notes in "Differentiate Instruction."

Use the following questions to guide students through question 2:

- *What strategy can be used to find the lengths of VM, CV, and CM?*
- *Given CM, how can the length of TM be determined?*

13.2.3 Try It: Circles on the Coordinate Plane (continued)

a. Complete the table.

Line Segment	Length
$\overline{VM}$	$d=\sqrt{(7-5)^2+(6-0)^2}=2\sqrt{10}\approx 6.325$
$\overline{CV}$	$d=\sqrt{(3-7)^2+(10-6)^2}=4\sqrt{2}\approx 5.657$
$\overline{CM}$	$d=\sqrt{(3-5)^2+(10-0)^2}=2\sqrt{26}$ ≈ 10.198
$\overline{MT}$	$\left(2\sqrt{10}\right)^2 = MT\left(2\sqrt{26}\right)$ $MT\approx 3.922$

b. What is the perimeter of ΔCMV?
Perimeter $= CV+VM+MC$
$CV+VM+MC\approx 22.18$ units

c. Complete the table.

Line	$\overleftrightarrow{GW}$	$\overleftrightarrow{CM}$
Slope	$\frac{-2}{3}$	-5
Equation in Point-Slope Form	$y-6=\frac{-2}{3}(x-1)$	$y-0=-5(x-5)$
Equation in Slope-Intercept Form	$y=\frac{-2}{3}x+\frac{20}{3}$	$y=-5x+25$

d. Use the slope-intercept form of both lines to find point T.
$\frac{-2}{3}x+\frac{20}{3}=-5x+25$
$(4.231, 3.846)$

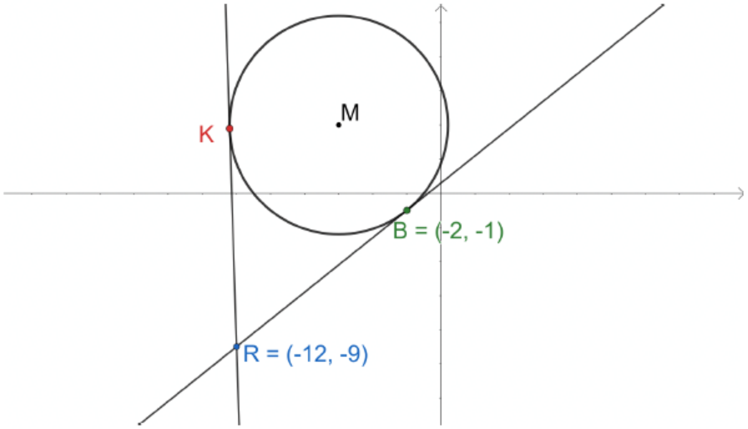
Common Misconception
Students may struggle with finding MT since it requires students to use their knowledge on finding segment lengths of secants and tangents of a circle.

Since point W is not a whole number, consider guiding students through finding the slope of GW.

Students may make errors with fractions when writing the equations of lines for line GW.

13.2.3 Try It: Circles on the Coordinate Plane (continued)

3. Circle M is shown with tangents $\overline{RK}$ and $\overline{RB}$.



a. What is the length of $\overline{RK}$?
$RB = RK$
$RB = \sqrt{(-2+12)^2 + (-1+9)^2} = 2\sqrt{41} \approx 12.806$

b. The center of the circle is located at $(-6, 4)$. Write the equation of circle M.
$r = \sqrt{(-6+2)^2 + (4+1)^2} = \sqrt{41}$
$(x+6)^2 + (y-4)^2 = 41$

c. What type of quadrilateral is $KMBR$? Justify your choice.
Quadrilateral $KMBR$ is a kite because the measure of two pairs of adjacent sides are equal.
$KM = MB$; $KR = BR$

d. What is the distance between point R and point M?
$\left(2\sqrt{41}\right)^2 + \left(\sqrt{41}\right)^2 = \sqrt{205}$; $RM \approx 14.318$

Consider asking the students the following questions to assist with question 3:

- *What segment is RK congruent to in the diagram?*
- *Why is RB congruent to RK?*
- *Given the center of the circle, how can the length of the radius be found to write the equation of the circle?*

What characteristics determine that quadrilateral $KMBR$ is a kite?

For a challenge, have students find the coordinates of point K.

13.2.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

13.2.4 Wrap-Up: Cool Down

1. Write a note to a classmate who might have missed today's lesson, in your own words, about circles on the coordinate plane.

   Answers will vary.

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.

13.3 Determining Points on a Circle

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals. MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.
Learning Targets	• I can determine whether a point is on the circle, inside the circle, or outside the circle. • I can find the center of a circle when given two points. • I can write the equation of a circle given the endpoints of a diameter.

Benchmark Coherence	Building On	Working Toward
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.	N/A

Essential Questions

- How can you find the center of a circle when given two points?
- What is the relationship between the radius and where a point lies within a circle?
- How can you determine whether a point lies on the circle, on the inside, or on the outside of the circle?
- How do you write the equation of a circle given the endpoints of the diameter?

Lesson Narrative

The lesson guides students on how to determine whether a given point is on the circle, inside the circle, or outside of the circle. Students are also expected to be able to write equations of circles and graph circles on the coordinate plane, referring to Unit 10. Students will also discover how to find the center of the circle when given two endpoints, and how to write the equation of a circle given the endpoints of the diameter.

Component Summary

13.3.1 Warm-Up (~5 min.)	Notice and Wonder *(SE p. 294)*	13.3.3 Try It (20 min.)	Practice on finding points on a circle *(SE p. 298)*
13.3.2 Exploration (20 min.)	Discovery on determining if a point lies inside, outside, or on a circle *(SE p. 295)*	13.3.4 Wrap-Up (~5 Min)	Ticket Out the Door *(SE p. 299)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• (Optional) Graphing utility	• If allowing students to use a graphing utility, prepare this before class starts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with simplifying equations to determine if the simplified equation is equal to, greater than, or less than the radius squared.
Students may struggle with using the distance formula to find the diameter in 13.2.2, on question 2a.</td><td>Consider providing graph paper, or allow students to use a graphing utility, to help visualize circles that are not graphed in the lesson, and/or that have radii that are not whole numbers.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
Lesson 13.3 has multiple opportunities for students to notice and wonder. 13.3.1 has a Notice and Wonder to start the lesson, and 13.3.2 gives multiple opportunities for students to notice and wonder to create the conjectures that students are expected to form.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Lesson 13.3 requires that students work both individually, and with other students. MTR 1 states that students should support one another and ask questions when working on a mathematical task. Students are expected to ask Notice and Wonder questions throughout the lesson, tying in the Literacy and Language Routine.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider allowing students to use a graphing utility for much of this lesson. Using a graphing utility can help students better determine the relationship between the endpoints of a circle and the center. Using a graphing utility can also help students determine where a point lies on the circle, inside the circle, and outside the circle.</td></tr>
<tr><td colspan="3">Lesson Notes</td></tr>
</table>

13.3.1 Warm-Up: Notice and Wonder

Component Overview: *Students read a prompt about the picture and make observations on things they notice and wonder.*

13.3.1 Warm-Up: Notice and Wonder

Castlerigg Stone Circle is one of the earliest stone circles to be found in Britain and is important in terms of megalithic astronomy and geometry. It is composed of 38 free standing stones, some measuring up to 10 feet high and dating back to the Neolithic period, 4000 to 5000 years ago.

1. What do you notice?
   Answers will vary. I notice there are no other rocks except the ones making the circle.

2. What do you wonder?
   Answers will vary. I wonder how they moved them and placed them in this location.

Consider allowing students to share their Notice and Wonder with their classmates, if time allows, as this is related to the Literacy and Language Instructional Routine for this lesson.

13.3.2 Exploration: Points on a Circle

Component Overview: *This Exploration guides students through determining if coordinate points are on a circle or are on the inside or outside of a given circle. Students will make conjectures about the relationships between the equation of a circle and the location of a point. Students will use this distance formula to find the diameter, and the midpoint formula to find the radius.*

13.3.2 Exploration: Points on a Circle

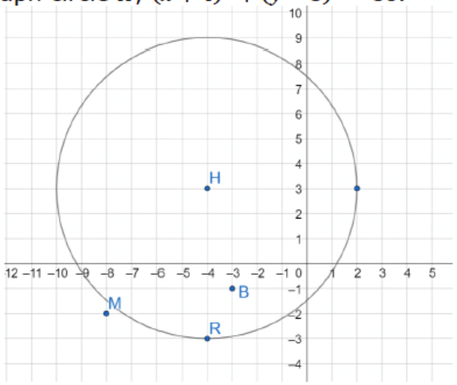
1. The equation of circle H is $(x+4)^2 + (y-3)^2 = 36$.

   a. Complete the table by substituting the given coordinates for x and y in the equation of circle H. Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$M(-8,-2)$	$(-8+4)^2 + (-2\ -\ 3)^2 = 36$ $41 \neq 36$	$41 = 36$
$B(-3,-1)$	$(-3+4)^2 + (-1\ -\ 3)^2 = 36$ $17 \neq 36$	$17 = 36$
$R(-4,-3)$	$(-4+4)^2 + (-3\ -\ 3)^2 = 36$ $36 = 36$	$36 = 36$

   b. What do you notice about the simplified equations?
   Answers will vary.
   Two of the simplified equations are false statements and one is a true statement.

   c. Graph circle H, $(x+4)^2 + (y-3)^2 = 36$.



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before releasing students to do 13.3.3 Try It. While circulating, look for students who may need assistance when substituting the given coordinate for x and y. Consider modeling the first coordinate for these students.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

d. Plot points $M(-8,-2)$, $B(-3,-1)$, and $R(-4,-3)$.
See graph.

e. What do you notice about the relationship between the location of the points and their simplified equations?
Answers will vary. If the equation simplifies to the radius squared, then the point is on the circumference of the circle. If it simplifies to a number less than the radius squared, it's inside the circle. If it simplifies to a number greater than the radius squared, then it's outside the circle.

f. Write a conjecture about the location of a point and the equation of the circle.
Answers will vary. If the coordinates are substituted into the equation and the equation is simplified to a true statement, then the point lies on the circumference of the circle.

g. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjecture. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Conjecture	Initials
Answers will vary.		

h. Review your conjecture with your partner. If necessary, make any changes.

For a visual approach, have students make a table with the conjectures. See the table below. This may help students organize their conjectures.

Radius Squared	*Location of Point*
> Radius squared	*Outside the circle*
< Radius squared	*Inside the circle*
= Radius squared	*On the circle*

Give students time to discuss the conjectures, as this is the guiding point of the lesson. Allowing time for students to develop the conjectures will also tie to MTR 1.

13.3.2 Exploration: Points on a Circle (continued)

2. The equation of circle K is $(x+6)^2+(y+5)^2=25$ with points $N(-6{,}0)$, $V(-3,-1)$, $T(-1,-5)$, and $R(-9,-9)$ on the circle.

   a. What is the length of the diameter of circle K?
   $r=\sqrt{(-6+6)^2+(-5+0)^2}=5;\ d=10$

   b. Complete the table by finding the distance between the given points.

Points	Work	Distance
$N(-6{,}0)$ and $R(-9,-9)$	$d=\sqrt{(-6+9)^2+(0+9)^2}$	$\sqrt{90}\approx 9.487$
$V(-3,-1)$ and $R(-9,-9)$	$d=\sqrt{(-3+9)^2+(-1+9)^2}$	$\sqrt{100}=10$
$T(-1,-5)$ and $R(-9,-9)$	$d=\sqrt{(-1+9)^2+(-5+9)^2}$	$\sqrt{80}\approx 8.944$

   c. What do you notice about the distances between the points?
   Answers will vary. Only one point has a distance from point R that is equal to the length of the diameter.

   d. What can you conclude about points V and R?
   Points V and R are endpoints of a diameter of the circle.

Some students may struggle with finding the diameter of the circle. To help students, have the students find the radius using the equation, and then multiply by 2.

Consider employing a jigsaw routine with question 2b and assigning students one of the three problems to complete. Then, have students share the results of their problems with the class.

While circulating, ask students to share their notice and conclusion with you.

13.3.2 Exploration: Points on a Circle (continued)

3. Circle W has a radius of $\sqrt{29}$ and points $S(2,-2)$, $D(5,1)$, and $H(12,-6)$.
   a. Determine which two points are the endpoints of a diameter of circle W. Show your work.
      Diameter $= 2(\sqrt{29})$
      $SH = \sqrt{(2-12)^2 + (-2+6)^2} = \sqrt{116} = 2\sqrt{29}$
      Points S and H are the endpoints of a diameter of circle W.
   b. Describe how to find the center of the circle using the two endpoints of the diameter.
      Use the midpoint formula to determine the midpoint of the diameter
   c. What is the center of the circle?
      $\left(\frac{2+12}{2}, \frac{-2-6}{2}\right) = (7,-4)$
   d. Write the equation for circle W.
      $(x-7)^2 + (y+4)^2 = 29$

4. The points $G\ (3,-1)$ and $H\ (3,3)$ are the endpoints of a diameter of circle M. Write the equation of circle M.
   $\left(\frac{3+3}{2}, \frac{-1+3}{2}\right) = (3,1)$

   $GH = \sqrt{(3-3)^2 + (-1-3)^2} = 4;\ r = 2$

   $(x-3)^2 + (y-1)^2 = 4$

Consider asking partner pairs the following question to guide students through question 3:
- *How can you determine which two points are the endpoints of the diameter?*

Consider allowing students to use graph paper or an online graphing utility as a visual for question 4.

Consider asking partner pairs the following question to guide students through question 4:
- *How can you determine the coordinates of the radius?*
- *How do you find the length of the radius when given the coordinate?*
- *What two pieces of information are needed to write the equation of a circle?*

13.3.3 Try It: Points on a Circle

Component Overview: *Students are working on skills that were completed during the 13.3.2 Exploration. Students should work on the questions individually.*

13.3.3 Try It: Points on a Circle

1. The equation of circle P is $(x-8)^2+(y-2)^2=13$.

   Determine whether the following points lie inside, outside, or on circle P.

Point	Work	Location
$(5,-1)$	$(5-8)^2+(-1-2)^2=18$ $18>13$	○ Inside circle P ○ Outside circle P ○ On circle P
$(6,5)$	$(6-8)^2+(5-2)^2=13$ $13=13$	○ Inside circle P ○ Outside circle P ○ On circle P
$(12,2)$	$(12-8)^2+(2-2)^2=16$ $16>13$	○ Inside circle P ○ Outside circle P ○ On circle P
$(11,0)$	$(11-8)^2+(0-2)^2=13$ $13=13$	○ Inside circle P ○ Outside circle P ○ On circle P
$(9,4)$	$(9-8)^2+(4-2)^2=5$ $5<13$	○ Inside circle P ○ Outside circle P ○ On circle P

Before releasing students to complete the questions, consider debriefing student questions, and addressing misconceptions you noted in 13.3.2 Exploration. Also, consider highlighting student voices to share rich discussions or comments overhead while they collaborated.

If students need further support, consider having students work with a partner or small group using the Instructional Routine, "Take Turns". Each student will take a turn in identifying the location of each point. The other student(s) will agree or disagree with the work, and can use this time to ask clarifying questions.

13.3.3 Try It: Points on a Circle (continued)

2. Circle T and circle R intersect at point H.
   Equation of circle T is $x^2 + (y-7)^2 = 20$
   Equation of circle R is $(x-7)^2 + (y-3)^2 = 25$

   a. Which coordinate pair is the location of point H?
      - $(-4,5)$
      - $(7,8)$
      - $(2,3)$
      - $(4,9)$

      $(2)^2 + (3 - 7)^2 = 20$
      $(2 - 7)^2 + (3 - 3)^2 = 25$

3. Circle S has a radius of 10 and points $F(-6,-6)$, $C(8,8)$, $W(10,2)$, and $M(-8,-4)$ on the circle.

   a. Determine which two of the given points are endpoints of a diameter of circle S.
      Diameter $= 20$
      $CM = \sqrt{(-8 - 8)^2 + (-4 - 8)^2} = 20$

      $\overline{\text{CM}}$ is a diameter of Circle T

   b. Find the center of circle S.

      $\left(\frac{8 - 8}{2}, \frac{8 - 4}{2}\right) = (0, 2)$

   c. Write the equation of circle S.
      $x^2 + (y - 2)^2 = 100$

- *What is the relationship between the location of the point and the equation of the circle?*
- *How can you determine if the point is on the circumference of a circle?*
- *How can you determine the point of intersection of the two circles?*

Consider allowing students to use a graphing utility for questions 2 and 3 to provide as a visual.

13.3.4 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a question to determine mastery of the lesson.*

13.3.4 Wrap-Up: Ticket Out the Door

1. The equation of circle W is $x^2 + (y + 4)^2 = 26$.

   Select all the points that lie on circle W.

   - ☐ $(3,0)$
   - ☐ $(5,-3)$ $(5)^2 + (-3 + 4)^2 = 26$
   - ☐ $(1,1)$ $(1)^2 + (1 + 4)^2 = 26$
   - ☐ $(-3,-2)$
   - ☐ $(-5,-5)$ $(5)^2 + (-5 + 4)^2 = 26$

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

For students who finish early, have students create an equation of a circle, and three coordinate points that lie on the circle.

13.4 Partitioning Line Segments

Lesson Introduction

Benchmark	MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Target	• I can use coordinate geometry to partition a line segment.

	Building On	Working Toward
Benchmark Coherence	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.	N/A

Essential Questions	• How can you use coordinate geometry to partition a line segment? • What is a partitioned line segment?
Lesson Narrative	Students are exploring how to find a point that is used to partition a directed line segment by a given ratio. The Exploration part of the lesson guides students into finding ratios of partitions of lines. Students construct a conjecture that will be used in later parts of the lesson. Students will also relate the midpoint to partitioning line segments.

Component Summary	13.4.1 Warm-Up (~5 mins.)	Heroic person for students *(SE p. 300)*	13.4.3 Guided Instruction (15 mins.)	Practice problems on partitioning line segments *(SE p. 302)*
	13.4.2 Exploration (20 mins.)	Guided Discovery on partitioning directed line segments by a given ratio *(SE p. 300)*	13.4.4 Try It (5 mins.)	Two questions on partitioning line segments *(SE p. 304)*
	13.4.5 Wrap Up (~5 mins.)	Students summarize their learning *(SE p. 305)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• N/A	• If necessary, prepare visuals for students before the lesson.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with visualizing part:part and part:whole ratios.
Students may struggle with writing ratios and identifying the denominators throughout the lesson.
Students may struggle with determining which endpoint belongs to which part of the ratio.</td><td>Consider providing visuals for students to demonstrate the concept of partitioning.
Students should not be required to memorize a formula. Relating the formula to student conjectures and to similar triangles will help students in understanding why the formula works.
If time is an issue, then consider assigning 13.4.4 Try It for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Math Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
Allow students to use the two methods of partitioning a directed line segment: the algebraic representation that is derived in the lesson, and the visual representation using slope. Have students make connections between the two methods, and have students demonstrate why both methods work.</td><td>MA.K12.MTR.2.1: Multiple Representations
Students use graphs and equations to partition directed line segments, emphasizing the use of multiple representations. Having students use both an algebraic and a visual representation of partitioning a segment helps students make connections between the two representations.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider using visuals to demonstrate how to partition a segment by using such graphing utilities as Desmos and GeoGebra. Allowing students to see visuals of partitioning may help students with determining the whole:part and part:part ratios in a directed line segment.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.4.1 Warm-Up: Heroic Person

Component Overview: *Students answer two questions regarding a personal hero.*

13.4.1 Warm-Up: Heroic Person

1. Who do you consider heroic and inspirational?
   Answers will vary.

2. Why did you pick this person?
   Answers will vary.

Consider having a few students who feel comfortable to share their responses with the class. Also, consider taking the time to read student choices, and write a short note to each student about their hero.

13.4.2 Exploration: Partitioning a Line Segment

Component Overview: *Students will begin working with a line segment and will use ratios to partition a directed line segment into a given ratio.*

13.4.2 Exploration: Partitioning a Line Segment

1. Directed line segment $\overline{SP}$ is shown.

   a. How many equal parts is $\overline{SP}$ partitioned into?
   5

   b. Make a mark where $\overline{SP}$ is partitioned into a $2:3$ ratio. Call this point R.
   See above.

   c. Complete the statement.

   Point R is [○ $\frac{2}{3}$ / ○ $\frac{2}{5}$ / ○ $\frac{3}{5}$] of the distance from S to P.

   d. Complete the equation that can be used to find the distance of $\overline{SR}$.

   $SR = \frac{2}{5} SP$

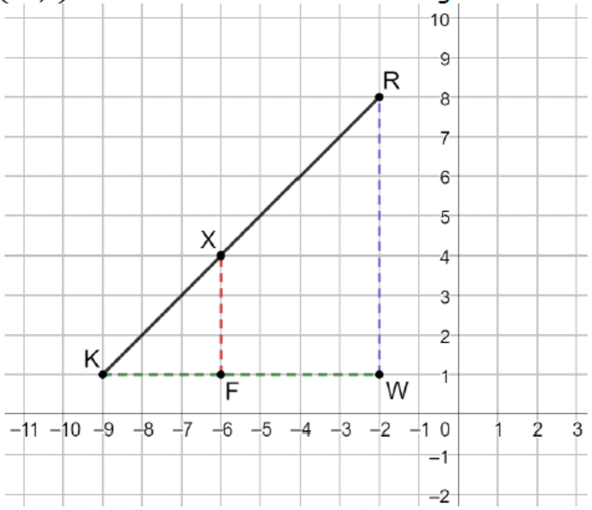
This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas and questions that you can ask them to share in the whole-group 13.4.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

- *What does it mean to partition something?*
- *How can the number line be used to determine a partition?*

At first, students may forget to include all parts of the segment as the denominator. For instance, some students may struggle to conceptualize that point R is 2/5 the distance between points S and P.

2. Point X partitions directed line segment KR in a $3:4$ ratio. Points $K(-9,1)$, $F(-6,1)$, $X(-6,4)$, $R(-2,8)$, and $W(-2,1)$ are shown on the coordinate grid.



a. Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$KF:FW$	$3:4$	○ part:part ○ part:whole
$KF:KW$	$3:7$	○ part:part ○ part:whole
$XF:RW$	$3:7$	○ part:part ○ part:whole
$KX:KR$	$3:7$	○ part:part ○ part:whole

b. What do you notice about the numeric ratios?
Answers will vary. The part:part ratio is different than the part:whole ratios. The part:whole ratios are all the same.

For a visual approach, have students label the lengths of all segments that are dotted in the graph.

Students may struggle with finding the part:part and the part:whole ratios. Having students label the lengths is important. In KX:KR, show students that the line goes through the coordinate plane evenly at each point, and that makes finding the ratio more accessible for students.

- *How is part to part different from part to whole?*
- *How would the ratio change if the letters of the segments were reversed?*

13.4.2 Exploration: Partitioning a Line Segment (continued)

c. Two students were discussing the directed line segment KR.

Nevaeh	Isabella
Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{4}$.	Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{7}$.

Work with your partner to explain why neither student has the correct scale factor. Include in your explanation what the correct scale factor should be so that the result of dilating at point K is point R.
Nevaeh tried to apply the part:part ratio when she should have used the part:whole ratio. Isabella recognized to use the part:whole ratio, but the scale factor is the reciprocal of what she used. The correct scale factor is $\frac{7}{3}$.

d. Discuss with your partner which ratios can be used to justify that $\Delta KXF \sim \Delta KRW$. Summarize your discussion.
Answers will vary. The ratios of the lengths of corresponding sides are all the same.

e. With your partner, write a conjecture on how point X could be found if only given the endpoints of the directed line segment KR and the ratio.
Answers will vary.
x-coordinate of X: $-9+\frac{3}{7}(-2--9) = -6$
y-coordinate of X: $1+\frac{3}{7}(8-1) = 4$

f. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials
Answers will vary.	

For a kinesthetic approach for collaboration for this activity, have students divide into two sides of the classroom, one side for Nevaeh, and the other for Isabella. Have each group discuss why neither student is correct on finding the scale factor. Then, have each group pick a spokesperson to share the group's findings.

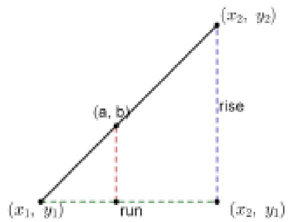
As a segue to 13.4.3 Guided Instruction, have a few students share their conjectures with the class to be able to see any commonalities and differences.

13.4.3 Guided Instruction: Partitioning a Line Segment

Component Overview: *Teachers solidify student understanding from 13.4.2 Exploration by bringing together student conjectures and apply the conjectures to solve problems.*

13.4.3 Guided Instruction: Partitioning a Line Segment

1. Generalize the conjecture in the Exploration if the given ratio is part:whole.

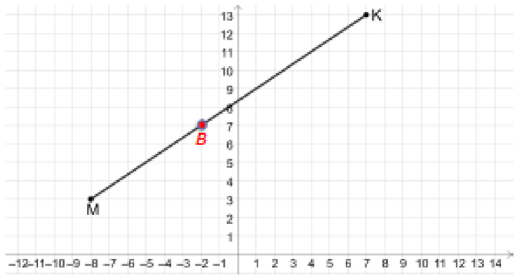


Generalization:

x-coordinate: $a = x_1 + (part/whole)(run)$

y-coordinate: $b = y_1 + (part/whole)(rise)$

2. Directed line segment MK is shown with endpoints at point $M(-8,3)$ and point $K(7,13)$.



Directed line segment MK is partitioned by point B in a ratio of $2:3$.

a. Write an equation that can be used to find the distance of $\overline{MB}$.

$MB = \frac{2}{5}MK$

Consider beginning this section by discussing student conjectures, common errors, and student observations from 13.4.2 Exploration. Relating the formula to student conjectures and to similar triangles will help them understand why the formula works. Students should not be required to memorize a formula. Some students will prefer to use slope or some other form of the formula without writing the formula. Some students may prefer an equivalent way of writing the generalized form of the conjecture. If you have more than one class, the conjectures may vary.

- *How would the equation change if the ratio is part:part?*
- *How do you know which endpoint of the line segment belongs to each part of the ratio?*

Question 2 breaks down the generalization so that students experience the generalization. After teaching through all five parts, refer to the conjecture and point out how each part of the question relates to the generalization.

13.4.3 Guided Instruction: Partitioning a Line Segment (continued)

b. Use your equation to find the horizontal distance of $\overline{MB}$.

$MB = \frac{2}{5}(15)$; *Horizontal distance* $= 6$

c. Use your equation to find the vertical distance of $\overline{MB}$.

$MB = \frac{2}{5}(10)$; *Vertical distance* $= 4$

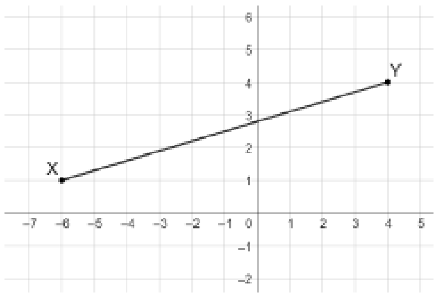
d. Use the horizontal distance and the vertical distance to plot point B on $\overline{MK}$.

See above.

e. What are the coordinates of point B?

$(-2, 7)$

3. Directed line segment XY has endpoints at point $X(-6,1)$ and point $Y(4,4)$.



Directed line segment XY is partitioned $\frac{1}{2}$ of the distance from X to Y at point W.

a. What is the horizontal distance of $\overline{XW}$?

$\frac{10}{2} = 5$

b. What is the vertical distance of $\overline{XW}$?

$\frac{3}{2} = 1.5$

c. What are the coordinates of point W?

$(-1, 2.5)$

Have students use different colored pens/pencils to label the horizontal and vertical distance for questions 2 and 3.

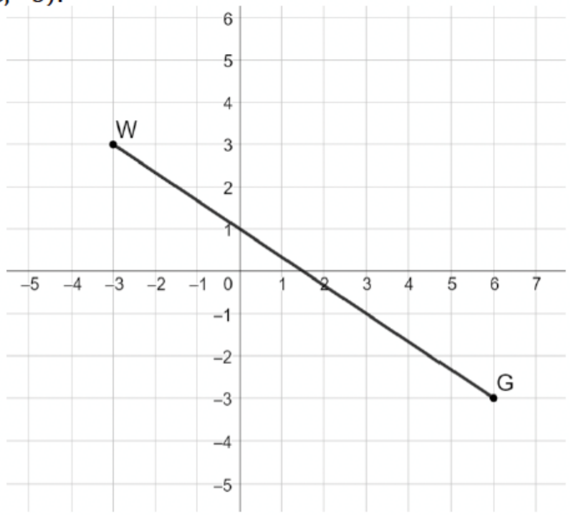
- *Why is point W the midpoint?*
- *How is a line segment partitioned $\frac{1}{2}$ of the distance different from a line segment partitioned in a ratio of $1:2$?*

13.4.4 Try It: Partitioning a Line Segment

Component Overview: *Students continue practicing how to find the point on a directed line segment by partitioning into a given ratio.*

13.4.4 Try It: Partitioning a Line Segment

1. Directed line segment WG has endpoints at $W(-3,3)$ and $G(6,-3)$.



Directed line segment WG is partitioned $\frac{2}{3}$ of the distance from W to G at point D.

a. What is the horizontal distance of $\overline{WD}$?

$9\left(\frac{2}{3}\right) = 6$

b. What is the vertical distance of $\overline{WD}$?

$6\left(\frac{2}{3}\right) = 4$

c. What are the coordinates of point D?

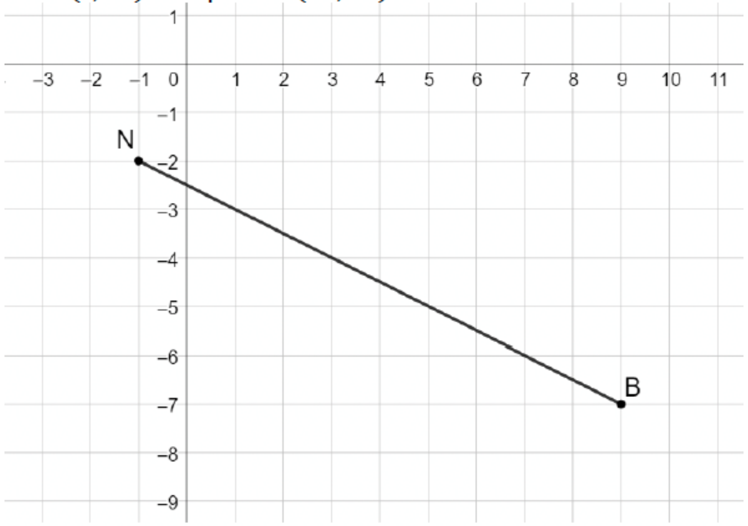
$(3,-1)$

How can slope be used to find point D?

If students finish this section early, have students create a directed line segment, and have students partition by several different ratios.

13.4.4 Try It: Partitioning a Line Segment (continued)

2. Directed line segment BN has endpoints at point $B(9,-7)$ and point $N(-1,-2)$.



Directed line segment BN is partitioned by point H at a $4{:}1$ ratio.

What are the coordinates of point H?

Horizontal distance: $10\left(\frac{4}{5}\right)=8$

Vertical distance: $5\left(\frac{4}{5}\right)=4$

Point H: $(9-8,-7+4)\rightarrow(1,-3)$

Consider having some students use slope to find the coordinate of point H, and having some students use the generalization derived earlier in the lesson to find point H.

13.4.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

13.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

   Answers will vary.

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

13.5 Weighted Averages

Lesson Introduction

Benchmark	MA.912.GR.3.1 Determine the weighted average of two or more points on a line.	Learning Targets	• I can determine the weighted average of two or more points on a line. • I can explain how changing the weight moves the weighted average of points.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can you determine the weighted average of two or more points on a line? • How can you explain how changing the weight moves the weighted average of points?
Lesson Narrative	This lesson applies knowledge from Lesson 13.4, to guide students through explorations on finding the weighted average of two or more points on a line. The lesson relates partitioning line segments to the definition of a weighted average, and how to find the weighted average on the coordinate plane using a given ratio.

Component Summary	13.5.1 Warm-Up (~5 mins.)	Stock market and growth of shares *(SE p. 306)*	13.5.4 Collaboration (5 mins.)	Using weighted averages to dilate polygons *(SE p. 312)*
	13.5.2 Exploration (10 mins.)	Discovery on weighted averages *(SE p. 307)*	13.5.5 Your Turn! (5 mins.)	Independent practice problems on weighted averages *(SE p. 313)*
	13.5.3 Guided Instruction (20 mins.)	Weighted averages on the coordinate plane *(SE p. 309)*	13.5.6 Wrap-Up (~5 mins.)	Ticket out the door *(SE p. 314)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • N/A Additional Terms: • Dilation	• (Optional): Fishing Weight • (Optional): Pencil to tie fishing weight to • (Optional): String to tie the fishing weight to pencil	• If choosing to do so, prepare the string, fishing weight, and pencil.

Teacher Tips

Common Misconceptions	Things to Consider
• On 13.5.3 Guided Instruction, students may struggle with writing expressions using given ratios with weighted averages. • Students may struggle with visualizing weighted averages on the graph in 13.5.3 Guided Instruction. • Students may struggle with dilating using segment partitioning in 13.5.4 Collaboration.	• Consider using a fishing weight and a pencil to demonstrate the idea of weighted averages in 13.5.3 Guided Instruction. • 13.5.4 Collaboration can be used as enrichment. • If time is an issue, then consider assigning 13.5.5 Your Turn! for homework. • Consider using the Desmos graph that is linked for question 5 in 13.5.3 Guided Instruction to move the point to help students further relate the weighted average to its graphical representation.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Three Reads The concept of weighted averages on a line can confuse students at first. To help guide students through any misconceptions, implement a Three Reads on 13.5.3 Guided Instruction. Allow students to read through the scenario one time, silently. Have students read again and highlight any key points or concepts in the second read. On the third read, students should read again, either silently or aloud. Then, have students summarize what they read with a partner.	**MA.K12.MTR.2.1: Multiple Representations** Lesson 13.5 requires that students view the problems from both a graph and an equation. Allow students to make connections between the graphical representations as well as the algebraic representations. Being able to successfully master this standard will require students to progress from modeling with graphs to using expressions.

Differentiate Instruction

Consider using a manipulative for modeling the concept of a weighted average. A ruler or pencil with a fishing weight can be used to model the concept of a weighted average. For a digital approach, consider finding a video or applet that models the concept of a weighted average.

Lesson Notes:

13.5.1 Warm-Up: Between Two Numbers

Component Overview: *Students begin by reading a scenario about the stock market and answer two questions.*

13.5.1 Warm-Up: Between Two Numbers

A stock exchange, or stock market, is a system for buying and selling bonds, securities, or stocks. A stock is a share in the ownership of a company. NASDAQ is short for National Association of Securities Dealers Automated Quotations, which is the world's largest electronic stock exchange. Each company that is traded has an abbreviation of one to four letters.

Jaxon's grandmother bought 100 shares of Netflix and 100 shares of Apple for him when he was born on March 29, 2003. She paid $1.37 per share for Netflix (NASDAQ: NFLX) and paid $0.21 per share for Apple (NASDAQ: AAPL).

On Jaxon's 18th birthday, Netflix's stock price was at $513.95 and Apple's stock price was at $121.39.

1. How much were the stocks worth on Jaxon's 18th birthday?
   $63,534.00

The stock price history of Netflix shows that at one point the stock had increased in worth by 45,000%, and Apple stock had increased in worth by 140,000%.

2. Write the decimal equivalent of each percentage increase.

Stock	Decimal Equivalent
Apple	1400
Netflix	450

The stock market is a real-world context for mathematics that can effectively garner student interest, with a variety of opportunities for meaningful connections to mathematics in the real world (MTR 7).

Consider one of the following strategies for embedding this lesson in real-world contexts:

- *Show a video on the stock market, or popular mainstream applications of stocks.*
- *Have students participate in stock market simulations as an extension project.*
- *Partner with a local company to provide students with real-life stocks to monitor. For example, some apps give free stock just for signing up.*

13.5.2 Exploration: Exploring Weighted Averages

Component Overview: *Students explore weighted averages by starting with midpoints (ratio 1:1); then, students discover the mathematics behind weighted averages using a graphical representation.*

13.5.2 Exploration: Exploring Weighted Averages

1. A number line is shown. Each grid space is 1 unit.

a. Find the midpoint of $\overline{HM}$ and label it B.

Assuming that points H and M have the same weight, the diagram represents how points H, B, and M are in balance.

b. Discuss with your partner why point B acts as a point of balance. Summarize your discussion.
Answers will vary. Point B is the midpoint of line segment HM so the distance from H to B is the same as the distance from B to M so point B will balance H and M.

c. Keyshawn told his partner that if the scale is drawn where the scale is centered on point K, then point H would be higher than point M. Describe or draw what would have to be done to balance the scale when it is centered at point K.
Answers will vary. The side of the scale with point H will need to be lowered. To lower it another point H could be added.

The ratio of $HK:KM$ is $3:9$, which can be written as $\frac{1}{3}$. The ratio, $\frac{1}{3}$, can be used to determine the location of point K using the steps shown.

$$\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } M) = \frac{3}{4}(3) + \frac{1}{4}(15) = \frac{9}{4} + \frac{15}{4} = \frac{24}{4} = 6$$

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 13.5.3 Guided Instruction. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 13.5.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

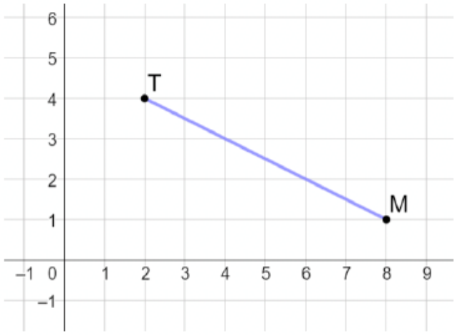
Students may struggle with answering question 1c. Demonstrating a scale, see-saw, or balance may help students visualize that more weight should be added. Students may not connect what that the weight should be H.

- *How do the fractions, $\frac{3}{4}$ and $\frac{1}{4}$, relate to the ratio $\frac{1}{3}$?*
- *How does the coefficient of H indicate that point H is weighted?*

13.5.2 Exploration: Exploring Weighted Averages (continued)

The ratio $\frac{1}{3}$ was used to find $\frac{3}{4}$ and $\frac{1}{4}$ by noting that the ratio indicates there are a total of 4 parts. To balance at point K, point H would be "heavier" than point M, so it is weighted as 3 of the 4 parts with point M weighted as 1 of the 4 parts.

2. Point T has coordinates $(2, 4)$, and point M has coordinates $(8, 1)$.



   a. What is the midpoint of line segment TM?
   $\left(\frac{2+8}{2}, \frac{4+1}{2}\right) = (5, 2.5)$

   b. Complete the table.

	x-value	y-value
$T(2,4)$	$\frac{1}{2}(2)$	$\frac{1}{2}(4)$
$M(8,1)$	$\frac{1}{2}(8)$	$\frac{1}{2}(1)$
Total	5	2.5

   c. Discuss with your partner how the table shows the process of finding the midpoint.
   Answers will vary. Find half of each point's coordinate value and then add them together.

- *Can you use the graph to find the midpoint? If so, how?*
- *What do the ½'s represent in the table?*

Consider allowing students times to write down things they notice and wonder about the table before discussing with a partner.

d. Complete the expression that can be used to find the point that partitions the line segment in a $1:1$ ratio. T represents the coordinates of point T, and M represents the coordinates of point M.

$$\frac{1}{2}T + \frac{1}{2}M$$

e. Find the point that partitions directed line segment TM in a $2:1$ ratio.
$(6, 2)$

f. Complete the table to calculate $\frac{1}{3}T + \frac{2}{3}M$.

	x-value	y-value
$T(2,4)$	$\frac{1}{3}(2)$	$\frac{1}{3}(4)$
$M(8,1)$	$\frac{2}{3}(8)$	$\frac{2}{3}(1)$
Total	6	2

g. What do you notice about the total in the table and the point found in part e that partitions the directed line segment in a $2:1$ ratio?
They are the same.

3. Line segment KR has endpoints at $K(-6,1)$ and $R(2,5)$.

a. Complete the expression for the point that partitions directed line segment KR in a $3:1$ ratio.

$$\frac{1}{4}K + \frac{3}{4}R$$

b. Use the expression to find the point that partitions directed line segment KR in a $3:1$ ratio.

$\frac{1}{4}(-6) + \frac{3}{4}(2) = 0$

$\frac{1}{4}(1) + \frac{3}{4}(5) = 4$

$(0,4)$

- *Why does the ratio of 1:1 create the equation of $\frac{1}{2}T + \frac{1}{2}M$?*
- *Which strategy works better for finding the point that partitions a directed line segment, the table or a formula? Explain your choice.*

As you monitor students, listen for misconceptions or confusion on why a ratio of 2:1 relates to the formula. Consider how to support student learning during 13.5.3 Guided Instruction. Students may benefit from reviewing the questions 2f, 2g, and 3 at the start of Guided Instruction.

13.5.3 Guided Instruction: Determining Weighted Averages

Component Overview: *Students are introduced to the concept of weighted averages on a line. Students are guided through a scenario about a seesaw, and how weight distribution on a see-saw works.*

13.5.3 Guided Instruction: Determining Weighted Averages

Weight is a measure of force pushing or pulling on an object due to gravity. This means that finding a weighted average puts more "weight" on one point than another.

If B is weighted twice as heavily as A, then during the calculation, B is to measure two times the value of A. In other words, B has two times the "gravitational pull" of A.

Think of it in terms of a seesaw. In the picture below, point B is weighted more heavily than point A.

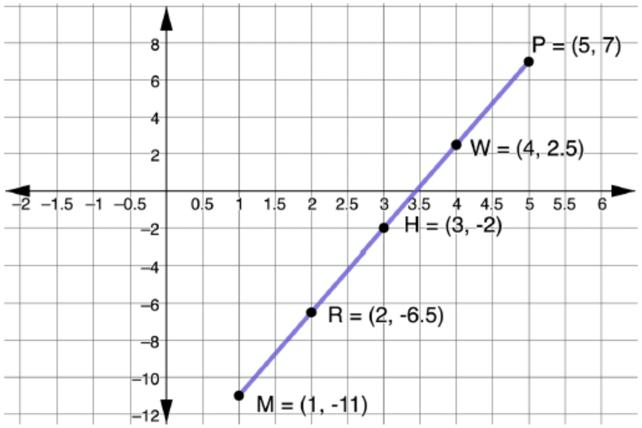
1. Discuss with your partner what the girls close to point B would have to do so that the seesaw would balance. Summarize your discussion.
   To balance the seesaw, the girls on the heavier end would need to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object.

Have students read this section 3 times. See the Language and Literacy Routine on page 2 for further guidance.

- *What do you notice about the picture?*
- *How does this picture relate to the lesson and standard?*

It may be helpful to demonstrate the balance of an object where one end is heavier than another. As in the seesaw picture, to balance the seesaw, the girls on the heavier end would have to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. Consider using a fishing weight and a pencil to demonstrate this idea. See "Differentiate Instruction" for other ideas.

Directed line segment MP is shown, where point M is at $(1,-11)$ and point P is at $(5,7)$. Point H is the midpoint of $\overline{MP}$, point W is the midpoint of $\overline{HP}$, and point R is the midpoint of $\overline{MH}$.



Since point H is the midpoint of $\overline{MP}$, the coordinates of point H can be determined by finding the average of points M and P.

$$\frac{M+P}{2}$$

Point W partitions the line segment MP so that point P weighs three times as much as point M. This means that point W is closer to point P because point P is "heavier" than point M. The given expression can be used to calculate this weighted average when point P weighs three times as much as point M.

$$\frac{M+P+P+P}{4}$$

Have students either underline or highlight the information at the top of this page. Consider doing another Three Reads on this page as well. See guidance on Language and Literacy Routine.

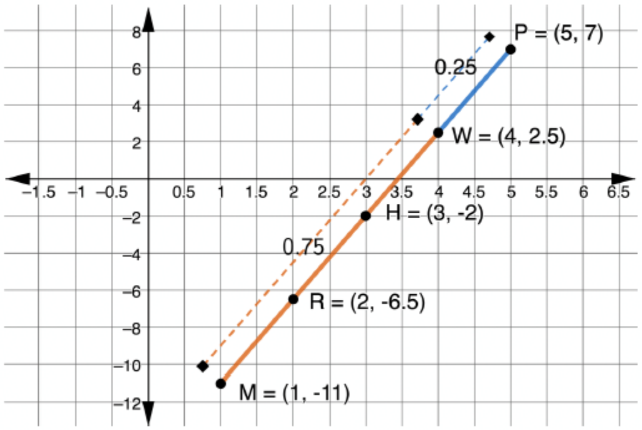
Discuss with students how the expression can also be written as $\frac{1}{2}M + \frac{1}{2}P$, and $\frac{1}{4}M + \frac{3}{4}P$.

Consider asking the following questions to help guide students through this page:

- *Why is the equation $\frac{M+P}{2}$?*
- *How was this equation derived?*
- *At the bottom of the page, why is the equation $\frac{M+P+P+P}{4}$?*
- *How was this equation derived?*

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

The graph shows the weighted averages.



2. Discuss with your partner why the distance from point W to point P is shorter if point P is "heavier" than point M. Summarize your discussion.
   As in the seesaw picture, to balance the seesaw the girls on the heavier end would need to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. Point W is the balancing point so would need to be closer to point P because point P is "heavier."

3. Point R partitions the line segment MP so that point M weighs three times as much as point P. Write an expression that shows this weighted average.
   $$\frac{M+M+M+P}{4}$$

A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

Students may struggle with conceptualizing the graph on this page. Consider providing students with additional visualization and explanation, if necessary.

Have a couple of students share their partner discussions with the class to get a variety of perspectives on this question.

How can you write an expression that models point M having three times as much weight as point P?

Consider revisiting question 1 in the Collaboration to make the connection to the balance questions.

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

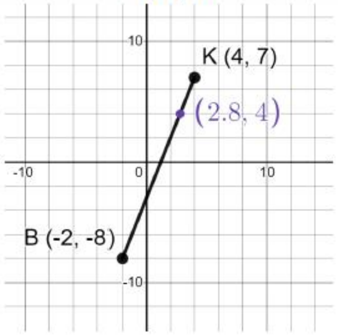
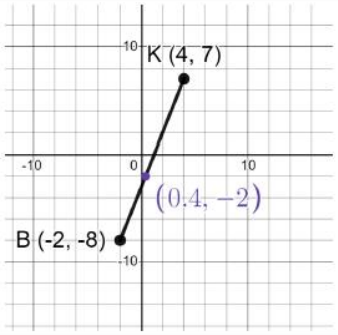
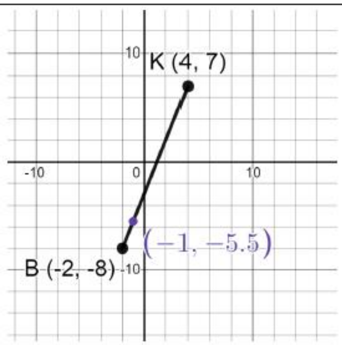
4. Write the expression that can be used to find the point that partitions $\overline{MP}$ in the given ratio, and find the endpoint that is the weighted point.

Ratio	Expression	Point	Weighted Point
$1:1$	$\frac{M+P}{2}$	$\frac{1+5}{2}, \frac{-11+7}{2} = (3,-2)$	*Neither*
$1:3$	$\frac{M+M+M+P}{4}$	$\frac{1+1+1+5}{4}, \frac{-11-11-11+7}{4} = (2,-6.5)$	M
$3:1$	$\frac{M+P+P+P}{4}$	$\frac{1+5+5+5}{4}, \frac{-11+7+7+7}{4} = (4,2.5)$	P

Students may struggle with writing the expression in the table on question 4. If necessary, write the expression and have students explain it. Emphasize the reason that a point is repeated is because it is the weighted point.

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

5. A line segment is shown in the table with three different weighted averages. Write the expression that can be used to find each weighted average.

Partition	Graph	Expression
4: 1		$\frac{B+K+K+K+K}{5}$ *or* $\frac{1}{5}B+\frac{4}{5}K$
2: 3		$\frac{B+B+B+K+K}{5}$ *or* $\frac{3}{5}B+\frac{2}{5}K$
1: 5		$\frac{B+B+B+B+B+K}{6}$ *or* $\frac{5}{6}B+\frac{1}{6}K$

Use the Desmos graph embedded in the digital Student Edition to show students the distances and ratios for the graphs in question 5 by moving the purple point along the line.

Remind students to write the expression by using the weighted point. Avoid providing students with a quick summary such as “use the second number of the ratio as the numerator of the first fraction, and the first number of the ratio as the numerator of the second fraction”. Students may find such a summary difficult to remember. Focus on students understanding why that summary works.

- *Why is B used as the first letter in the expression?*
- *Why is K used as the second letter in the expression?*
- *What is your thought process when writing the expression?*

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

6. Discuss with your partner how changing the weights moves the weighted average on the number line. Summarize your discussion.
   As in the seesaw picture, to balance the seesaw the girls on the heavier end would have to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. The weighted average shifts closer to whichever endpoint is "heavier" according to the ratio.

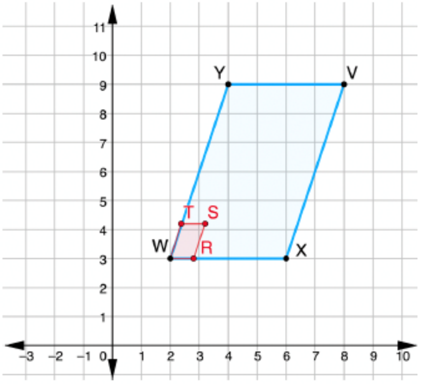
Allow students to share their discussions with the class to hear a variety of student perspectives.

13.5.4 Collaboration: Weighted Averages and Dilations

Component Overview: *Students find the coordinates of the image of given polygons by using weighted averages and segment partitioning.*

13.5.4 Collaboration: Weighted Averages and Dilations

1. $VXWY$ has vertices at $V(8,9)$, $X(6,3)$, $W(2,3)$, and $Y(4,9)$.



   a. Find the point that partitions $\overline{WX}$ in a $1:4$ ratio. Label it R.

   $\frac{4}{5}(2)+\frac{1}{5}(6)=2.8$

   $\frac{4}{5}(3)+\frac{1}{5}(3)=3$

   $(2.8,3)$

   b. Find the point that partitions $\overline{WY}$ in a $1:4$ ratio. Label it T.

   $\frac{4}{5}(2)+\frac{1}{5}(4)=2.4$; $\frac{4}{5}(3)+\frac{1}{5}(9)=4.2$

   $(2.4,4.2)$

   c. Find the point that partitions $\overline{WV}$ in a $1:4$ ratio. Label it S.

   $\frac{4}{5}(2)+\frac{1}{5}(8)=3.2$; $\frac{4}{5}(3)+\frac{1}{5}(9)=4.2$

   $(3.2,4.2)$

   d. Is $WTSR$ a dilation of $WYVX$? Justify your answer.

   Yes, it is a dilation of $\frac{1}{5}$ about the point W.

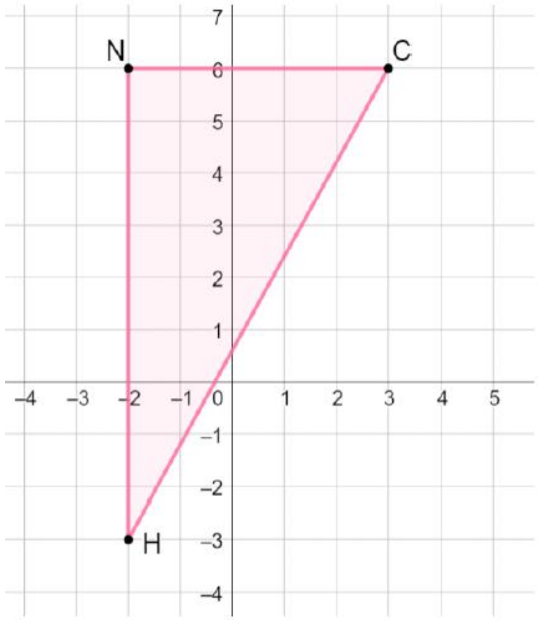
Consider using this component for enrichment. Consider using Stepping Stones in place of this component for students who struggle with weighted averages.

Alternatively, this component can be completed by assigning different students parts a, b, and c. Then, pair students with a different part to have each partner explain how they determine the answer. Then, have each pair do the part that neither was assigned.

How can an equation be set up to find the partitioned points?

- *How can you determine whether $WTSR$ is a dilation of $WYVX$?*
- *How can you find the scale factor of the dilation?*

2. Triangle NCH has vertices at $N(-2,6)$, $C(3,6)$, and $H(-2,-3)$.



Use segment partitioning to dilate ΔNCH using center C and scale factor $\frac{3}{4}$.

N': $\frac{3}{4}N + \frac{1}{4}C \rightarrow (-0.75, 6)$

H': $\frac{3}{4}H + \frac{1}{4}C \rightarrow (-0.75, -0.75)$

Students may struggle with using segment partitioning to find the scale factor. Allow students to use the graph to plot points. Consider using the guiding questions below to help students.

If using this component for enrichment provide students the opportunity to explore dilations using online geometry tools such as GeoGebra. Pair students together and have each create a triangle and apply a dilation using a scale factor of their choice or of your choosing. The center of dilation should be a vertex. Each pair writes down the coordinates of the original triangle and of the dilation and switches the information with another pair. The other pair uses the coordinates to determine the center of dilation and the scale factor. Students may choose to do this by trial and error or algebraically using $\frac{a}{b}$ to represent the scale factor. For example, from question 2 the equation $\frac{a}{b}(-2) + \left(1 - \frac{a}{b}\right)(3) = -0.75$ can be used to determine the scale factor $\frac{a}{b}$,

Use the following questions to guide students through dilating ΔNCH, using center C and a scale factor of ¾

- *How long is line segment NC? (5)*
- *What is ¾ of 5? (3.75)*
- *What is the coordinate that is 3.75 units from point C? (-0.75,6)*
- *How long is line segment NH? (9)*
- *What is ¾ of 9? (6.75)*
- *What is the point that is 6.75 units away from point N'? (-0.75, -0.75)*

13.5.5 Your Turn!

Component Overview: *Students are given two problems to work independently.*

13.5.5 Your Turn!

1. Line segment MK has endpoints at $M(-2,1)$ and $K(8,7)$.

   a. Write an expression for the point that partitions directed line segment MK in a $2:3$ ratio.

   $\frac{3}{5}M + \frac{2}{5}K$

   b. Use the expression to find the point that partitions directed line segment MK in a $2:3$ ratio.

   $\frac{3}{5}(-2) + \frac{2}{5}(8) = 2$

   $\frac{3}{5}(1) + \frac{2}{5}(7) = 3.4$

   $(2, 3.4)$

Students may struggle with writing the expression for the point that partitions the directed line segment for the given ratio. Consider using guided questions:

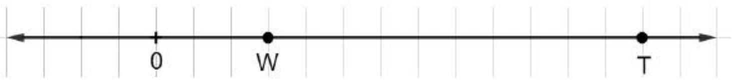
- *Which point weighs more?*
- *How many total parts does line segment MK have?*

13.5.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a Ticket Out the Door.*

13.5.6 Wrap-Up: Ticket Out the Door

1. A number line is shown. Each grid space is 1 unit.



   a. Determine the point on the number line that satisfies the expression shown. Label the point M.

$$\frac{2}{5}(\text{point } W) + \frac{3}{5}(\text{point } T)$$

   b. What is the ratio $WM : MT$?
   6: 4 or 3: 2

How well students perform on the Ticket Out the Door, coupled with what was observed during instruction, can provide information for differentiating instruction and supporting student learning using small group instruction. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

13.6 Classifying Triangles

Lesson Introduction

Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify triangles in a mathematical or real-world context. • I can draw a triangle on a coordinate plane using given definitions, properties, or theorems.

	Building On	Working Toward
Benchmark Coherence	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.	N/A

Essential Questions	• How can coordinate geometry be used to classify triangles on the coordinate plane? • How can the Pythagorean Theorem be used to find the distance between two points on the coordinate plane? • What are the characteristics of equilateral, isosceles, and scalene triangles?
Lesson Narrative	Lesson 13.6 guides students through the process of using the Pythagorean Theorem to calculate distance on the coordinate plane, building from content knowledge from 8th grade. The lesson requires students to graph and use the distance formula and/or Pythagorean Theorem to find the measures of lengths to identify types of triangles. Students are expected to recall terms such as equilateral, scalene, and isosceles, and identify their characteristics.

Component Summary	13.6.1 Warm-Up (~5 mins.)	Notice and Wonder *(SE p. 315)*	13.6.3 Exploration (15 mins.)	Identifying types of triangles on coordinate plane *(SE p. 316)*
	13.6.2 Refresher (5 mins.)	Refresher on using the Pythagorean Theorem *(SE p. 315)*	13.6.4 Your Turn! (20 mins.)	Practice on identifying triangles on the coordinate plane *(SE p. 319)*
	13.6.5 Wrap-Up (~5 mins)	Students summarize their learning *(SE p. 321)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Equilateral Triangle • Isosceles Triangle • Scalene Triangle	• Calculators • (Optional) Graphing Utility • (Optional) Colored pencils/pens	• If necessary, prepare a graphing utility, such as Desmos or GeoGebra.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may incorrectly set up the Pythagorean Theorem, and may forget to find the square root of c^2. • Students may incorrectly identify whether a triangle is equilateral, isosceles, or scalene. • Students may incorrectly use the Pythagorean Theorem to find distance.	• Consider allowing students who need extra guidance and support to use a graphing utility, such as GeoGebra, to help students see the relationships between the triangles. • Consider allowing students to use colored pencils/pens during this lesson to label the sides of the triangles. • Determine if your students need the support of all or part of the Refresher. • In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Anticipate, Monitor, Select, Sequence, Connect The five practices for orchestrating mathematical discussion can be implemented effectively in Lesson 13.6. Anticipate the different strategies and interpretations and pay attention to students' thinking processes. Select which students can share their solutions, and sequence by common misconception or strategy. Finally, help students connect their thinking and notice patterns of different solutions and processes.	**MA.K12.MTR.3.1: Mathematical Fluency** Students will have to select efficient and appropriate methods for solving problems. This will require students to maintain flexibility and accuracy, while performing the procedures necessary to solve the problems. Students will have to adapt their knowledge of the Pythagorean Theorem to apply it to new contexts.
Differentiate Instruction	Consider differentiating the problems in the Your Turn! section of the lesson. Give each problem a point value that is equivalent to the number. For example, question 1 is worth 1 point, question 2 is worth 2 points, and so on. Students are required to earn a minimum of five points. Structuring the Your Turn! using this format gives students an opportunity to have choice and will require students to do a mixture of lower and higher-level problems.	
Lesson Note		

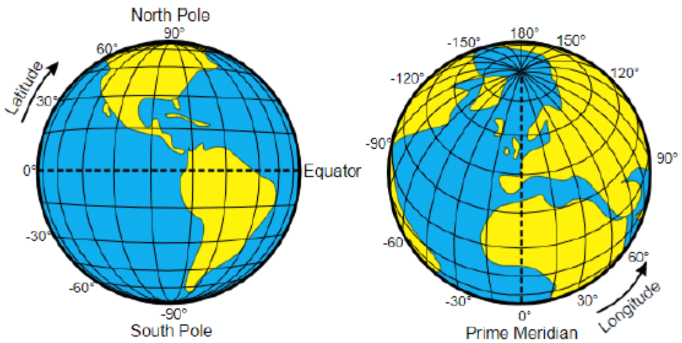
13.6.1 Warm-Up: Latitude and Longitude

Component Overview: *Students read the prompt about latitude and longitude and are given a table of the latitude and longitude of several cities in Florida. Students are asked to notice and wonder.*

13.6.1 Warm-Up: Latitude and Longitude

In ancient times, people used landmarks to identify locations, but travelers were not familiar with local landmarks, making traveling difficult. Hipparchus, a Greek astronomer, was the first to specify location using latitude and longitude as coordinates. Hipparchus applied his knowledge of spherical angles to the problem of denoting locations on the Earth's surface.

Latitude is the measurement of the globe north to south, and longitude is the measurement of the globe east to west.



The table shows the latitude and longitude of different Florida cities.

City	Coordinates	City	Coordinates
Panama City	$30.2N, 85.7W$	Miami	$25.8N, 80.2W$
Tallahassee	$30.46N, 84.28W$	Tampa	$28.0N, 82.5W$
Lake Worth	$26.6N, 80.1W$	Orlando	$28.5N, 81.4W$
Jacksonville	$30.3N, 81.7W$	Gainesville	$29.7N, 82.3W$

1. What do you notice? What do you wonder?
   Answers will vary. The latitude values are all close to 30, and the longitude values are near 80. How far apart are the cities in miles from each other?

Give students an opportunity to share their Notice and Wonder and see if there are any commonalities.

Consider having students take an educated guess on the longitude and latitude of your school, using the information given.

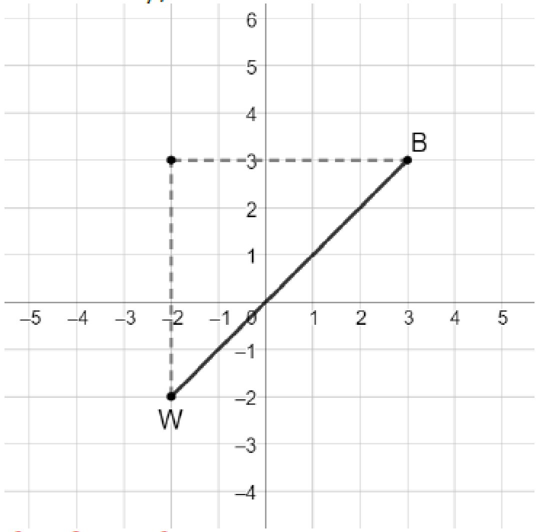
13.6.2 Refresher: Finding Distance

Component Overview: *Students review how to use the Pythagorean Theorem to find the distance on the coordinate plane.*

13.6.2 Refresher: Finding Distance

Work with your partner to complete the following.

1. Find the length of $\overline{WB}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



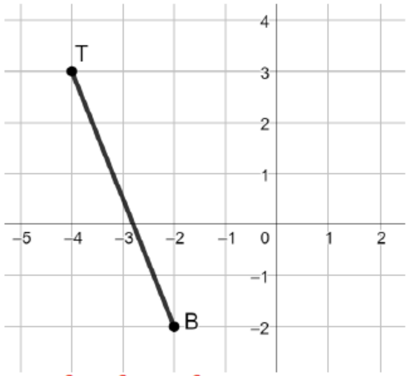
$5^2 + 5^2 = WB^2$
$25 + 25 = WB^2$
$50 = WB^2$
$\sqrt{50} = WB$
$7.071 = WB$

Begin by asking students what they remember about the Pythagorean Theorem, which is used extensively in Unit 7, as well as other units throughout the course. Consider using the following questions:

- *What is the Pythagorean Theorem used for?*
- *How can you use the Pythagorean Theorem to find the length of a side of a right triangle?*

13.6.2 Refresher: Finding Distance (continued)

2. Use the Pythagorean Theorem to find the length of $\overline{TB}$. If necessary, round to the nearest thousandth.



$5^2 + 2^2 = TB^2$
$25 + 4 = TB^2$
$29 = TB^2$
$\sqrt{29} = \sqrt{TB^2}$
$5.385 = TB$

Consider opportunities to provide differentiation using a visual approach. Have students draw a right triangle on the coordinate plane on question 2 to help better visualize how the Pythagorean Theorem can be used to find distance on the coordinate plane.

3. Explain how the Pythagorean Theorem can be used to find the distance between two points.
Answers will vary. Find the horizontal and vertical distance between the two endpoints. This can be done by using the gridlines on the coordinate grid or by finding the difference between the x-values and the difference between the y-values of the endpoints. Then, substitute the horizontal distance and vertical distance between the points for a and b in the Pythagorean Theorem, $a^2 + b^2 = c^2$. Solve to find the length of the line segment, which will be the hypotenuse of the right triangle.

Tie question 3 to the MLR of Anticipate, Monitor, Select, Sequence, and Connect. Consider the ways that students will answer this question and select a couple of responses to discuss as a lead into using the Pythagorean Theorem to find distance and length on the coordinate plane.

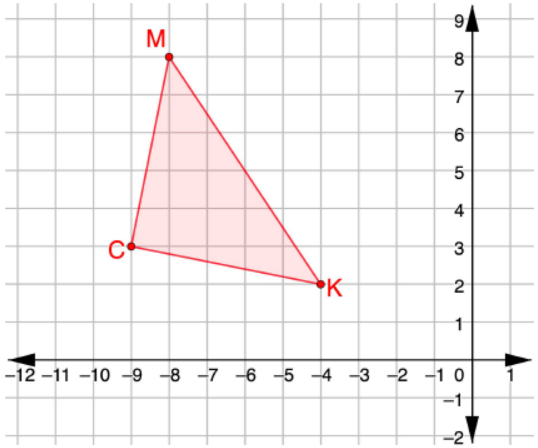
13.6.3 Exploration: Classifying Triangles

Component Overview: *This Exploration has students graph triangles on the coordinate plane to find the distances of each side. Students use the distances to classify the triangles as equilateral, isosceles, and scalene, based on the number of congruent sides.*

13.6.3 Exploration: Classifying Triangles

1. ΔMKC has vertices at $M(-8,8)$, $K(-4,2)$, and $C(-9,3)$.

   a. Graph ΔMKC on the coordinate plane.



   b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{CM}$	5.099
$\overline{MK}$	7.211
$\overline{KC}$	5.099

   c. What do you notice about the side lengths of ΔMKC?
   Two sides have the same lengths.

Consider allowing students to use a tool such as GeoGebra to graph the triangles. GeoGebra can be used to help visualize the relationship between the values of the segments and the type of triangle that is formed.

13.6.3 Exploration: Classifying Triangles (continued)

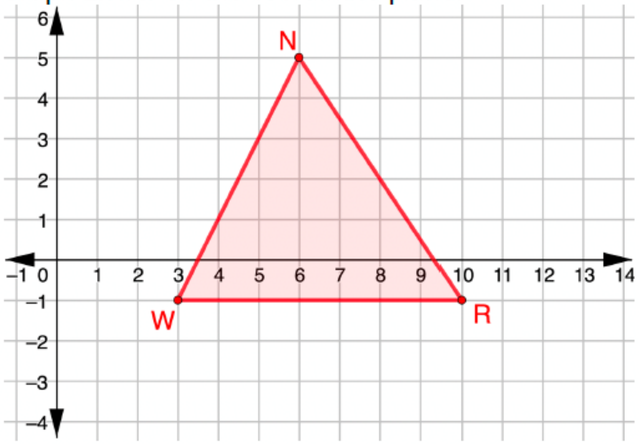
d. Complete the statement.

ΔMKC is classified as a(n) ○ equilateral ○ isosceles ○ scalene triangle

because ΔMKC has ○ no ○ two ○ three equal side lengths.

2. ΔNWR has vertices at $N(6,5)$, $W(3,-1)$, and $R(10,-1)$.

a. Graph ΔNWR on the coordinate plane.



b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{NR}$	7.211
$\overline{RW}$	7
$\overline{WN}$	6.708

Have students make a prediction about which type of triangle is on the coordinate plane before students complete the table. Doing this will tie into the Literacy and Language Routine by having students looking at several different responses and reasonings.

Be careful that students do not rely on their prediction and choose to not support it mathematically.

c. What do you notice about the side lengths of ΔNWR?
None of the sides have the same length.

d. Complete the statement.

ΔNWR is classified as a(n)
- ○ equilateral
- ○ isosceles
- ○ scalene

triangle

because ΔNWR has
- ○ no
- ○ two
- ○ three

equal side lengths.

Consider polling students for their prediction on what type of triangle ΔNWR is. Because the measures of the side lengths are close, the triangle does not appear to be scalene. It may be helpful for students to understand why their predictions may be incorrect.

3. ΔMSG has vertices at $M(-3, 4)$, $S(2.2, 1)$, and $G(-3, -2)$.

a. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{MS}$	6.003
$\overline{SG}$	6.003
$\overline{GM}$	6

b. Complete the statement.

ΔMSG is classified as a(n)
- ○ equilateral
- ○ isosceles
- ○ scalene

triangle

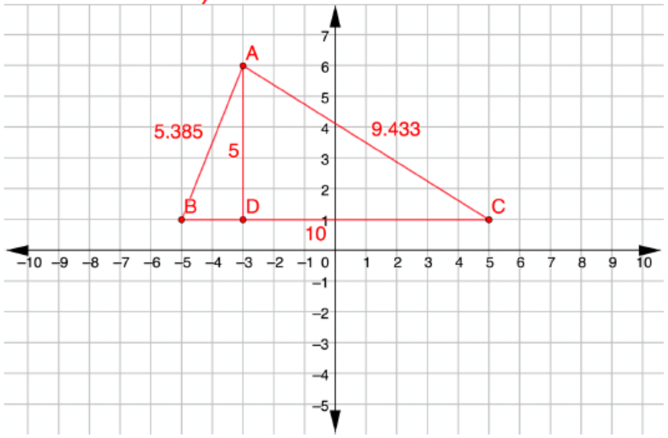
because ΔMSG has
- ○ no
- ○ two
- ○ three

equal side lengths.

13.6.3 Exploration: Classifying Triangles (continued)

4. Draw a scalene triangle with one vertex at $(-3,6)$ and with a base that is twice the length of its height.

Answers will vary.



Use the following questions to help students construct their triangle for question 4:

- *What step would be completed first to begin constructing the triangle?*
- *How many units do you want to draw the height of the triangle?*
- *How can you find the length of the base?*

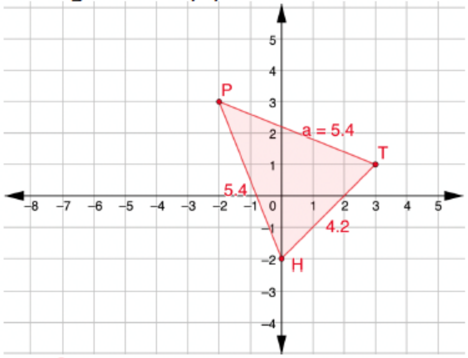
Have students compare their triangles with a partner or another student in the classroom. If time allows, consider sharing a few examples with the class.

13.6.4 Your Turn!

Component Overview: *Students work independently to practice classifying triangles on the coordinate plane.*

13.6.4 Your Turn!

1. ΔPTH has vertices at $P(-2,3)$, $T(3,1)$, and $H(0,-2)$. Determine if ΔPTH is an equilateral, isosceles, or scalene triangle. Justify your answer with math.



$3^2 + 3^2 = HT^2$
$9 + 9 = HT^2$
$18 = HT^2$
$\sqrt{18} = \sqrt{HT^2}$
$4.243 \approx HT$

$2^2 + 5^2 = PT^2$
$4 + 25 = PT^2$
$29 = PT^2$
$\sqrt{29} = \sqrt{PT^2}$
$5.385 \approx PT$

$2^2 + 5^2 = PH^2$
$4 + 25 = PH^2$
$29 = PH^2$
$\sqrt{29} = \sqrt{PH^2}$
$5.385 \approx PH$

PH and PT are the same length. This is an isosceles triangle.

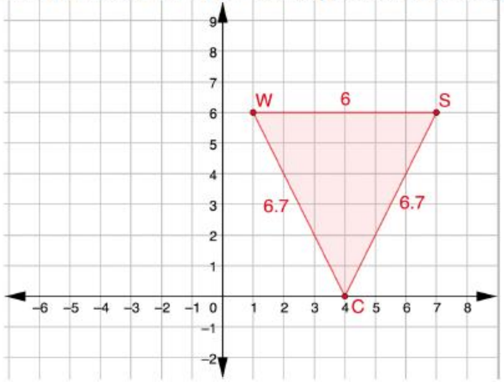
Consider having students work in groups of three. Each student in the group is responsible for finding the length of one side of the triangle. Together as a group, they will review each other's work, and ask clarifying questions, if needed. Then, the group will determine the type of triangle that is given.

Consider having students create a table to organize their work for question 1.

13.6.4 Your Turn! (continued)

2. Given: ΔSCW with $W(1,6)$, $S(7,6)$, and $C(4,0)$

   a. Show: ΔSCW is **NOT** an equilateral triangle



$3^2 + 6^2 = CS^2$
$9 + 36 = CS^2$
$45 = CS^2$
$\sqrt{45} = \sqrt{CS^2}$

$6.708 \approx CS$

$3^2 + 6^2 = CW^2$
$9 + 36 = CW^2$
$45 = CW^2$
$\sqrt{45} = \sqrt{CW^2}$
$6.708 \approx CW$

$WS = 6$

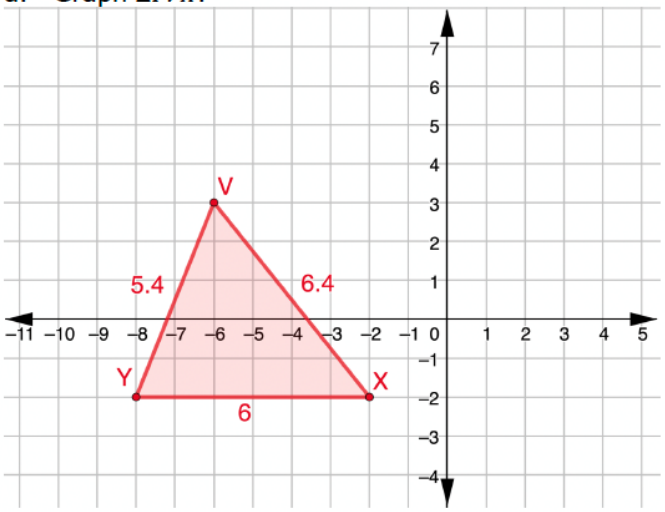
The triangle is not equilateral because all of the sides are not the same.

   b. What type of triangle is ΔSCW?
   It is an isosceles triangle because two of the sides have the same length.

Students may be tempted to not show work for this problem. Recommend that students show work on question 2.

3. Diego is creating a triangular garden and uses the ΔYVX with vertices at $Y(-8,-2)$, $V(-6,3)$, and $X(-2,-2)$ to design the garden.

   a. Graph ΔYVX.



   b. Complete the table. If necessary, round to the nearest thousandth.

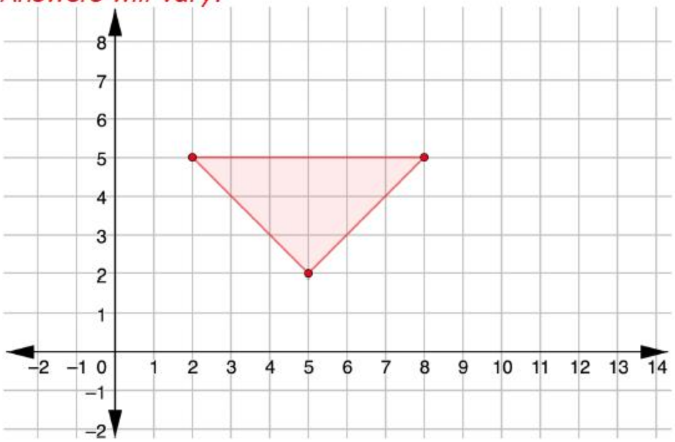
Line Segment	Side Length
$\overline{YV}$	5.385
$\overline{VX}$	6.403
$\overline{XY}$	6

   c. Classify the type of triangular garden Diego is creating.
   It is a scalene triangular garden.

13.6.4 Your Turn! (continued)

4. Draw an isosceles triangle whose base is three more than the height of the triangle and has one vertex at (2,5).

Answers will vary.



Use Anticipate, Monitor, Select, Sequence, and Connect for question 4, and have students share their triangles to see if students created different triangles.

Consider allowing students to try and create their triangle in GeoGebra as a safe place, before drawing it on the coordinate plane provided.

For students who finish early, challenge them to create their own triangles, and have other students identify whether the triangle is equilateral, isosceles, or scalene using coordinate geometry.

13.6.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

13.6.5 Wrap-Up: Cool Down

1. Identify **two** strategies you can use to classify triangles using coordinate geometry.
   Answers will vary.

Consider the online resources for additional enrichment opportunities for students to practice or test their abilities.

13.7 Centroids of Triangles

Lesson Introduction

Benchmark	MA.912.GR.3.2 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Target	• I can use coordinate geometry to identify properties of a triangle.

	Building On	Working Toward
Benchmark Coherence	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.	N/A

Essential Questions	• How do you find the centroid of a triangle by graphing the medians? • What is a median in a triangle? • What is a centroid in a triangle? • How do you find the centroid using the coordinates of the midpoints of the three sides of the triangle?
Lesson Narrative	Lesson 13.7 guides students to find the centroid of a triangle. The Exploration has students finding the midpoints of each side of the triangles. Then, students use the steps to construct the medians in order to find the centroid of the given triangle. In the Collaboration activity, students discover that the process for finding the centroid involves finding the averages of the x-coordinates and the y-coordinates. Students work with both strategies in the lesson.

Component Summary	13.7.1 Warm-Up (~5 mins)	First question to come to mind *(SE p. 322)*	13.7.3 Collaboration (10 mins)	Partner/group exercise on finding the centroid *(SE p. 325)*
	13.7.2 Exploration (20 mins)	Discovery activity on centroids of triangles *(SE p. 322)*	13.7.4 Your Turn! (10 mins)	Independent practice finding the centroid *(SE p. 325)*
	13.7.5 Wrap-Up (~5 mins)	Ticket Out the Door *(SE p. 326)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Centroid • Midpoint • Median	• Straightedge • Calculators • (Optional): Graphing utility • (Optional): Patty paper • (Optional): Colored pens/pencils	• If planning on using patty paper, graph paper, and/or colored pens and pencils, have these materials ready for students to use.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with finding the centroid when decimals are involved.
Students may struggle with finding the third vertex of the triangle on question 2 in 13.7.4 Your Turn!.</td><td>Consider allowing students to use colored pencils/pens when graphing the medians to find the centroid.
Consider allowing students to use patty paper or graph paper on certain problems.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
Lesson 13.7 is structured to help students make connections between two various strategies for finding the centroid of a triangle. Execute this lesson by implementing strategies that help students compare and connect the methods for finding the centroid of a triangle.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Lesson 13.7 is designed to provide students with the opportunity to make a connection between the x- and y-coordinates to find the centroid. The lesson also provides students with scaffolding necessary to discover patterns relating to finding the centroid of a triangle. Encourage students to find and look for the patterns when finding the centroid of a triangle.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider using visuals to highlight the concept of centroids. There are several graphing utilities that students can use to discover the relationships of centroids of triangles. Another approach is to have students draw a triangle on patty paper. Students find the midpoints on each side of the triangle, and then fold each of the three segments at the midpoints to form the medians. Once the three medians are visible, have students find the point where the three medians meet to discover the centroid.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.7.1 Warm-Up: Human Hamster Wheel

Component Overview: *Students examine a picture and write down the first question that comes to their mind. Students record two things that they noticed in the picture.*

13.7.1 Warm-Up: Human Hamster Wheel

A human hamster wheel is shown.

1. What is the first question that comes to your mind?
   Answers will vary.
   What is the diameter of this? Will the man be able to hold on if it starts rolling faster? How many of the white boxes is this made up of?

2. What are two things you notice about the human hamster wheel?
   Answers will vary.
   It is very large and round. The white boxes making up the wheel appear to be equal in size.

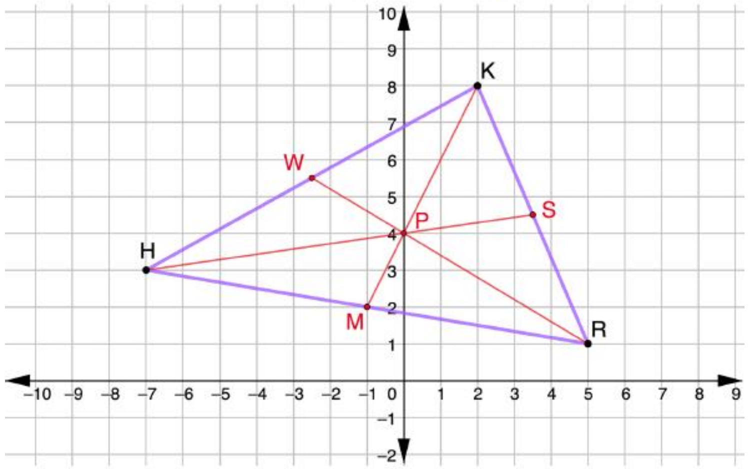
Consider having students share their questions and notices with the class to hear a variety of viewpoints.

13.7.2 Exploration: Centroids of Triangles

Component Overview: *This Collaboration guides students through the steps for finding the centroid of a triangle. Students start by finding midpoints of each side of the triangle, and then are guided to find the centroid using the averages of the x-coordinates and the y-coordinates.*

13.7.2 Exploration: Centroids of Triangles

1. ΔHKR has vertices at $H(-7,3)$, $K(2,8)$, and $R(5,1)$.



a. Complete the table by finding the midpoint of each line segment.

Line Segment	Work	Midpoint
$\overline{HK}$	$\left(\frac{-7+2}{2}, \frac{3+8}{2}\right)$ $\left(\frac{-5}{2}, \frac{11}{2}\right)$	$(-2.5, 5.5)$
$\overline{KR}$	$\left(\frac{2+5}{2}, \frac{8+1}{2}\right)$ $\left(\frac{7}{2}, \frac{9}{2}\right)$	$(3.5, 4.5)$
$\overline{RH}$	$\left(\frac{5-7}{2}, \frac{1+3}{2}\right)$ $\left(\frac{-2}{2}, \frac{4}{2}\right)$	$(-1, 2)$

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Consider where to take the opportunity to summarize and synthesize what is discovered before the next component. Allow students to revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group when you summarize.

Additional scaffolding ideas and questions are provided, if needed, but this should be a primarily student-directed Exploration.

Consider having students plot the midpoints on the graph after finding the results in the table.

As students are working, ask the following questions to prompt student thinking:

- *What is the midpoint of a line segment?*
- *Can the graphs be used to find the midpoints? If so, how?*

13.7.2 Exploration: Centroids of Triangles (continued)

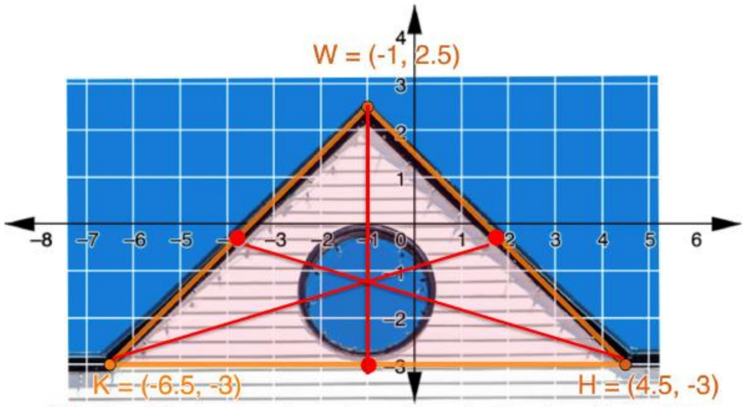
b. Complete the steps to find the centroid of triangle HKR.
Shown on given triangle.

1	Graph the midpoint of line $\overline{HK}$ and label it W.
2	Using a straightedge, draw the median from W to the opposite vertex, $\angle HRK$.
3	Graph the midpoint of line $\overline{KR}$ and label it S.
4	Using a straightedge, draw the median from S to the opposite vertex, $\angle KHR$.
5	Graph the midpoint of line $\overline{RH}$ and label it M.
6	Using a straightedge, draw the median from M to the opposite vertex, $\angle HKR$.
7	Label the centroid of triangle HKR point P.

c. What are the coordinates of the centroid of ΔHKR?
$(0, 4)$

Consider having students use three separate colors to draw the medians, so that each median is a different color. This strategy may be helpful for students to see the medians and to find the centroid.

2. The façade of a house is shown on the coordinate grid. The façade is represented by ΔWHK with vertices $W(-1, 2.5)$, $H(4.5, -3)$, and $K(-6.5, -3)$.



Work with your partner to estimate the location of the centroid of the triangular façade.

Midpoint of $KW = \left(\frac{-6.5-1}{2}, \frac{-3+2.5}{2}\right)$
$= \left(\frac{-7.5}{2}, \frac{-0.5}{2}\right)$
$= (-3.75, -0.25)$

Midpoint of $KH = \left(\frac{-6.5+4.5}{2}, \frac{-3-3}{2}\right)$
$= \left(\frac{-2}{2}, \frac{-6}{2}\right)$
$= (-1, -3)$

Midpoint of $WH = \left(\frac{-1+4.5}{2}, \frac{2.5-3}{2}\right)$
$= \left(\frac{3.5}{2}, \frac{-0.5}{2}\right)$
$= (1.75, -0.25)$

See coordinate gird for example student work.
Answers will vary.
Centroid is approximately $(-1, -1.2)$.

Students may struggle with this problem, as there are multiple decimal coordinates in the diagram.

Consider having students use patty paper over the diagram to better see and find the centroid.

Also, consider having students make a table similar to the table in question 1a to organize their work.

13.7.2 Exploration: Centroids of Triangles (continued)

3. Use the centroids for ΔHKR and ΔWHK to complete the table.
   Answers for the centroid of ΔWHK will vary because students estimated the location.

Triangle	Centroid	Average of the x –coordinates of the Vertices	Average of the y –coordinates of the Vertices
ΔHKR	$(0,4)$	0	4
ΔWHK	$(-1,-1.2)$	-1	-1.17

4. Work with your partner to make a conjecture about the relationships of the centroid and the average of the x and y coordinates.
   The average of the x-coordinates of the vertices of the triangle gives the x-value of the centroid. The average of the y-coordinates of the vertices of the triangle gives the y-value of the centroid.

5. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjectures. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
   Answers will vary.

Conjecture	My Summary of Their Conjecture	Initials

6. Review your conjecture with your partner. If necessary, make any changes.
   Answers will vary.

Which representation made it easier to find the centroid, the picture, or the table? Why?

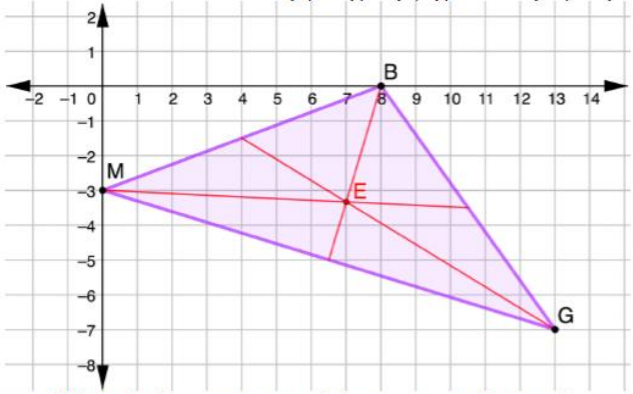
Provide students with the opportunity to share their conjectures with the class, as well as with the two classmates. Look for any commonalities in the conjectures. Use the Literacy and Language Routine to have students compare their conjectures, and connect to the first method for finding the centroid.

13.7.3 Collaboration: Centroids of Triangles

Component Overview: *Students work together to find the centroids of triangles. Allow students to work with partners and/or small groups during this component of the lesson.*

13.7.3 Collaboration: Centroids of Triangles

1. ΔMBG has vertices at $M(0,-3)$, $B(8,0)$, and $G(13,-7)$.



   a. What is the average of the x –coordinates?
   7

   b. What is the average of the y –coordinates?
   $\frac{-10}{3}$

   c. What are the coordinates of the centroid?
   $\left(7,-\frac{10}{3}\right)$

   d. Verify the coordinates of the centroid by drawing the median of each side.
   Shown above.

If your students struggled creating a conjecture in 13.7.2 Exploration, then consider using 13.7.3 as a Guided Instruction or to work with a small group of students who need further support.

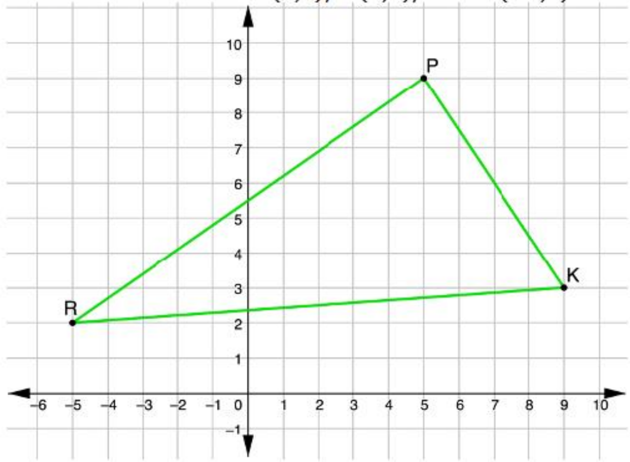
Encourage students to interact and discuss while completing this section of the lesson by asking clarifying questions and explaining their thought process. Have students use MLR5 to connect the patterns for the methods for finding the centroid. These questions give students the opportunity to make connections.

13.7.4 Your Turn!

Component Overview: *Students work independently to practice finding the centroids of triangles using different representations of the given triangle.*

13.7.4 Your Turn!

1. ΔPKR has vertices at $P(5,9)$, $K(9,3)$, and $R(-5,2)$.



What are the coordinates of the centroid of ΔPKR?

Centroid is $\left(3, \frac{14}{3}\right)$.

Monitor which strategy students are using to find the centroid. Allow students to either find the centroid by finding the average of the x and y-coordinates, or by having students find the midpoint of each side of the triangle and graphing the medians.

13.7.4 Your Turn! (continued)

2. ΔNHT has vertices at $N(-8,-3)$ and $H(3,-1)$ and a centroid at $(-2,-4)$.

   What are the coordinates of vertex T?

   $\frac{-8+3+x}{3}=-2$

   $x=-1$

   $\frac{-3-1+y}{3}=-4$

   $y=-8$

   Vertex T is $(-1,-8)$.

3. Artemio is finding the centroid of ΔWYV with vertices at $W(-8,5)$, $Y(7,8)$, and $V(-2,-3)$.

   His work is shown.

Artemio's Work	
$x=\frac{8+7+2}{3}$ $x=\frac{17}{3}$	$y=\frac{5+8+3}{3}$ $y=\frac{16}{3}$
Centroid $\left(\frac{17}{3},\frac{16}{3}\right)$	

   a. Write a note to Artemio explaining his mistake.
   There should be negative signs in front of the 8 and 2 on the x side and in front of the 3 on the y side.

   b. What are the coordinates of the centroid of ΔWYV?
   $\left(-1,\frac{10}{3}\right)$

Consider opportunities to have students use a visual graphing utility to help assist with question 2.

Ask the following questions to students who are struggling with question 2:

- *What is the pattern that you discovered in the lesson?*
- *How can you set up an equation to find the centroid?*

Have a couple of students share their notes to see a variety of viewpoints.

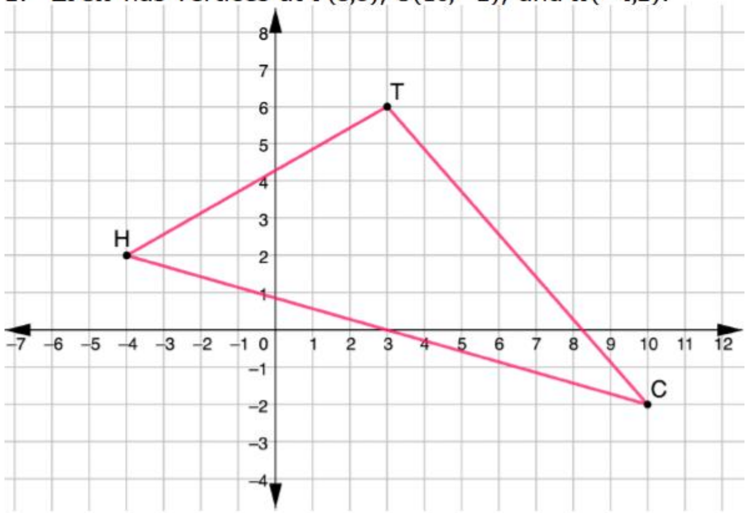
For students who finish early, challenge them to create three coordinates and graph on a coordinate plane, or use a graphing utility. Then, have students find the centroid using either method.

13.7.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a question based on the learning target of the lesson.*

13.7.5 Wrap-Up: Ticket Out the Door

1. ΔTCH has vertices at $T(3,6)$, $C(10,-2)$, and $H(-4,2)$.



What are the coordinates of the centroid of ΔTCH?
Centroid is $(3,2)$.

Use student understanding to inform your instruction, and to determine where, during other lessons, to work with students who struggle with determining the centroid. Make a list of misconceptions, and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

13.8 Classifying Quadrilaterals – Part 1

Lesson Introduction

Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context. • I can draw a quadrilateral on a coordinate plane using given definitions, properties, or theorems.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can quadrilaterals be classified according to their properties? • What justifications can be given to prove that a shape is a certain type of quadrilateral? • How do you find the lengths of sides and diagonals of quadrilaterals?
Lesson Narrative	Throughout Lesson 13.8, students work on classifying quadrilaterals on a coordinate plane. This lesson guides students to use the Pythagorean Theorem to find the lengths of the sides and diagonals of quadrilaterals. Students will use slope to determine if lines are parallel, perpendicular, or neither. Students will then classify the quadrilateral using the specific properties of each type of quadrilateral.

Component Summary				
	13.8.1 Warm-Up (~5 mins)	Comparison of measurements *(SE p. 327)*	13.8.4 Guided Practice (15 mins)	Drawing, Classifying, and Justifying Quadrilaterals *(SE p. 331)*
	13.8.2 Exploration (10 mins)	Properties of Quadrilaterals *(SE p. 328)*	13.8.5 Your Turn! (5 mins)	Classifying quadrilaterals Independently *(SE p. 332)*
	13.8.3 Exploration (10 mins)	Classifying Quadrilaterals *(SE p. 329)*	13.8.6 Wrap-Up (~5 mins)	Error Analysis *(SE p. 333)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Quadrilateral • Vertices	• Calculators • (optional) Graphing utility	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may make an error when using the Pythagorean Theorem or Distance Formula to find the lengths of sides or diagonals of a quadrilateral.
Students may forget the properties for each quadrilateral.</td><td>13.8.2 Exploration spirals content from Unit 8. Consider using information from assessments and observational data to understand what misconceptions from unit 8 may perpetuate in this unit. Consider how to address these misconceptions, and where small group instruction may be helpful.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Discussion Supports
There are several places within the lesson that students are asked to discuss with their partners. While going through the Guided Practice, consider having students turn to their partner to discuss what they just learned. This opens the lesson to have students justify their work with the discussion. Use this with 13.8.2 Exploration, questions 2-3, and 13.8.4 Guided Practice, question 4. Listen to students’ conversations and consider asking higher order questions to get the students thinking about their justifications.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students will be working on classifying quadrilaterals in the coordinate plane. As students explore, they can ask their partners clarifying questions to deepen their understanding. The Explorations and Guided Practice provide help and support for students with the new concept. In 13.8.5 Your Turn! students participate in individual, effortful learning, while working through complex problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Visuals of all types of quadrilaterals can help a learner see what properties they are applying to what shape, to classify it correctly. The use of GeoGebra, as a class or on student devices, can help the students to see the drawings and be able to identify some of the properties applied to the shapes, to help with classification of all types of quadrilaterals. Discussions throughout the lesson will assist students who benefit from auditory entry points.</td></tr>
<tr><td colspan="3">Lesson Notes</td></tr>
</table>

13.8.1 Warm-Up: Would You Rather?

Component Overview: Students will identify which amount of gold they would rather have, where they have the choice between a 14 m cube of gold or two cubes — a 12 m cube and the other an 8 m cube — combined.

13.8.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of 1 ounce of gold was nearly $2000.

1. Would you rather have one cube made of solid gold measuring 14 meters (m) on each side, or two cubes made of solid gold, one measuring 12 m on each side and one measuring 8 m on each side?

Answers will vary.

2. Explain your choice.
Answers will vary. I would take the 14-meter square because it is the biggest square. I would take the two 12 meter and 8-meter squares because they have more surface area.

Ask the following question to connect the Warm-Up to the lesson:

- To compare the weight of the gold, is it better to find the surface area or the volume?

13.8.2 Exploration: Classifying Quadrilaterals Using Properties

Component Overview: *The Exploration has students using a word bank to classify quadrilaterals by specific properties. Students will discuss with their partner about certain properties to see if it can be used to be classified as a quadrilateral or not.*

13.8.2 Exploration: Classifying Quadrilaterals Using Properties

1. Complete the table by filling in the blanks using the word bank. Words may be used more than once, or they may not be used.

Word Bank	
bisect(s)	congruent
parallel	perpendicular

Classification	Property
Parallelogram	Opposite sides are ___*parallel*___ and ___*congruent*___.
Parallelogram	Diagonals ___*bisect*___ each other.
Rectangle	Diagonals are ___*congruent*___.
Rhombus	Diagonals are ___*perpendicular*___.
Square	Diagonals are ___*perpendicular*___ and ___*congruent*___.
Kite	Exactly one diagonal ___*bisects*___ the other, and diagonals are ___*perpendicular*___.
Isosceles Trapezoid	Diagonals are ___*congruent*___ and at least one pair of sides are ___*parallel*___.

2. Discuss with your partner if knowing two consecutive sides are congruent is sufficient to classify a quadrilateral. Summarize your discussion and justify your decision.
Answers will vary.
No. There is more than one type of quadrilateral that has congruent consecutive sides. For example, this is the case for both kites and squares.

Allow students to review their notes and Unit 8 to recall the properties. Research shows that this type of spaced recall helps students solidify the information.

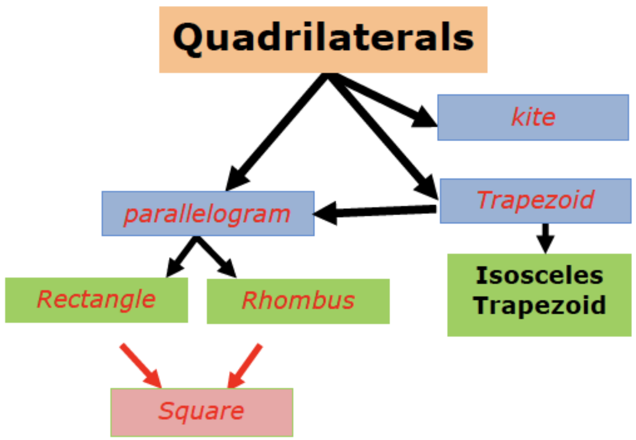
While students are working, consider collecting observational data, using a checklist of the quadrilaterals. Specifically, focus on what attributes students struggled remembering or finding. Use the observations from 13.8.2 and 13.8.3 to inform what differentiation or small-group instruction may be helpful.

Consider having drawings of the quadrilaterals to help in classifying them, according to specific properties.

13.8.2 Exploration: Classifying Quadrilaterals Using Properties (continued)

3. Discuss with your partner if knowing a pair of opposite sides are congruent is sufficient to classify the quadrilateral. Summarize your discussion and justify your decision.
   Answers will vary.
   No. Many types of quadrilaterals have opposite sides that are congruent. Rectangles, squares, and rhombi are a few examples.

4. Complete the diagram using the different types of quadrilaterals. Each blue box is a subcategory of quadrilaterals. Each green box is a subcategory of the blue box that it is connected to.



5. Which quadrilateral was not included in the diagram?
   Square

6. Draw a box and any necessary arrows to include the missing quadrilateral in the diagram.
   Shown above.

Consider asking students who finish early to think of other ways to display the relationships and hierarchy of quadrilaterals.

It might be beneficial to have students draw each quadrilateral and label the properties next to the category names to help visual learners know what each quadrilateral looks like.

After students complete the diagram, consider putting the diagram on display in the classroom for students to use as a reference.

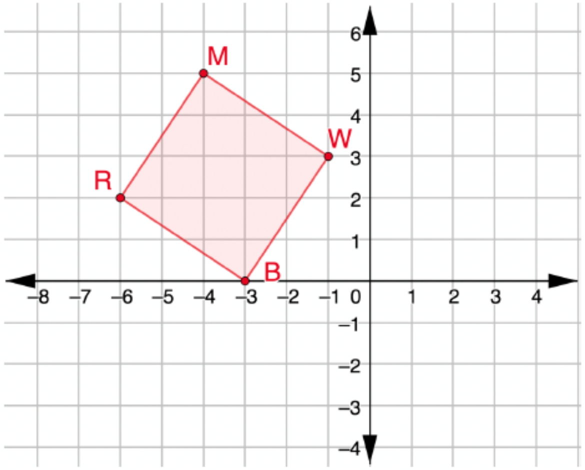
13.8.3 Exploration: Classifying Quadrilaterals

Component Overview: *In this Exploration, students will use the properties of quadrilaterals to classify the quadrilateral given.*

13.8.3 Exploration: Classifying Quadrilaterals

1. $RMWB$ has vertices at $R(-6,2)$, $M(-4,5)$, $W(-1,3)$, and $B(-3,0)$.

   a. Graph $RMWB$ on the coordinate plane.



   b. Complete the table.

Line Segment	Length
$\overline{RM}$	$\sqrt{13}$
$\overline{MW}$	$\sqrt{13}$
$\overline{WB}$	$\sqrt{13}$
$\overline{BR}$	$\sqrt{13}$

   c. What do you notice about the side lengths of $RMWB$?
   All the sides lengths are equal.

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.8.4 Guided Practice. This could include highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class.

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

d. Select all of the classifications that apply to $RMWB$.
- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid

e. Complete the table.

Diagonal Line Segment	Length
$\overline{RW}$	$\sqrt{26}$
$\overline{MB}$	$\sqrt{26}$

f. What do you notice about the diagonal lengths of $RMWB$?
The diagonals are congruent.

g. Complete the statement.

Since $RMWB$ ○ has / ○ does not have congruent diagonals, $RMWB$ is classified as a(n) _square or rectangle_.

2. $NTCV$ has vertices at $N(2,3)$, $T(3,5)$, $C(6,6)$, and $V(4,2)$.
   a. Complete the table.

Line Segment	Length
$\overline{NT}$	$\sqrt{5}$
$\overline{TC}$	$\sqrt{10}$
$\overline{CV}$	$2\sqrt{5}$
$\overline{VN}$	$\sqrt{5}$

Consider using this question to revisit the formal definition of a trapezoid as a quadrilateral with at least one pair of parallel sides. Use guiding questions such as, "Which quadrilaterals can also be classified as trapezoids?", (i.e., square, rectangle, rhombus, parallelogram).

- *How can you determine the length of the line segment?*
- *What are the properties used to help in the classification of the quadrilateral?*
- *What information is best used in classifications with side measurements and the diagonal?*

13.8.3 Exploration: Classifying Quadrilaterals (continued)

b. What do you notice about the side lengths of $NTCV$?
Two adjacent sides are the same measurements.
$NT = VN$

c. Complete the table.

Line Segment	Slope
$\overline{NT}$	2
$\overline{TC}$	$\frac{1}{3}$
$\overline{CV}$	2
$\overline{VN}$	$-\frac{1}{2}$

d. What do you notice about the slopes of $NTCV$?
Answers will vary. Quadrilateral $NTCV$ has one set of parallel sides and two set of perpendicular sides.

e. Select all of the classifications that apply to $NTCV$.

- ☐ Isosceles trapezoid
- ☐ Kite
- ☐ Parallelogram
- ☐ Rectangle
- ☐ Rhombus
- ☐ Square
- ☐ Trapezoid

- *What information do the slopes of the line segments tell us about the quadrilateral?*
- *Why do you need to find the lengths and slopes of the sides for quadrilateral $NTCV$, whereas with quadrilateral $RMWB$, you needed side lengths and diagonals?*
- *Do you need to find the diagonals of quadrilateral $NTCV$? Why or why not?*

13.8.4 Guided Practice: Classifying Quadrilaterals

Component Overview: *Students work independently to practice justifying quadrilaterals as specific shapes, and how to use the coordinate plane to help in the justification.*

13.8.4 Guided Practice: Classifying Quadrilaterals

1. $GSPD$ has vertices at $G(-2,-3)$, $S(3,-2)$, $P(5,-4)$, and $D(0,-5)$.

   Justify that $GSPD$ is a parallelogram but not a rectangle. Show your work.

   $1^2+5^2=GS^2$
   $\sqrt{26}=GS$

   $2^2+2^2=SP^2$
   $2\sqrt{2}=SP$

   $1^2+5^2=DP^2$
   $\sqrt{26}=DP$

   $2^2+2^2=GD^2$
   $2\sqrt{2}=GD$

   Quadrilateral $GSPD$ is a parallelogram because opposite sides are congruent.

   $GP=\sqrt{(5+2)^2+(-4+3)^2}$
   $GP=\sqrt{50}$

   $DS=\sqrt{(0-3)^2+(-5+2)^2}$
   $DS=\sqrt{18}$

   Since the diagonals are not congruent, the parallelogram is not a rectangle.

2. $HMTK$ has vertices at $H(-2,4)$, $M(-2,0)$, $T(-6,-1)$, and $K(-6,5)$. Classify $HMTK$ and justify your classification using mathematics.

   $\overline{HM}$ and $\overline{KT}$ both have undefined slopes thus making them parallel. $\overline{KH}$ has a slope of $-\frac{1}{4}$ and $\overline{TM}$ has a slope of $\frac{1}{4}$, making them not parallel.

   $KH=\sqrt{1^2+4^2}=\sqrt{17}$
   $TM=\sqrt{1^2+4^2}=\sqrt{17}$

   Since the non-parallel sides are equal in length, the quadrilateral is an isosceles trapezoid.

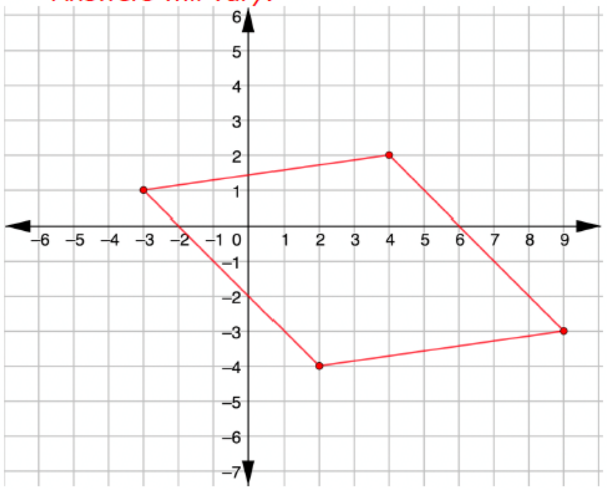
Consider allowing students to use graph paper or an online graphing utility with question 1 to provide entry points for visual learners.

Consider using Instructional Routine, "Take Turns," during question 1. Students work with a partner or a small group. Students will take turns finding the length of sides or the diagonals. Students will be explaining, justifying, and asking clarifying questions within their group.

Consider providing students who might be struggling with a table to stay organized, and to have a better visual of the connections between side lengths and slopes.

13.8.4 Guided Practice: Classifying Quadrilaterals (continued)

3. Use the coordinate grid to complete the following.
   a. Draw a quadrilateral for which the side lengths are all equal, but the diagonals are not congruent.
   Answers will vary.



   b. Complete the statement.
   This quadrilateral is classified as a ___*rhombus*___.

4. Discuss with your partner how another classmate could use the coordinates of your quadrilateral to verify that your classification is correct. Summarize your discussion.
   Answers will vary.
   They could use the Pythagorean theorem to find the side measures of the quadrilateral to see if all sides measure the same. Then, they could use the Pythagorean theorem to find the length of the diagonals to determine that they are not equal.

Consider having students trade their quadrilateral with a partner to clarify the quadrilateral drawn is a rhombus, and to critique their partner's work, if needed.

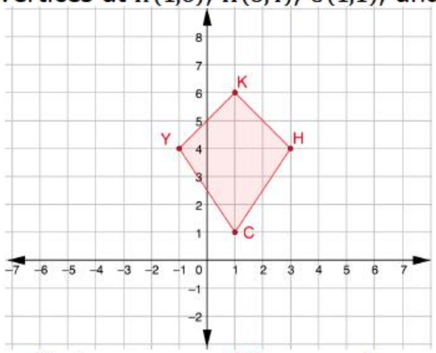
For students who finish early, consider challenging them to create a quadrilateral, then trade quadrilaterals with a partner to see if their partner can classify their quadrilateral with justifications.

13.8.5 Your Turn!

Component Overview: *Students work independently on drawing quadrilaterals to classify them by identifying the properties. Students use previous knowledge from the Exploration and Guided Practice areas to work independently.*

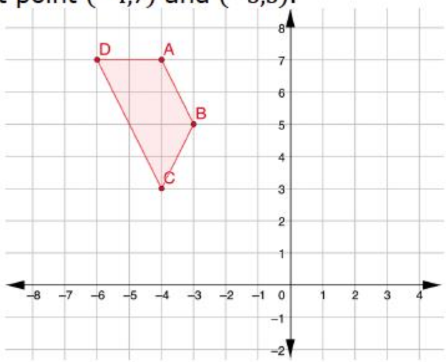
13.8.5 Your Turn!

1. $KHCY$ has vertices at $K(1,6)$, $H(3,4)$, $C(1,1)$, and $Y(-1,4)$.



$KHCY$ is classified as a ___*kite*___ because ...
exactly two pairs of adjacent sides are congruent, but opposite sides are not. Diagonals are perpendicular.

2. Draw a quadrilateral with one set of parallel sides and vertices at point $(-4,7)$ and $(-3,5)$.



This quadrilateral is classified as a ___*trapezoid*___ because ... *there is one set of parallel sides.*

Consider providing a table for students who may be struggling to keep their work organized, and to have a better visual on the connections between the side lengths, diagonals, and/or slopes.

Consider using guided questions to help students classify the quadrilateral:

- *What do you notice about the side lengths of the quadrilateral?*
- *What do you notice about the slopes of the side lengths of the quadrilateral?*

Have students share the quadrilateral they drew with a partner. Each partner should verify that the quadrilateral classified matches the quadrilateral drawn.

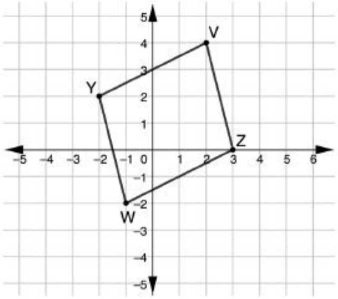
13.8.6 Wrap-Up: Error Analysis

Component Overview: *Students work through the Wrap-Up individually to decide if they agree or disagree with the given classification, and to justify why they are using properties and mathematical calculations.*

13.8.6 Wrap-Up: Error Analysis

1. $YVZW$ has vertices at $Y(-2,2)$, $V(2,4)$, $Z(3,0)$, and $W(-1,-2)$.

   Aryana graphed quadrilateral $YVZW$ and classified the quadrilateral as a rhombus.



   a. Do you agree with Aryana?
   No.

   b. Justify your answer using properties and showing your work.

   $4^2 + 1^2 = WY^2$
   $\sqrt{17} = WY$

   $4^2 + 1^2 = VZ^2$
   $\sqrt{17} = VZ$

   $4^2 + 2^2 = WZ^2$
   $2\sqrt{5} = WZ$

   $4^2 + 2^2 = YV^2$
   $2\sqrt{5} = YV$

   All four sides do not have equal length.

Students' answers on this activity can provide valuable data as to whether students understand the content of this lesson.

13.9 Classifying Quadrilaterals – Part 2

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context. • I can use coordinate geometry to solve geometric problems involving quadrilaterals in a mathematical or real-world context.

Benchmark Coherence	**Building On**	**Working Toward:**
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.	N/A

Essential Question	• How can you classify a quadrilateral in the real-world?
Lesson Narrative	Lesson 13.9 continues work with quadrilaterals, where students classify a quadrilateral that is in real-world context. Students will classify and justify their classification using properties.

Component Summary	13.9.1 Warm-Up (~5 mins)	Between Two Numbers *(SE p. 334)*	13.9.3 Your Turn! (10 mins)	Classifying quadrilaterals independently *(SE p. 338)*
	13.9.2 Collaboration (25 mins)	Real-World Quadrilaterals *(SE p. 335)*	13.9.4 Wrap-Up (~5 mins)	Self-Reflection *(SE p. 339)*

Resources	**Glossary**	**Materials**	**Additional Preparation**
	Key Terms: N/A Additional Terms: • Quadrilateral	• Calculators • (Optional) Graph paper or Graphing Utility	• If wanted, teachers can have the shapes graphed using GeoGebra ahead of the lesson on the board or on the internet, so it can be shown as a visual.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may get the properties mixed up according to what quadrilateral they belong to because some overlap.
Students may miscalculate the length measurements by not leaving them in radical form or rounding as stated.
Students may round values too early.</td><td>Consider if, during 13.9.2 Collaboration, students may benefit from you pulling the class back as a whole group. If so, then determine after which questions this would be helpful.
Consider if there are students who could benefit from working some of the questions in 13.9.2 Collaboration with you in a small group.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Three Reads
Questions 3a and 3b in 13.9.2 Collaboration allows students to engage in a collaboration by taking turns reading the problem to each other, identifying key information to share with each other, and discussing their reasonings of classifications mathematically using the quadrilateral properties.</td><td>MA.K12.MTR.6.1: Reasonableness
Throughout the lesson, reasonableness is used in justifications of shapes using properties of quadrilaterals. In 13.9.2 question 3, as students decide what person to choose, they must be precise and explain their choice. Within length measurements in the unit, checking the calculations is important when determining if there are congruent measurements or not.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Allow students to use a graphing utility to help them when plotting points in determining classifications and properties of quadrilaterals. Students may need to have their properties list with them if they are lower-level learners who need that guidance to help with classifying the quadrilaterals. These visuals will assist students who benefit from visual entry points. Consider assigning partners for 13.9.2 questions 3a and 3b to have them engage in collaboration with one another on choosing the appropriate choice of who worked the problem correctly, and to justify classifications of the properties to the quadrilateral correctly. These discussions will help deepen the understanding of the lesson for auditory learners.</td></tr>
<tr><td colspan="3">Lesson Notes</td></tr>
</table>

13.9.1 Warm-Up: Between Two Numbers

Component Overview: *Students will formulate their own values to substitute into the equation to make it reasonable.*

13.9.1 Warm-Up: Between Two Numbers

Sigsbee Deep	Britton Hill	Panorama Tower
Dwight Sigsbee led the first expedition to chart the floor of the Gulf of Mexico. The Sigsbee Deep, the deepest region of the Gulf of Mexico, was named after him. The exact maximum depth is not known, but is estimated to be between 3,750 meters (m) and 4,384 m.	Britton Hill, located in Walton County, is the highest point in the state of Florida. Britton Hill's elevation is 105 m.	Panorama Tower in Miami is the tallest building in Florida. It is 265 m tall.

Consider asking students if they have ever been to Britton Hill or the Panorama Tower.

1. Complete the inequality using the digits 0 – 9, without repeating, where $B =$ number of Britton Hills and $P =$ number of Panorama Towers.
   Answers will vary.

   $$3750 < \boxed{0}\boxed{9}B + \boxed{1}\boxed{2}P < 4384$$

2. Use a different set of digits, without repeating.
   Answers will vary.

   $$3750 < \boxed{1}\boxed{8}B + \boxed{0}\boxed{9}P < 4384$$

Consider extending this information into a discussion with the class to have students present their ideas.

- *What is your method of choosing the values?*
- *With which question was it easier for you to find the values?*

13.9.2 Collaboration: Real-World Quadrilaterals

Component Overview: *Students are asked to find lengths of segments in real-world situations, and classify the given quadrilateral using its properties.*

13.9.2 Collaboration: Real-World Quadrilaterals

1. Francesco is using computer software to design the layout of a house. The base of the house creates $DMHT$ with coordinates $D(-4,1)$, $M(4,5)$, $H(6,1)$, and $T(-2,-3)$.

   a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DM}$	$\sqrt{(4+4)^2+(5-1)^2}$ $\sqrt{(8)^2+(4)^2}$ $\sqrt{64+16}$ $\sqrt{80}=4\sqrt{5}$
$\overline{MH}$	$\sqrt{(6-4)^2+(1-5)^2}$ $\sqrt{(2)^2+(-4)^2}$ $\sqrt{4+16}$ $\sqrt{20}=2\sqrt{5}$
$\overline{HT}$	$\sqrt{(-2-6)^2+(-3-1)^2}$ $\sqrt{(-8)^2+(-4)^2}$ $\sqrt{64+16}$ $\sqrt{80}=4\sqrt{5}$
$\overline{TD}$	$\sqrt{(-4+2)^2+(1+3)^2}$ $\sqrt{(-2)^2+(4)^2}$ $\sqrt{4+16}$ $\sqrt{20}=2\sqrt{5}$

This Collaboration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.9.3 Your Turn!. This could include highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Collaboration.

Consider using the Instructional Routine, "Take Turns." Each member of the group will take a turn at finding the length of the line segment. Together, the group will critique each other's work, ask clarifying questions, and make corrections, if needed.

b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DH}$	$\sqrt{(6+4)^2+(1-1)^2}$ $\sqrt{(10)^2+(0)^2}$ $\sqrt{100+0}$ $\sqrt{100}$ 10
$\overline{MT}$	$\sqrt{(-2-4)^2+(-3-5)^2}$ $\sqrt{(-6)^2+(-8)^2}$ $\sqrt{36+64}$ $\sqrt{100}$ 10

c. Finish the statement.
$DMHT$ is classified as a ___*rectangle*___ because …

the diagonals are congruent with opposite sides being congruent and parallel.

d. If 1 unit on the coordinate plane equals 5 feet (ft), what are the dimensions of the base of the house? Round to the nearest thousandth.
$22.361\ ft \times 44.721\ ft$

2. Jillian is determining which type of tile to place on the floor. She wants a tile in the shape of a parallelogram. The vertices of the tile options are shown.

Tile A	Tile B
$(-11,6.5)$, $(-10,3.5)$, $(-6,6.5)$, and $(-7,9.5)$	$(3,9.5)$, $(7,6)$, $(9,8.5)$, and $(5,11)$

a. Which tile should Jillian pick if she wants a tile in the shape of a parallelogram?
Tile A

b. Explain your choice by listing the properties of parallelograms that were applied.
Answers may vary. I found the slope of the sides and determined that opposite sides are parallel for Tile A.

For students who finish early on question 1, challenge them to create a new house design, making sure their points make a house base. Students can swap with another peer and answer the same question as in question 1, and think about how the new problem relates to the original in the solutions of lengths or classifications.

Provide graph paper or a graphing utility for students on question 2, where they can graph the points for the tiles to determine which one will work and fit the description. Some students may also need a properties sheet, depending upon their fluency with it.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

3. Aharon and Shreya classified $RNPF$ with vertices at $R(-3,7)$, $N(-1,7)$, $P(-3,2)$, and $F(-5,4)$. Their classifications are shown.

Aharon's Classification	Shreya's Classification
$RNPF$ is a square or rectangle because the diagonals are congruent.	Quadrilateral $RNPF$ has congruent diagonals but cannot be classified as a parallelogram.

a. Whose classification do you agree with? Justify your answer using properties.
Shreya because more than just diagonals of a shape are needed to identify what type of quadrilateral it is.

b. Write a note to the person whose classification is incorrect, explaining their mistake.
Answers will vary.
Side measurements of the shape should be considered to be able to identify the shape and not just the diagonals.

Consider guiding students through by asking the following questions:
- *What are the properties for a square and a rectangle?*
- *Do you need all of those properties to classify a quadrilateral as a square or rectangle? Why or why not?*

Consider providing struggling students with a table to keep their work organized and help them better see the connections between side lengths.

4. $KVSW$ has vertices at $K(10,-4)$, $V(9,-8)$, $S(1,-4)$, and $W(6,-2)$.

   a. Show mathematically that $KVSW$ is a trapezoid, but not an isosceles trapezoid.

   Slope of $\overline{WK} = \frac{-2-(-4)}{6-10} = -\frac{2}{4} = -\frac{1}{2}$
   Slope of $\overline{SV} = \frac{-8-(-4)}{9-1} = -\frac{4}{8} = -\frac{1}{2}$
   Slope of $\overline{KV} = \frac{-4-(-8)}{10-9} = \frac{4}{1} = 4$
   Slope of $\overline{SW} = \frac{-4-(-2)}{1-6} = \frac{-2}{-5} = \frac{2}{5}$
   Quadrilateral $KVSW$ is a trapezoid since it has one set of parallel sides.

   $SW = \sqrt{(6-1)^2 + (-2+4)^2}$
   $SW = \sqrt{29}$

   $KW = \sqrt{(6-10)^2 + (-2+4)^2}$
   $KW = 2\sqrt{5}$

   $VK = \sqrt{(9-10)^2 + (-8+4)^2}$
   $VK = \sqrt{17}$

   $SV = \sqrt{(9-1)^2 + (-8+4)^2}$
   $SV = 4\sqrt{5}$

   Two sides are not congruent so the trapezoid is not an isosceles.

   b. List the properties of an isosceles trapezoid that did not apply to $KVSW$.
   The legs of an isosceles trapezoid are congruent.
   The diagonals of an isosceles trapezoid are congruent.

- *What property does Quadrilateral $KVSW$ have to make it a trapezoid?*
- *What would Quadrilateral $KVSW$ need in order for it to be an isosceles trapezoid?*

Consider providing students with graph paper or a graphing utility to provide as a visual.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

5. Ariyon was graphing square $MGNR$, but she mixed up her vertices. She knows that point M has coordinates $(3,5)$ and point G has coordinates $(7,7)$. Her two options for points N and R are shown.

Point N	Point R
(8,3)	(5,1)
(9,3)	(4,1)

a. Which coordinates are correct for points N and R? Justify your answer or show your work.
$N = (9,3)$ and $R = (5,1)$.

Slope of $\overline{MG}$ is $\frac{1}{2}$
Slope of $\overline{RN}$ is $\frac{1}{2}$
Slope of $\overline{GN}$ is -2
Slope of $\overline{MR}$ is -2

$MG = \sqrt{(7-3)^2 + (7-5)^2} = 2\sqrt{5}$

$RN = \sqrt{(5-9)^2 + (1-3)^2} = 2\sqrt{5}$

$MR = \sqrt{(3-5)^2 + (5-1)^2} = 2\sqrt{5}$

$GN = \sqrt{(7-9)^2 + (7-3)^2} = 2\sqrt{5}$

b. Which properties of a square did you use to determine the correct points N and R?
A square has opposite sides that are parallel and adjacent sides that are perpendicular and all four sides congruent.

- *What other properties can you use to determine the coordinates for points N and R?*
- *How did you determine which coordinates are correct for points N and R?*

13.9.3 Your Turn!

Component Overview: *Students are classifying quadrilaterals and find missing measurements of quadrilaterals using the vertices given.*

13.9.3 Your Turn!

1. $SCWK$ has vertices at $S(-3,-4)$, $C(0,-2)$, $W(0,-10)$, and $K(-3,-8)$.
   $SCWK$ is classified as a(n) _*isosceles trapezoid*___
   because … *the bases are parallel and the two legs are congruent.*

2. Ziquan is designing a template for his 3D printer. He created two quadrilaterals using his software.

$TGHN$	$MWRB$
$T(16,6)$, $G(18,4)$, $H(17,-2)$, and $N(14,4)$	$M(4,5)$, $W(7,4)$, $R(8,-3)$, and $B(3,2)$

   a. Which quadrilateral can be classified as a kite? Show your work.
   Quadrilateral $MWRB$ is a kite.

   $MB = MW$
   $MW = \sqrt{(7-4)^2 + (4-5)^2} = \sqrt{10}$
   $MB = \sqrt{(3-4)^2 + (2-5)^2} = \sqrt{10}$

   $BR = WR$
   $BR = \sqrt{(3-8)^2 + (2+3)^2} = 5\sqrt{2}$
   $WR = \sqrt{(7-8)^2 + (4-(-3))^2} = 5\sqrt{2}$

   b. List the properties of a kite that you used.
   A kite has exactly two pairs of adjacent congruent sides.

   c. If 1 unit on the coordinate plane equals 6 inches (in), what are the measurements of the sides of the kite? Round to the nearest thousandth.
   $MB = 6(\sqrt{10}) = 18.974$
   $MW = 6(\sqrt{10}) = 18.974$
   $WR = 6(\sqrt{50}) = 42.426$
   $BR = 6(\sqrt{50}) = 42.426$

Question #1, students could put "trapezoid" instead of "isosceles trapezoid" because the bases are parallel, and the other two sides are not congruent, but this would not be precise.

Some students may need graph paper or a graphing utility. Also consider providing a table for students for their work.

Consider asking guiding questions for students who may be struggling:
- *What are the properties of a kite?*
- *What do you notice about the side lengths of the quadrilaterals?*

13.9.4 Wrap-Up: Reflection

Component Overview: *Students reflect on how they feel about what they have learned.*

13.9.4 Wrap-Up: Reflection

1. Place an **X** on the line in relation to how you feel about today's learning target.
   Answers will vary.

   ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

Additional questions to assess students learning and understanding:

- *What is something that became clearer today about classifying quadrilaterals?*
- *What area of quadrilaterals within the lesson did you struggle with?*

13.10 Classifying Quadrilaterals – Part 3

Lesson Introduction

Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.	Learning Target	• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.

	Building On	Working Toward
Benchmark Coherence	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through the given point.	N/A

Essential Questions	• Which quadrilateral properties can be used to classify the shape? • How can the shape be determined by the slope, side lengths, and/or diagonal lengths? • What verifications best fit the information found per card given in the activity?
Lesson Narrative	Students complete an activity using stations where they apply coordinate geometry to determine the type of quadrilateral the information represents.

Component Summary	13.10.1 Warm-Up (~5 Min)	Would You Rather? *(SE p. 340)*	13.10.2 Activity (40 Min)	Stations *(SE p. 340)*
	13.10.3 Wrap-Up (~5 Min)	Students summarize their learning *(SE p. 344)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• 13.10.2 Station Activity Cards • (Optional) Calculators • (Optional) Graphing Utility or Graph Paper	• N/A

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may struggle with remembering the properties of all the quadrilaterals. • Students could plot the vertices incorrectly, and lead to wrong calculations.	• Consider how best to organize the activity for your students. The activity can be completed by students individually, with partners, or in a small group. • If using small groups, consider making one of the stations a small group for students who need further support to work with you. • A calculator is not necessary for 13.10.2 Activity because students can leave values in radical form. Consider making 13.10.2 no calculator, and then allowing a calculator in the activity in 13.11.2
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Students can complete the 13.10.2 Activity with a partner or small group. Students can form their own group or be placed by the teacher. Students can be dealt one card or several as a group. Each group can read the cards to each other, and think about justifications of the problem while working it out. Students then can share with each other or the class as to what justifications they found and give reasoning including measurements, shape sketches, and properties to use in order to classify the quadrilateral.	**MA.K12.MTR.2.1: Multiple Representations** Students are completing an activity for 13.10.2, where they are asked to classify and justify the quadrilateral that is created with the information calculated and found per card. Students will be given multiple representations of each quadrilateral, and are asked to determine different information to classify the given quadrilateral, or to identify the missing coordinate for the quadrilateral given.
Differentiate Instruction	For struggling learners, consider handing the students the cards one a at time to complete. This will help keep students from getting overwhelmed, and help to focus on one problem at a time. Consider offering graphing paper or a graphing utility to assist visual learners. Auditory learners will benefit from working in a group to engage in discussions about classifying each quadrilateral. Placing the task cards around the room will help kinesthetic learners engage in the lesson.	

Lesson Notes

13.10.1 Warm-Up: Would You Rather?

Component Overview: *Students are asked to give their opinion on if they would rather go to the movies or watch tv shows.*

13.10.1 Warm-Up: Would You Rather?

1. Would you rather ...

Give up watching movies?	OR	Give up watching TV shows?

   Answers will vary.

2. Explain your choice.
   Answers will vary.

- *Why would the choice be movies over watching tv shows?*
- *Why would the choice be watching tv shows over choosing to go to the movies?*
- *What were the thoughts that came to mind as the question was read?*

13.10.2 Activity: Stations

Component Overview: *Students will be given cards that act as stations, and use the information on a card to classify the quadrilateral that is represented.*

13.10.2 Activity: Stations

Complete the table for each question card.

Question	Work		Answer
1	Slopes	Slopes of $\overline{RB}$ and $\overline{KM} = 1$ Slopes of $\overline{BM}$ and $\overline{RK} = -1$	**Properties** Opposite sides are parallel. Adjacent sides are perpendicular. All four sides are congruent. Diagonals are congruent.
	Side Lengths	$RB = \sqrt{18} = 4.243$ $BM = \sqrt{18} = 4.243$ $KM = \sqrt{18} = 4.243$ $RK = \sqrt{18} = 4.243$	
	Diagonal Lengths	$RM = 6$ $BK = 6$	**Quadrilateral** Square

Question	Work		Answer
2	Slopes	Slopes of $\overline{NH}$ and $\overline{DP}$ are undefined. Slopes of $\overline{ND}$ and $\overline{HP} = \frac{-3}{2}$	**Properties** Opposites sides are parallel and congruent.
	Side Lengths	$NH = 3$ $DP = 3$ $ND = \sqrt{13} = 3.606$ $HP = \sqrt{13} = 3.606$	
	Diagonal Lengths	$NP = \sqrt{40} = 6.325$ $HD = 2$	**Quadrilateral** Parallelogram

There are 8 cards that have information on them for the student to use to classify the quadrilateral that the information represents. Students record their work and answer in the workbook. The activity offers flexibility, so determine what works best for your students. If utilizing all 8 cards, then divide your class into 8 or 9 groups. The ninth group could be where students work with you to either have you check their work to that point, or to work on common misconceptions you have noticed. Set up the groups and the order so that only those students who need further support will visit your station. Eight cards are provided so that you have the flexibility to use some or all of them. Consider using a timer to note when it is time for students to rotate. Additional questions are provided for you to consider.

Consider using these suggestions for the Stations Activity:

- *Consider providing students with graph paper or a graphing utility, if needed.*
- *Lower-level students could only be given a few cards at a time to complete.*
- *Middle-level students could be given about 5-6 cards to complete for the class time.*
- *Upper-level students could be given all 8 cards to complete.*
- *The teacher can group students according to peer partners or small groups, depending on the teacher knowing how the students work best.*

13.10.2 Activity: Stations (continued)

<table>
<tr><th>Question</th><th colspan="2">Work</th><th>Answer</th></tr>
<tr><td rowspan="4">3</td><td rowspan="2">Length of $\overline{WH}$ and $\overline{HC}$</td><td rowspan="2">$WH = \sqrt{5} = 2.236$
$HC = \sqrt{10} = 3.162$</td><td>$(-10,1)$</td></tr>
<tr><td>Verification</td></tr>
<tr><td>Slopes of $\overline{WH}$ and $\overline{HC}$</td><td>Slope of $WH = \frac{1}{2}$
Slope of $HC = -\frac{1}{3}$</td><td rowspan="2">$WS = \sqrt{10}$
$= 3.162$

$(-10,1)$ would be point S because $\overline{WH}$ has the same slope as $\overline{CS}$ and $\overline{WS}$ is congruent to $\overline{HC}$.</td></tr>
<tr><td>Slope of $\overline{CS}$ Using Each Point S</td><td>$(-6,-2) = 1$
$(-10,1) = \frac{1}{2}$
$(-12,3) = \frac{1}{8}$</td></tr>
</table>

<table>
<tr><th>Question</th><th colspan="2">Work</th><th>Answer</th></tr>
<tr><td rowspan="4">4</td><td rowspan="2">Slopes</td><td rowspan="2">Slopes of $\overline{MT}$ and $\overline{FP} = 1$
Slopes of $\overline{MP}$ and $\overline{TF} = -\frac{3}{4}$</td><td>Properties</td></tr>
<tr><td rowspan="2">Opposite sides are parallel and congruent.

Diagonals are not congruent.</td></tr>
<tr><td>Side Lengths</td><td>$MT = \sqrt{18} = 4.243$
$FP = \sqrt{18} = 4.243$
$MP = \sqrt{25} = 5$
$TF = \sqrt{25} = 5$</td></tr>
<tr><td>Diagonal Lengths</td><td>$MF = \sqrt{37} = 6.083$
$TP = 7$</td><td>Who is correct?

Marissa</td></tr>
</table>

- *How can the lengths and slopes help you determine the coordinates for Point S?*
- *If quadrilateral $WHCS$ was just a trapezoid, how would your thought process change?*

- *What are the methods that was taken on card #4 to determine who was correct and who was not?*

13.10.2 Activity: Stations (continued)

Question	Work		Answer
5	Slopes	Slope of $\overline{KR} = 4$ Slope of $\overline{KC} = \frac{2}{5}$ Slope of $\overline{RN} = \frac{1}{4}$ Slope of $\overline{NC} = \frac{5}{2}$	**Properties** *Exactly two pairs of adjacent sides are congruent.*
	Side Lengths	$KR = \sqrt{17} = 4.123$ $KC = \sqrt{29} = 5.385$ $RN = \sqrt{17} = 4.123$ $CN = \sqrt{29} = 5.385$	
	Diagonal Lengths	$KN = \sqrt{18} = 4.423$ $RC = \sqrt{72} = 8.485$	**Quadrilateral** *Kite*

- *When you drew the quadrilateral on the coordinate grid, what quadrilateral did you think it was?*
- *What information about the quadrilateral, if any, did the slopes provide?*

Question	Work		Answer
6	Slopes	Slopes of $\overline{BR}$ and $\overline{WH} = -2$ Slopes of $\overline{WB}$ and $\overline{HR} = \frac{3}{5}$ Slopes of $\overline{TF}$ and $\overline{NS} = \frac{1}{2}$ Slopes of $\overline{TN}$ and $\overline{FS} = -2$	**Which quadrilateral is a rectangle?** *Quadrilateral $TFSN$ is a rectangle*
	Side Lengths	$BR = \sqrt{20} = 4.472$ $WH = \sqrt{20} = 4.472$ $WB = \sqrt{34} = 5.831$ $HR = \sqrt{34} = 5.831$ $TF = \sqrt{20} = 4.472$ $NS = \sqrt{20} = 4.472$ $TN = \sqrt{45} = 6.708$ $FS = \sqrt{45} = 6.708$	**Properties** *Opposite sides are parallel and adjacent sides are perpendicular.* *Opposite sides are congruent.* *Diagonals are congruent.*
	Diagonal Lengths	$WR = \sqrt{50} = 7.071$ $BH = \sqrt{58} = 7.616$ $TS = \sqrt{65} = 8.062$ $FN = \sqrt{65} = 8.062$	

- *Could you easily determine which quadrilateral was the rectangle from the graph?*
- *What information about the quadrilateral, if any, did the diagonals provide?*

13.10.2 Activity: Stations (continued)

Question	Work		Answer
			Why is $VWXY$ not a square?
7	**Slopes**	*Slopes of* $\overline{XY}$ *and* $\overline{WV} = 2$ *Slopes of* $\overline{YV}$ *and* $\overline{XW} = \frac{1}{2}$	*The slopes of adjacent sides are not opposite reciprocals to indicate perpendicular lines.*
	Side Lengths	$XY = \sqrt{20} = 4.472$ $WV = \sqrt{20} = 4.472$ $YV = \sqrt{20} = 4.472$ $XW = \sqrt{20} = 4.472$	*All sides are congruent proving that the quadrilateral is a rhombus.*
	Diagonal Lengths	$YW = \sqrt{8} = 2.828$ $XV = \sqrt{72} = 8.485$	*Diagonals are not congruent proving that the rhombus is not a square.*

- *What qualities does a rhombus have that a square does not?*
- *How is a square different than a rhombus?*

13.10.2 Activity: Stations (continued)

<table>
<tr><th>Question</th><th colspan="2">Work</th><th>Answer</th></tr>
<tr><td rowspan="3">8</td><td rowspan="2">Slope and Length of $\overline{KD}$</td><td rowspan="2">$m = 1$

$KD = \sqrt{18}$
$= 4.243$</td><td>$(7,3)$ would be the point S.</td></tr>
<tr><th>Verification</th></tr>
<tr><td>Slope of $\overline{MS}$ Using Each Point S</td><td>$(4,5) = -\frac{3}{2}$
$(6,3) = -\frac{5}{4}$
$(7,3) = -1$</td><td>The slope of $\overline{KD}$ is 1 and the slope of $\overline{MS}$ is -1 making the diagonals perpendicular.

$MD = 3$
$MK = 3$
$DS = \sqrt{29}$
$KS = \sqrt{29}$

A kite has exactly two pairs of adjacent sides congruent.</td></tr>
</table>

- *Could you easily determine where point S should be located using the graph?*
- *Which, if any, information was most useful to help determine which point S was correct?*
  - *Slope of KD*
  - *Length of $\overline{KD}$*

Consider challenging students who finish early by having them to create their own activity card to share with another person in their group/class to answer.

13.10.3 Wrap-Up: Cool Down

Component Overview: *Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

13.10.3 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

   Answers will vary.

13.11 What Is the Figure?

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify triangles and quadrilaterals. • I can use coordinate geometry to justify definitions, properties, or theorems of triangles and quadrilaterals. • I can use coordinate geometry to solve mathematical problems involving triangles and quadrilaterals.

Benchmark Coherence	**Building On**	**Working Toward**
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.	N/A

Essential Questions	• How can we use a quadrilateral's property to help justify that we have correctly identified them? • What can help us distinguish the difference between multiple quadrilaterals?
Lesson Narrative	In this lesson, students continue to use coordinate geometry to justify and classify polygons based on their properties through a Card Sort. Students are tasked to match each image with its correct classification, coordinates, and property, while also justifying their matches.

Component Summary	13.11.1 Warm-Up (~10 mins)	Bell Work *(SE p. 345)*	13.11.2 Activity (35 mins)	Card Sort *(SE p. 346)*
	13.11.3 Wrap-Up (~5 mins)	Students summarize their learning *(SE p. 347)*		

Resources	**Glossary**	**Materials**	**Additional Preparation**
	<u>Key Terms</u>: N/A <u>Additional Glossary</u>: N/A	• 13.11.2 Activity Card Sort cards • Calculators	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

- Students may get the properties mixed up because some properties of polygons overlap.
- Students may miscalculate the length measurements when using the Pythagorean Theorem or the Distance Formula.
- Students may complete matches by the look of the polygon, instead of justifying with math.

</td><td>

- Consider having students who are more visual learners to have graph paper or a graphing tool for the 13.11.1 Warm-Up.
- Consider having an anchor chart with the various type of quadrilaterals and polygons, along with an image for students who need help, during the Card Sort when matching images with names of polygons.
- Consider using information from the previous lessons to form a small group of students who are struggling with using coordinate geometry to classify figures. Facilitate these students as they work through 13.11.2 Activity.
- If time permits, consider utilizing 13.11.2 Activity over two days. The content that is spiraled in the activity utilizes concepts found in the three benchmarks noted, as well as concepts from MA.912.GR.1.4 and MA.912.GR.1.5.

</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>

Anticipate, Monitor, Select, Sequence, Connect

In the 13.11.12 Card Sort, students are anticipating, sequencing, and making connections between the multiple representations of polygons and how they are each represented. Students create a sequence on how they organize and choose their cards, and can form the connection between these representations through their justifications.

</td><td>

MA.K12.MTR.2.1: Multiple Representations

In the 13.11.2 Card Sort, students are building their understanding of coordinate geometry through using graphs, coordinate points, and properties of polygons. Students are expressing these connections between the concepts and representations using justifications.

</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">

In the 13.11.1 Warm-Up, consider that visual learners may have difficulties with identifying the stage that is in the shape of a trapezoid, so consider providing a graphing tool or graph paper for those students. Throughout the entire lesson, also consider having an anchor chart near the front of the class where students can reference the various types of quadrilaterals and polygons.

Additional differentiation opportunities are denoted throughout the lesson guide.

</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.11.1 Warm-Up: Bell Work

Component Overview: *Students are comparing two different quadrilateral stages and determining which one to use as a stage.*

13.11.1 Warm-Up: Bell Work

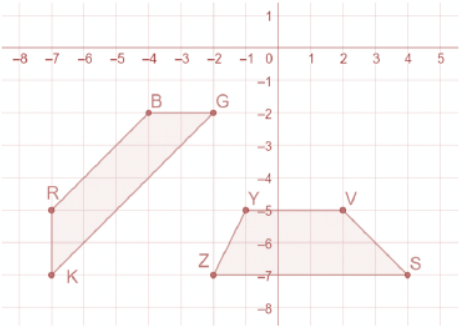
Jonah is designing a stage in the shape of a trapezoid. Jonah has created two designs.

VSZY	*BGKR*
$V(2,-5)$, $S(4,-7)$, $Z(-2,-7)$, and $Y(-1,-5)$	$B(-4,-2)$, $G(-2,-2)$, $K(-7,-7)$, and $R(-7,-5)$

1. Determine whether Jonah can use either or both designs for the trapezoidal stage.

   - Show your work in an organized way.
   - Write your conclusion explaining the properties used to determine your answer.

   Answers will vary. Both shapes create a trapezoid. The definition states that a quadrilateral is a trapezoid if at least one pair of sides are parallel. In trapezoid $RBGK$ both $\overline{RB}$ and $\overline{KG}$ have slopes equal to 1, making them parallel. In trapezoid $YVSZ$ both $\overline{YV}$ and $\overline{ZS}$ have slopes equal to 0, making them parallel.



Use student understanding to inform your instruction, and to determine where further student support could be implemented. Make a list of misconceptions, and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

Consider that visual learners may have difficulties with identifying the stage that is in the shape of a trapezoid, so consider providing a graphing tool or graph paper for those students.

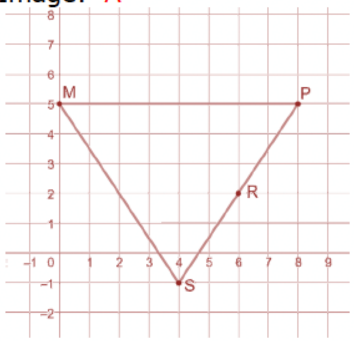
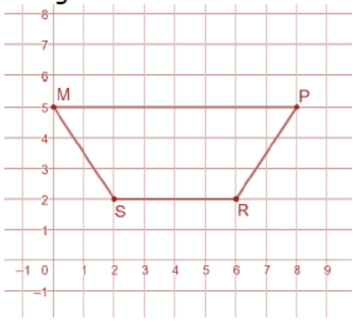
Consider also having an anchor chart near the front of the class where students can reference the various types of quadrilaterals, especially when trying to determine which stage Jonah should pick.

13.11.2 Activity: Card Sort

Component Overview: *Students are completing a Card Sort where they are matching each image, classification, and property that represents the same polygon. To make sure students are sure in their choices, they are then asked to provide justification with each match.*

13.11.2 Activity: Card Sort

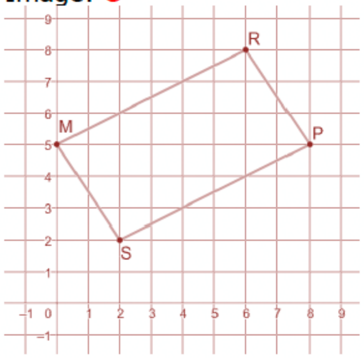
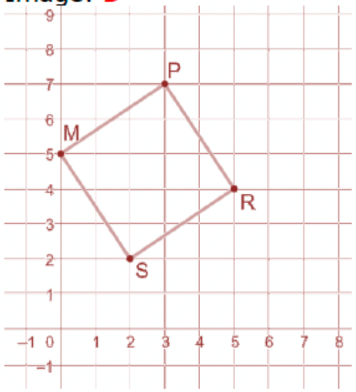
Match each image, classification, coordinates, and property that represent the same polygon. Record your matches in the table by writing the letter or number of each card and showing your justifications.
Order of matches may vary. Both the letter/number and answer are shown below.

Match 1	Match 2
Image: A 	Image: B 
Coordinates: 16 M(0,5), P(8,5), R(6,2), and S(4, −1)	Coordinates: 13 M(0,5), P(8,5), R(6,2), and S(2,2)
Classification: 4 *Isosceles triangle*	Classification: 7 *Isosceles trapezoid*
Property: 21 *Two adjacent sides are congruent*	Property: 23, 26 or 30 and 34 (also 22) *Diagonals are congruent. One pair of sides are congruent.*
Justification: *Answers will vary.* SP and $MS = \sqrt{52}$	Justification: *Answers will vary.* RP and $SM = \sqrt{13}$ SP and $RM = \sqrt{45}$

Students will match each image, classification, coordinate, and property(s) that represents the same polygon. Students record their matches in the table by writing the letter or number in their workbook. Consider having students work with a partner or small group. Ten cards are provided. This provides you with flexibility to determine how best to use the cards for your students. If students are struggling with certain figures, consider choosing to focus on these figures. A few different ways to utilize the Card Sort are:

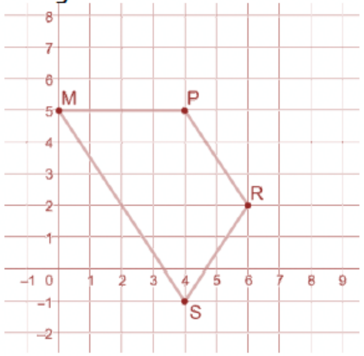
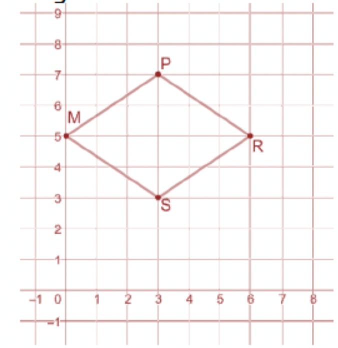
- *Have students work individually on 2-3 cards that represent figures your students have struggled with. Then, have students check their work with a partner. Allow students to adjust their answers. Then, consider giving students a copy of the remaining cards to complete on their own.*
- *Have students work in small groups on 4-5 cards that your students struggled with the most. Then, consider providing students a copy of the remaining cards to complete on their own.*
- *Utilize stations where the image or the set of coordinates is the station's anchor, and each station has all of the classifications, properties, coordinates or images, and justifications to sort through.*
- *Utilize the cards as a Gallery Walk where the image or the set of coordinates is displayed. Students then sort through the cards to add one to what is already displayed. Once 4 rotations are completed, then students rotate to verify the posted cards from the previous groups, adding a post-it note that states their agreement or their suggested change.*

13.11.2 Activity: Card Sort (continued)

Match 3	Match 4
Image: C 	Image: D 
Coordinates: 19 M(0,5), P(8,5), R(6,8), and S(2,2)	Coordinates: 17 M(0,5), P(3,7), R(5,4), and S(2,2)
Classification: 10 *Parallelogram*	Classification: 9 *Square*
Property: 25, 35 *Diagonals bisect each other.* *Opposite sides are congruent.* **Card 35 is also sufficient on its own.*	Property: 23, 26 or 30 and 27, 29, or 32 *Diagonals are congruent.* *Diagonals are perpendicular.*
Justification: *Answers will vary.* RM *and* $PS = \sqrt{45}$ MS *and* $RP = \sqrt{13}$ *Midpoint of* $\overline{MP}$ *is* $(4,5)$ *Midpoint of* $\overline{RS}$ *is* $(4,5)$	Justification: *Answers will vary.* MR *and* $PS = \sqrt{26}$ *Slope of* $\overline{MR} = -\frac{1}{5}$ *Slope of* $\overline{PS} = 5$

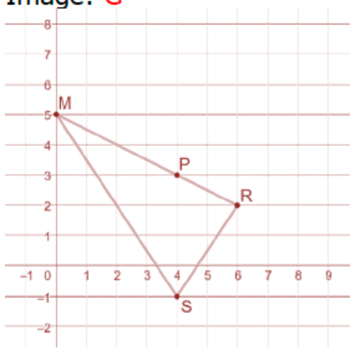
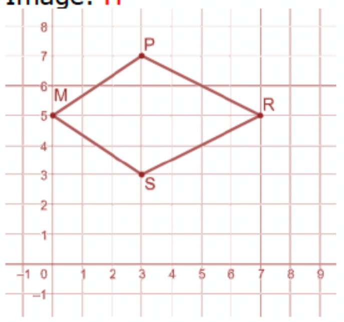
Consider having an anchor chart near the front of the class with the name and image of a variety of polygons and quadrilaterals. This can be especially helpful to visual learners. To challenge students, consider removing the image from the Card Sort.

13.11.2 Activity: Card Sort (continued)

Match 5	Match 6
Image: E 	Image: F 
Coordinates: 11 *$M(0,5)$, $P(4,5)$, $R(6,2)$, and $S(4,-1)$*	Coordinates: 20 *$M(0,5)$, $P(3,7)$, $R(6,5)$, and $S(3,3)$*
Classification: 3 *Trapezoid*	Classification: 5 *Rhombus*
Property: 22 *One pair of parallel sides.*	Property: 27, 29, or 32 and 28 or 31 *Diagonals are perpendicular. All sides are congruent.*
Justification: *Answers will vary.* *Slope of $\overline{PR} = -\frac{3}{2}$* *Slope of $\overline{MS} = -\frac{3}{2}$*	Justification: *Answers will vary.* *Slope of $\overline{MR} = 0$* *Slope of $\overline{PS}$ is undefined* $MP = PR = RS = SM = \sqrt{13}$

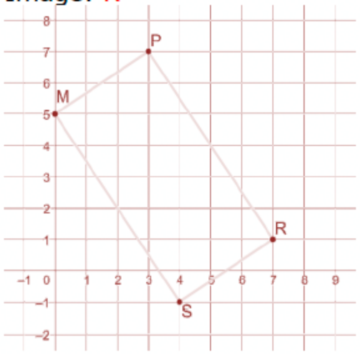
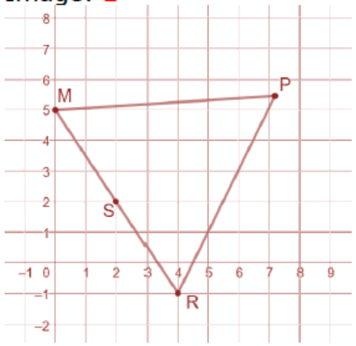
Consider that students will get match 4 (square) and match 6 (rhombus) mixed up, since, visually, they can look similar. Consider making sure students calculate the slope and length of the diagonals in their justification, so that they are sure about their matches.

13.11.2 Activity: Card Sort (continued)

Match 7	Match 8
Image: G 	Image: H 
Coordinates: 14 *$M(0,5)$, $P(4,3)$, $R(6,2)$, and $S(4,-1)$*	Coordinates: 18 *$M(0,5)$, $P(3,7)$, $R(7,5)$, and $S(3,3)$*
Classification: 1 *Scalene triangle*	Classification: 8 *Kite*
Property: 24 *No sides are equal.*	Property: 27, 29, or 32 and 33 *Diagonals are perpendicular. Exactly one diagonal bisects the other.*
Justification: *Answers will vary.* $MR = \sqrt{45}$ $RS = \sqrt{13}$ $MS = \sqrt{52}$	Justification: *Answers will vary.* *Slope of $\overline{PS}$ is undefined* *Slope of $\overline{MR} = 0$* *Midpoint of $\overline{PS}$ is $(3,5)$* *Midpoint of $\overline{MR}$ is $(3.5,5)$* *The equation of line $\overline{MR}$ is $y = 5$ and the midpoint of $\overline{PS}$ is a point on that line.* *The equation of line $\overline{PS}$ is $x = 3$, but the midpoint of $\overline{MR}$ does not lie on that line.*

- *How are match 6 (rhombus) and match 8 (kite) different?*
- *How can we determine the difference between the two polygons?*

13.11.2 Activity: Card Sort (continued)

Match 9	Match 10
Image: K 	Image: L 
Coordinates: 15 $M(0,5)$, $P(3,7)$, $R(7,1)$, and $S(4,-1)$	Coordinates: 12 $M(0,5)$, $P(7.5,5.5)$, $R(3.3,11.7)$, and $S(2,9)$
Classification: 2 *Rectangle*	Classification: 6 *Equilateral triangle*
Property: 23, 26 or 30 *Diagonals are congruent*	Property: 28 or 31 *All sides are congruent*
Justification: *Answers will vary.* $RM = PS = \sqrt{65}$	Justification: *Answers will vary.* $RM = PR = PM = \sqrt{56.5}$

Consider that students will get match 10 (equilateral triangle) and match 1 (isosceles triangle) mixed up, since, visually, they can look similar. Consider making sure students calculate the side lengths in their justification, so that they are sure about their matches.

Consider that students will get match 3 (parallelogram) and match 9 (rectangle) mixed up, since, visually, they can look similar. Consider making sure students look at the diagonals in their justification, so that they are sure about their matches.

- *How are match 3 (parallelogram) and match 9 (Rectangle) similar?*
- *How are they different?*
- *How can we tell the difference between them?*

For students who finish early, challenge them to find 3 different matches that they think other students in the class may have difficulties with and write an explanation why.

13.11.3 Wrap-Up: Cool Down

Component Overview: *Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

13.11.3 Wrap-Up: Cool Down

1. Explain how you can use coordinate geometry to classify triangles and quadrilaterals.

   Answers will vary.

13.12 Congruence and Similarity

Lesson Introduction

Benchmark	MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Target	• I can use coordinate geometry to justify congruence and similarity in polygons.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2 Understand similarity and congruence using models and transformations.	N/A

Essential Questions	• How do we determine if two figures are congruent, similar, or neither? • If we are given only coordinate points, what ways can we determine if two figures are congruent? Similar? Neither?
Lesson Narrative	In this lesson, students explore whether polygons are congruent, similar, or neither by finding the length of each of the sides, given only coordinate points. In the 13.12.3 Guided Practice and 13.12.4 Your Turn!, students apply their understanding of congruent and similar polygons by justifying their choice using definitions, theorems, or properties.

Component Summary				
	13.12.1 Warm-Up (~5 mins)	Notice and Wonder *(SE p. 348)*	13.12.3 Guided Practice (10 mins)	Determining if Polygons are congruent, similar, or neither *(SE p. 351)*
	13.12.2 Exploration (15 mins)	Congruence and Similarity in Polygons *(SE p. 349)*	13.12.4 Your Turn (10 mins)	Practice determining if polygons are congruent, similar, or neither *(SE p. 353)*
	13.12.5 Wrap-Up (~5 mins)	Summarizing the Lesson *(SE p. 354)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Glossary: N/A	• Calculators • (Optional) Graph paper / Graphing utility • (Optional)Patty paper	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may have difficulties identifying the difference in congruence and similarity in two figures, which either can be a result in them not understanding their definitions, and/or not being able to distinguish between the two when given two figures.</td><td>Consider having a list of various definitions, theorems, and properties that students have learned throughout the unit and previous units that could be used as justification when working through the 13.12.3 Guided Practice.
As a cumulating end of the course activity, consider having students also utilize their knowledge of transformations by noting the transformations that show congruence or similarity for all figures. Watch for students who miss important details, such as the center of dilation.
If time is an issue, then consider assigning 13.12.4 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In the 13.12.1 Warm-Up, students are shown an image and they are asked about what they notice and wonder. Consider opportunities and ways that students can share what they notice and wonder with their peers, such as in Gallery Walks or as a class discussion.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In the 13.12.2 Explorations and 13.12.3 Guided Practice, students are focusing on the relevant details within a problem, while also applying previously learned theorems, properties, and definitions to justify whether two figures are congruent or similar. Students are also connecting the similarities among figures that are similar and congruent, and how that will better help them when solving future problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For students who are struggling in the 13.12.2 Exploration with finding the side lengths given only the coordinate points, consider providing them with graph paper or a graphing tool so that they can see the shape and determine which side lengths they are calculating. Also, consider providing graph paper and patty paper when students are identifying the transformations in the same section.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.12.1 Warm-Up: Notice and Wonder

Component Overview: *Students are noticing and wondering about a sculpture seen in Orlando, Florida. Consider opportunities for students to share their ideas with their peers, either in a Gallery Walk, partner share, or class discussion.*

13.12.1 Warm-Up: Notice and Wonder

Brickley, the LEGO sea serpent in Lake Buena Vista at Disney Springs near Orlando, Florida, was made in 1997 from 170,000 LEGO bricks and is 30 feet long.

1. What do you notice?
   Answers will vary. The sculpture has a lot of symmetry in the points on the tail, face and chest of the sculpture.

2. What do you wonder?
   Answers will vary. How often do Legos have to be replaced?

Consider opportunities for students to share what they notice and wonder with their peers. Consider doing a simple partner share where they discuss with their partner. Or, consider a Gallery Walk where students can see multiple classmates' responses, and possibly even write down one thing that they notice and one thing that they wonder from another student.

13.12.2 Exploration: Congruence and Similarity

Component Overview: *In this Exploration, students are exploring if two polygons are congruent, similar, or neither based off of their side lengths, which they are expected to find using the given coordinate points. Consider providing students who are having difficulties visualizing the polygon with graph paper or a graphing tool for them to find the side lengths.*

13.12.2 Exploration: Congruence and Similarity

1. Parallelogram $DKTS$ has vertices at $D(-2,4)$, $K(-4,1)$, $T(-9,3)$, and $S(-7,6)$.

   Complete the table.

Line Segment	Length
$\overline{DK}$	$DK = \sqrt{13} \approx 3.606$
$\overline{KT}$	$KT = \sqrt{29} \approx 5.385$
$\overline{TS}$	$TS = \sqrt{13} \approx 3.606$
$\overline{SD}$	$SD = \sqrt{29} \approx 5.385$

2. Parallelogram $WRCN$ has vertices at $W(0,1)$, $R(2,-2)$, $C(7,0)$, and $N(5,3)$.

   Complete the table.

Line Segment	Length
$\overline{WR}$	$WR = \sqrt{13} \approx 3.606$
$\overline{RC}$	$RC = \sqrt{29} \approx 5.385$
$\overline{CN}$	$CN = \sqrt{13} \approx 3.606$
$\overline{NW}$	$NW = \sqrt{29} \approx 5.385$

3. What do you notice about the lengths of the line segments in parallelograms $DKTS$ and $WRCN$?
   The corresponding side lengths are congruent.

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. The concepts of congruence and similarity are not new to the student. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.12.3 Guided Practice. Consider highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

For students who are struggling with finding the side lengths given only the coordinate points, consider providing them with graph paper or a graphing tool so that they can see the shape and determine which side lengths they are calculating.

4. Complete the statement.

   Parallelogram $DKTS$ may be
   - congruent
   - similar

   to parallelogram $WRCN$ because all corresponding sides are
   - proportional.
   - congruent.

5. Work with your partner to describe how the parallelograms are congruent or similar using transformations. State which image you used as the pre-image.
   Answers will vary. Parallelogram $DKTS$ is the pre-image. When parallelogram $DKTS$ is reflected across the y-axis, horizontally translated left 2 units, and vertically translated down 3 units, the result is parallelogram $WRCN$.

6. Isosceles trapezoid $MPHN$ has vertices at $M(2,-6)$, $P(3,-3)$, $H(5,-2)$, and $N(8,-3)$.

   Complete the table.

Line Segment	Length
$\overline{MP}$	$\sqrt{10} \approx 3.162$
$\overline{PH}$	$\sqrt{5} \approx 2.236$
$\overline{HN}$	$\sqrt{10} \approx 3.162$
$\overline{NM}$	$\sqrt{45} \approx 6.708$

- *If parallelogram DKTS was similar to parallelogram WRCN, what would the side lengths look like?*
- *How would the side lengths change?*

13.12.2 Exploration: Congruence and Similarity (continued)

7. Isosceles trapezoid $WVYZ$ has vertices at $W(4,12)$, $V(6,6)$, $Y(10,4)$, and $Z(16,6)$.
Complete the table.

Line Segment	Length
$\overline{WV}$	$\sqrt{40} \approx 6.325$
$\overline{VY}$	$\sqrt{20} \approx 4.472$
$\overline{YZ}$	$\sqrt{40} \approx 6.325$
$\overline{ZW}$	$\sqrt{180} \approx 13.416$

8. What do you notice about the lengths of the line segments in isosceles trapezoids $MPHN$ and $WVYZ$?
The line segments in isosceles trapezoid $WVYZ$ are the side lengths of the isosceles trapezoid $MPHN$ multiplied by 2.

9. Complete the statement.
Isosceles trapezoid $MPHN$ is
   - o congruent
   - o similar

   to isosceles trapezoid $WVYZ$ because all corresponding sides are
   - o proportional.
   - o congruent.

10. Work with your partner to describe how the trapezoids are congruent or similar using transformations. State which image you used as the pre-image.
Answers will vary. Isosceles trapezoid $MPHN$ has been reflected across the x-axis and then dilated by a scale factor of 2 about the origin to create isosceles trapezoid $WVYZ$.

For students struggling with finding the transformations, provide students with graph paper and patty paper so that they can visually see the correct sequence of transformations.

For students that are struggling with identifying transformations, consider providing students with graph paper and patty paper.

- *How can we determine if polygons are similar or congruent?*
- *What characteristics do shapes that are congruent have?*
- *What characteristics do shapes that are similar have?*

13.12.3 Guided Practice: Congruence and Similarity

Component Overview: *Students are continuing their understanding of similarity and congruence from the previous section by determining if two polygons are congruent and similar, but instead of only calculating the side length, they are also asked to justify their choice by using definitions, properties, and theorems that they have learned.*

13.12.3 Guided Practice: Congruence and Similarity

1. Right triangle CKR has vertices at $C(-2,5)$, $K(-8,5)$, and $R(-8,2)$. Right triangle DPM has vertices at $D(-2,-4)$, $P(4,-4)$, and $M(4,-1)$.

   a. Determine if ΔCKR and ΔDPM are congruent, similar, or neither.
   The triangles are congruent because the side lengths are equal.

   b. What definition, theorem, or property justifies your conclusions?
   Answers will vary. The SSS congruence theorem.

   c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.
   Answers will vary. Let triangle CKR be the pre-image. When you rotate ΔCKR $180°$ about point C and then translate down 9 units the result is triangle DPM.

   d. Ask a classmate for their description. Record their description and write your summary of their description. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Description	My Summary of Their Description	Initials
Answers will vary.		

Allow informal observation throughout 13.12.2 Exploration to dictate how "guided" this Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

Consider having an anchor chart with multiple definitions, theorems, and properties that were learned in this unit and in earlier units so that students could reference when justifying whether two polygons are congruent or similar.

What is another justification, different than your original one, that also supports why triangle CKR and DPM are congruent?

For students who need a challenge, consider having them find another series of transformations in question 1c that would also show that the two figures are congruent.

13.12.3 Guided Practice: Congruence and Similarity (continued)

2. Parallelogram $HGNC$ has vertices at $H(5,-1)$, $G(2,3)$, $N(9,3)$, and $C(12,-1)$. Parallelogram $VYZW$ has vertices at $V(-1,1.5)$, $Y(0.5,4)$, $Z(-3,4)$, and $W(-4.5,1.5)$.

   a. Shin and Kendal are determining if parallelograms $HGNC$ and $VYZW$ are congruent, similar, or neither. Their work is shown.

Shin's Work	**Kendal's Work**
Parallelograms $HGNC$ and $VYZW$ are similar because $\frac{ZY}{NG}=\frac{VW}{HC}$	Parallelograms $HGNC$ and $VYZW$ are neither because $\frac{ZY}{NG}\neq\frac{YV}{GH}$

   b. Whose work do you agree with?
   Kendal's work.

   c. Write a note to the person who is incorrect explaining their mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

Two pairs of sides being proportional does not prove that two parallelograms are similar. More information about the other sides and the angles formed is needed.

Consider spending time reviewing the transformations that students wrote. Help students recall information that they may have forgotten. Including transformations in this lesson was purposely done so that students are using a spaced recall, which research shows helps students remember concepts. This could help students with end-of-year summative assessments.

For students who are struggling with determining whose work they agree with, consider having students first find the side lengths and/or graph the shapes before they make their decision.

Consider having students share their notes. Encourage students to be thorough in their explanations.

13.12.4 Your Turn!

Component Overview: *Students continue their understanding that was bult in the last two sections by determining if two figures are congruent, similar, or neither, given only their coordinate points.*

13.12.4 Your Turn!

Complete the table by determining if the given polygons are congruent, similar, or neither.

1. ΔSTK: $S(-6,8)$, $T(-10,5)$, and $K(-6,1)$
   ΔNRC: $N(3,7.5)$, $R(-3,3)$, and $C(3,-3)$
   - Congruent
   - Similar
   - Neither

2. Trapezoid $MPWY$: $M(3,4)$, $P(7,6)$, $W(10,5)$, and $Y(4,2)$
   Trapezoid $LKGH$: $L(-2,6)$, $K(-8,9)$, $G(-14,7)$, and $H(-4,2)$
   - Congruent
   - Similar
   - Neither

3. Kite $XZWY$: $X(-7,6)$, $Z(-6,9)$, $W(-3,10)$, and $Y(-1,4)$
   Kite $FGDK$: $F(2,3)$, $G(1,6)$, $D(-2,7)$, and $K(-4,1)$
   - Congruent
   - Similar
   - Neither

For students who are having difficulties visualizing the figures, consider providing graph paper or a graphing tool so that they can determine if the two figures are congruent, similar, or neither.

What are the ways to determine if two figures are congruent? Similar? Neither?

For students who finish early, consider having them do the following:

- *Create a set of coordinate points of a different figure in question 1 that is congruent to the given figure.*
- *Create a set of coordinate points of a different figure in question 2 that is similiar to the given figure.*
- *Create a set of coordinate points of a different figure in question 3 that is similar to the given figure.*

Also, consider having students create their own figure, and then create 3 different sets of coordinate points that are congruent, similar, and neither to their created figure.

13.12.5 Wrap-Up: Cool Down

Component Overview: *Students write a note to a friend, summarizing the ideas and concepts that they took away from this lesson.*

13.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to teach them how to use coordinate geometry to justify congruence and similarity in polygons.

Answers will vary.

Consider what responses and topics you are looking for students to have in their note that you would consider a student to have understood this lesson. Consider creating a small rubric that would help you when reading over the "notes," and what you would be looking for with a student who completely understood the lesson, versus a student who may need help.

13.13 Finding Perimeter and Area on the Coordinate Plane

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.</td><td>Learning Target</td><td>• I can find the perimeter and area of polygons and composite figures drawn on the coordinate plane.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• How can we find the area of a composite figure?
• How can we find the perimeter of a composite figure?
• Why is the perimeter of a composite figure not the same as adding the two separate perimeters of the figures it is made of?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">The lesson first has students calculating area and perimeter of two shapes separately, and then introduces them to the idea of composite figures. Students then explore how the area of a composite figure is the same as adding the area of the two figures, while that is not the case with the perimeter.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>13.13.1 Warm-Up (~5 mins)</td><td>Writing questions about the Dancing House (SE p. 355)</td><td>13.13.3 Try It (15 mins)</td><td>Practice finding the perimeter and area of composite figures (SE p. 358)</td></tr>
<tr><td>13.13.2 Exploration (25 mins)</td><td>Perimeter and area of composite figures (SE p. 355)</td><td>13.13.4 Wrap-Up (~5 mins)</td><td>Reflection (SE p. 359)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u>
• Composite Figure

<u>Additional Glossary:</u> N/A</td><td colspan="2">• Calculators</td><td>• N/A</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may think that to find the area of a parallelogram, it is the same as finding the area of a rectangle, especially since the side lengths could be used for a rectangle.
Students may add the perimeter of each shape of a composite figure, instead of calculating the perimeter of the figure.</td><td>Consider that this lesson is towards the end of the course and that the concepts of finding area and perimeter are not new, therefore, encourage students to use their resources to remember anything that they may have forgotten. This spaced recall can benefit the student by having them struggle to recall the concepts they have previously learned.
In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
In this lesson, students are identifying and comparing their approach to finding side lengths with that of other students in the class, while also determining which method is best. Students are also making connections to finding the perimeter and area of individual shapes, and how that can help us find the area and perimeter of composite figures.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students are analyzing problems in a way that makes sense for them given the task, such as when finding the area and perimeter of a composite figure. While it is hoped that students see the pattern to find the area and perimeter, students are still encouraged to analyze the problem on their own. This allows students to stay positive and engaged throughout the lesson.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider providing anchor charts, or reviewing the various area formulas and what each variable represents, so that students have that extra support in this lesson. Also, consider providing students with ways to find the length of diagonal lines, and consider performing an example so that students have an example to go off of when calculating the side lengths of shapes in this lesson.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.13.1 Warm-Up: The Dancing House

Component Overview: *Students are given an opportunity to observe a photo and create two questions that come to their minds. One question is what first comes to their minds, and the other question is one they know a 5th grader could answer.*

13.13.1 Warm-Up: The Dancing House

The Dancing House is located in Prague, Czech Republic, and was designed in 1992. The style of this building is known as deconstructivist architecture due to its unusual shape.

1. What is the first question that comes to your mind?
   Answers will vary. Is the structure of the twisted building as stable as a building that is not twisted?

2. Write a question that a 5th grader could answer using the photo.
   Answers will vary. What is an appropriate estimation for the number of windows in the twisted structure?

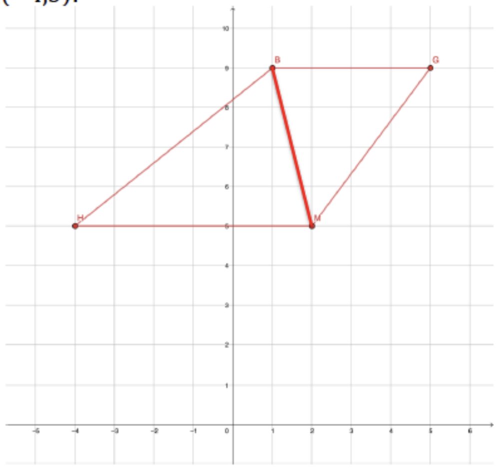
Consider an opportunity to have a discussion with students about how they created the question for questions. The discussion can include how they knew what topics to ask in their questions, such as which topics were too easy for a 5th grader versus which questions would be too difficult.

13.13.2 Exploration: Perimeter and Area

Component Overview: *In this section, students are finding the area and perimeter of a trapezoid and parallelogram separately, and then they are given a composite shape that is created from the same parallelogram and trapezoid and are asked to find the perimeter and area of the shape. The purpose of this section is to introduce students to composite figures, and to notice that the area of a composite figure is the same as adding the area of the two shapes, while the perimeter is not calculated the same way.*

13.13.2 Exploration: Perimeter and Area

1. Trapezoid $BGMH$ has vertices at $B(1,9)$, $G(5,9)$, $M(2,5)$, and $H(-4,5)$.



   a. Graph $BGMH$.
   Image found on graph.

   b. Complete the table.

Line Segment	Length
$\overline{BG}$	$BG = 4$
$\overline{GM}$	$GM = 5$
$\overline{MH}$	$MH = 6$
$\overline{HB}$	$HB = \sqrt{41}$

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Students utilize skills and concepts learned previously in the course. Since this is content that is spiraled, allow students to utilize their own resources to recall formulas. Monitor while students are working to determine if there are any questions that need to be debriefed before 13.3.3 Try It. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before releasing students to complete 13.3.3 Try It.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

c. What is the perimeter of $BGMH$?
21.403 units

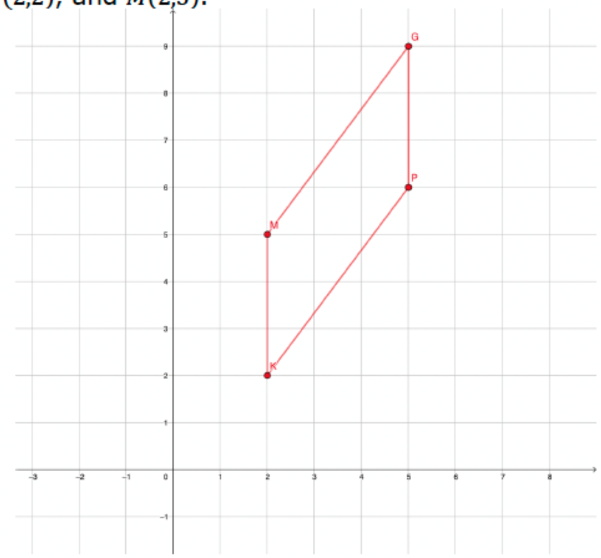
d. Draw a diagonal line from B to M to make ΔBMH and ΔBGM.
Image found on graph.

e. Find the area of ΔBMH and ΔBGM.

$\Delta BMH =$ ____12 *units*^{2}____ $\Delta BGM =$ ____8 *units*^{2}____

f. What is the area of $BGMH$?
20 *square units*

2. Parallelogram $GPKM$ has vertices at $G(5,9)$, $P(5,6)$, $K(2,2)$, and $M(2,5)$.



a. Graph $GPKM$.
Image found on graph.

- *Why do you think it is needed to separate BGMH into two separate triangles?*
- *If $BGMH$ was a rectangle, would separating it into two separate triangles to find the area be the same as using the equation to find the area of a rectangle?*
- *What other quadrilaterals do you think it will be beneficial to separate into two triangles in order to find the area?*

13.13.2 Exploration: Perimeter and Area (continued)

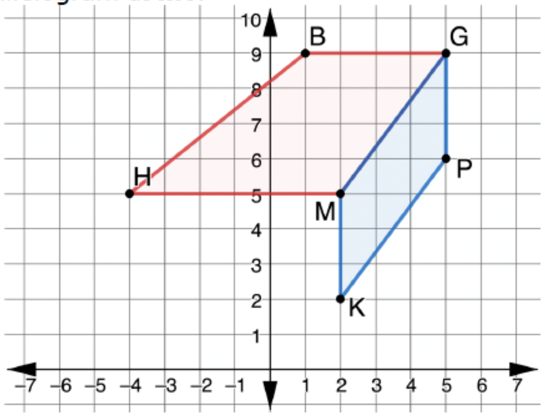
b. Complete the table.

Line Segment	Length
$\overline{GP}$	3
$\overline{PK}$	5
$\overline{KM}$	3
$\overline{MG}$	5

c. What is the perimeter of $GPKM$?
16 *units*

d. What is the area of $GPKM$?
9 *squared units*

3. The figure shown is created by trapezoid $BGMH$ and parallelogram $GPKM$.



a. What do you notice about this figure?
Answers will vary. Trapezoid $BGMH$ and parallelogram $GPKM$ share the side $\overline{GM}$

From your observations of students on question 1, consider whether some students could benefit from a check-in with you, or if utilizing a small group to work through question 2 with you could be beneficial to students who struggled with question 1.

A ***composite figure*** is a two- or three-dimensional figure that can be decomposed into smaller figures.

b. Explain why this figure is considered a composite figure.
Answers will vary. The figure is composed of trapezoid $BGMH$ and parallelogram $GPKM$ and can be broken up into the two shapes.

c. How could you find the perimeter of a composite figure?
Answers will vary. To find the perimeter of a composite figure, add up all of the side lengths that are on the perimeter of the figure, but do not include any sides they share like $\overline{GM}$ in the figure above.

d. What is the perimeter of composite figure $BGPKMH$?
27.403 units

e. Explain why the perimeter of the composite figure is not the sum of the perimeters of the parallelogram and the trapezoid.
Answers will vary. The perimeter of trapezoid $BGMH$ and parallelogram $GPKM$ both include the side length $\overline{GM}$ which is no longer considered an edge of the composite figure.

f. What is the area of $BGPKMH$?
29 square units

- *Why can't we find the area of parallelogram GPKM using the equation, $b \cdot h$?*
- *Why isn't the area of a parallelogram and rectangle the same?*
- *Why does it not produce the same result?*

Consider an opportunity to take all of the student responses and create a Gallery Walk or class discussion so that students can share what they notice with others. If turning this into an activity, consider having students write down 1 – 2 things that other students noticed that they did not.

Could we create a composite figure with trapezoid $BGMH$ and parallelogram $GPKM$ that would have the same perimeter as adding up the perimeters of trapezoid $BGMH$ and parallelogram $GPKM$?

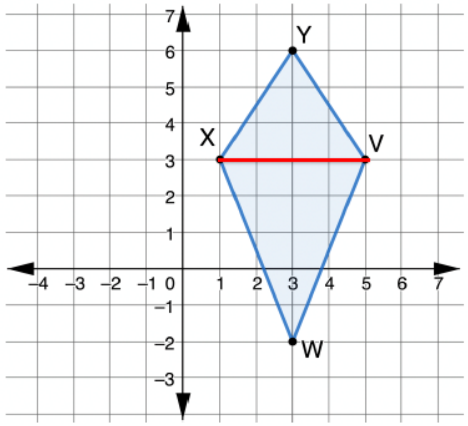
13.13.3 Try It: Perimeter and Area

Component Overview: *Students are extending the idea from the Exploration to another problem involving composite figures.*

13.13.3 Try It: Perimeter and Area

Use observational data from the 13.13.2 Exploration to determine how much guidance is needed to bring together key ideas before releasing students to complete this component. If necessary, consider modeling Question 1 to address common misconceptions shown in the Exploration.

1. Kite $YVWX$ is shown with vertices at $Y(3,6)$, $V(5,3)$, $W(3,-2)$, and $X(1,3)$.



a. Complete the table.

Line Segment	Length
$\overline{YV}$	$YV = \sqrt{13} \approx 3.606$
$\overline{VW}$	$VW = \sqrt{29} \approx 5.385$
$\overline{WX}$	$WX = \sqrt{29} \approx 5.385$
$\overline{XY}$	$XY = \sqrt{13} \approx 3.606$

b. What is the perimeter of $YVWX$?
17.981 *units*

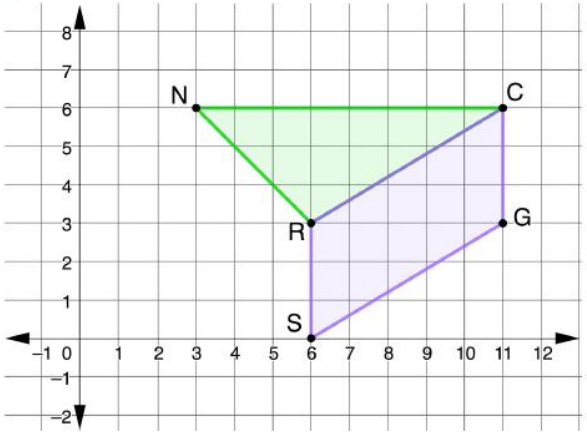
c. Draw line XV to create ΔYVX and ΔVWX.
Image found on graph.

d. Find the area of each triangle.
$\Delta YVX =$ 6 *units*^{2} $\Delta VWX =$ 10 *units*^{2}

e. What is the area of $YVWX$?
16 *square units*

13.13.3 Try It: Perimeter and Area (continued)

2. The composite figure shown is created by ΔNCR and parallelogram $RCGS$. $N(3,6)$, $C(11,6)$, $G(11,3)$, $S(6,0)$, and $R(6,3)$.



a. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
ΔNCR	*18.074 units*	*12 square units*
$RCGS$	*17.662 units*	*15 square units*

b. What is the perimeter of the composite figure?
24.074 units

c. What is the area of the composite figure?
27 square units

- *Why is dividing Kite $YVWX$ into two triangles helpful?*
- *What other ways could Kite $YVWX$ be divided?*
- *What is the perimeter of triangle YVX and VWX?*
- *Why is the perimeter of the kite YVWX not the same as adding the perimeter of triangle YVX and VWX?*

If working with students who are struggling, consider providing students a few minutes to write down things they notice and wonder about the composite figure. This will allow students to make observations before finding the perimeter and area.

For students who finish early, challenge them to create a third shape on the composite figure and find the area and perimeter of the additional shape (alone), and then the area and perimeter of the new composite function.

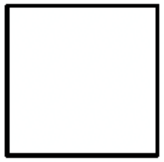
13.13.4 Wrap-Up: Cool Down

Component Overview: *Students complete a Cool Down that has a theme with shapes. Consider an opportunity to use this to collect data about where students are struggling and what they understand.*

13.13.4 Wrap-Up: Cool Down

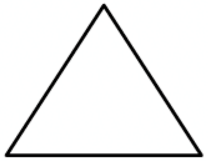
Consider creating a rubric with ideas that you would expect students to have written down in these 3 questions, that you would consider them to have mastered and understood this lesson.

1. What is something that is now "squared away" in your mind about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



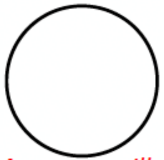
Answers will vary.

2. What are three "points" that are important about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



Answers will vary.

3. What is something that is still "circling" your brain about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



Answers will vary.

13.14 Using Coordinate Geometry to Solve Real-World Problems

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmarks</td><td colspan="2">MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.</td><td>Learning Targets</td><td>• I can use coordinate geometry to classify triangles and quadrilaterals.
• I can use coordinate geometry to solve real-world problems on the coordinate plane involving perimeter or area of polygons.
• I can use coordinate geometry to solve real-world geometric problems involving circles, triangles, and quadrilaterals.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On:</td><td colspan="2">Working Toward:</td></tr>
<tr><td colspan="2">MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• How can you use coordinate geometry to find the perimeter and area of composite figures in a real-world context?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">This is a cumulating lesson for unit 13. Students will be applying many of the skills and concepts to real-world problems.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>13.14.1 Warm-Up (~5 mins)</td><td>Bell Work (SE p. 360)</td><td rowspan="2">13.14.3 Wrap-Up (~5 mins)</td><td rowspan="2">Reflection on Unit 13 (SE p. 365)</td></tr>
<tr><td>13.14.2 Collaboration (40 mins)</td><td>Real-World Problems using Coordinate Geometry (SE p. 361)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms</u>: N/A

<u>Additional Glossary</u>: N/A</td><td colspan="2">• Calculators
• (Optional) Graph paper or Graphing utility</td><td>• Consider preparing small groups or partners for the Collaboration.</td></tr>
</table>

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle with question 3 because not all the coordinate points are whole numbers. • Students may struggle with finding the distance of two points in order to find the area and/or perimeter. • Students may add the perimeter of each shape of a composite figure, instead of calculating the perimeter of the figure.	• Students may not recall secant, tangent, and inscribed angles for question 3 in 13.14.2 Collaboration. Encourage students to utilize their resources to find the information they need to complete the problem. • Consider providing graphing paper or a graphing utility to use as a visual for the beginning of question 2. • Consider working with students who are struggling during part of 13.14.2 Collaboration. • In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
MLR1: Stronger and Clearer Each Time Encourage students to revise and refine both their ideas and their verbal and written output for questions in 13.4.2 Collaboration. Suggest to students to first think or write individually about a response, then to refine and clarify the response through conversation with their partner, and then finally revise their original written response.	**MA.K12.MTR.7.1: Real-World Contexts** Students connect the mathematical concepts used in Unit 13 to solve real-world problems. Students use graphical models to understand, represent, and solve the problems. Teachers should monitor students for accuracy in both their models and their methods.

Differentiate Instruction

Collaboration 13.14.2 provides both visual and auditory entry points for learners to engage with the content. Engaging the class in discussions by having students explain their justifications and how they found the area and perimeter will help auditory learners deepen their understanding. The coordinate planes provide students with a visual to see each section of a composite figure. Consider placing the questions around the classroom to help kinesthetic learners be engaged in the lesson.

Additional differentiation opportunities are denoted throughout the lesson guide.

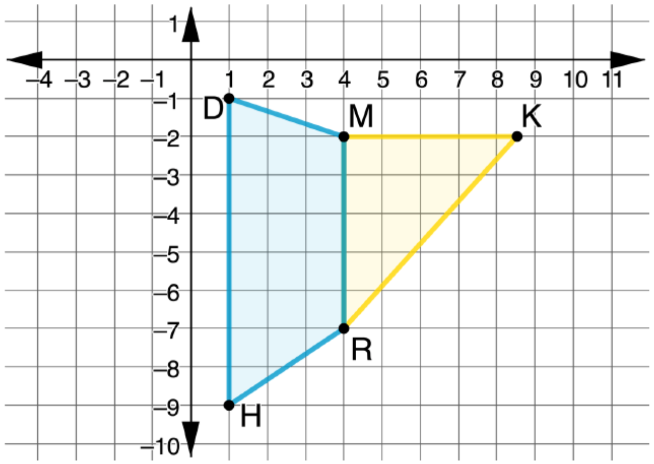
Lesson Notes:

13.14.1 Warm-Up: Bell Work

Component Overview: *Students are finding the area and perimeter of a composite figure that is created by a triangle and a trapezoid. Students should complete the questions independently.*

13.14.1 Warm-Up: Bell Work

1. The composite figure shown is created by ΔMKR and trapezoid $DMRH$. $M(4,-2)$, $K(8.5,-2)$, $R(4,-7)$, $H(1,-9)$, and $D(1,-1)$.



a. Complete the table. If necessary, round to the nearest thousandth.

Figure	Perimeter	Area
ΔMKR	16.227 *units*	11.25 *square units*
$DMRH$	19.768 *units*	19.5 *square units*

b. What is the perimeter of the composite figure? If necessary, round to the nearest thousandth.
25.995 *units*

c. What is the area of the composite figure? If necessary, round to the nearest thousandth.
30.75 *square units*

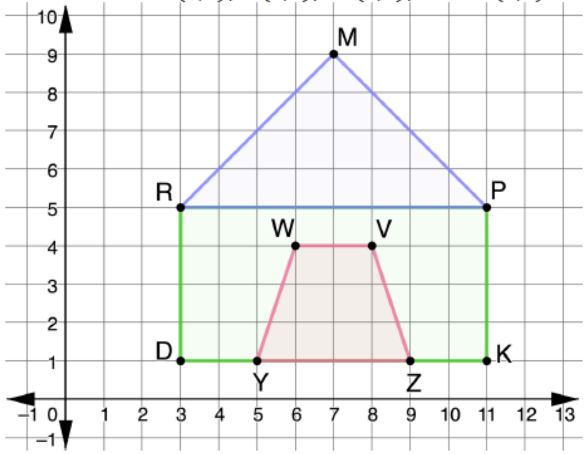
Use student understanding to inform today's Collaboration. Students who struggled on the Bell Work may benefit from working with you in a small group on part of the Collaboration. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.

13.14.2 Collaboration: Real-World Problems

Component Overview: *This Collaboration provides students with multiple opportunities to work with composite figures in the real world. Throughout each real-world prompt, students are asked to classify polygons, find the area and perimeter of polygons and composite figures, and find additional information while justifying their answers along the way.*

13.14.2 Collaboration: Real-World Problems

1. Lana is designing the front of a doghouse. The computer software generated coordinate points where each unit equals a foot. The roof has vertices at $M(7,9)$, $P(11,5)$, and $R(3,5)$. The front wall has vertices at $P(11,5)$, $K(11,1)$, $D(3,1)$, and $R(3,5)$. The door entrance has vertices at $Z(9,1)$, $Y(5,1)$, $W(6,4)$, and $V(8,4)$.



   a. What type of triangle is the roof, ΔMPR? Justify your answer.
   Isosceles triangle because $MR = MP = \sqrt{32}$

   b. What is the area of ΔMPR?
   16 square feet

   c. Clay tiles cost $\$8$ per square foot. How much will it cost Lana to put clay tiles on the given face of the doghouse's roof?
   $\$128.00$

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify. Consider having an answer key available for students to verify their answers when they have completed all of the parts of a question. Consider requiring students to receive your signature before moving on to the next question. This provides some accountability, while also allowing you to ensure students are understanding. As you review student work, consider which parts students should revise before moving on to the next question.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Collaboration.

d. What type of quadrilateral is the door, $WVZY$? Justify your answer.
An isosceles trapezoid because both $\overline{WV}$ and $\overline{YZ}$ have a slope of 0 so $\overline{WV} \parallel \overline{YZ}$ and $WY = VZ = \sqrt{10}$.

e. Lana is placing a frame around the door, $WVZY$. How many feet of wood will she need? If necessary, round to the nearest thousandth.
12.325 feet

f. What type of quadrilateral is the front wall, $RPKD$. Justify your answer.
$RPKD$ is a rectangle because $RD = PK = 4$ and $RP = DK = 8$, and the diagonals are congruent, $DP = RK = 4\sqrt{5}$.

g. Lana is painting the front wall, $RPKD$, excluding the door, $WVZY$. What is the area of the front wall Lana painting?
23 square feet

h. Paint costs $\$0.55$ per square foot. Excluding the door, how much will it cost to paint the front wall?
$\$12.65$

2. Justin is creating a layout for his garden in the shape of a quadrilateral, $GWDP$, with vertices $G(1,8)$, $W(11,8)$, $D(11,0)$, and $P(1,0)$.

   a. Classify $GWDP$. Justify your answer.
   Quadrilateral $GWDP$ is a rectangle because $GW = DP = 10$ and $WD = GP = 8$, and the diagonals are congruent, $DG = PW = 2\sqrt{41}$.

Consider challenging students by adding an extension to question 1h:

- *Lana found another company that sells paint for $\$0.05$ per square inch. If Lana is covering 23 square feet, is this a better deal than $\$0.55$ per square foot? Justify your answer. (Answer: $\$0.05$ per square inch is $\$0.60$ per square foot, so it is not a better deal.)*

Struggling students may benefit from using graphing paper or a graphing utility for question 2. This will provide students with a visual and could help them to see the three sections of the composite figure.

13.14.2 Collaboration: Real-World Problems (continued)

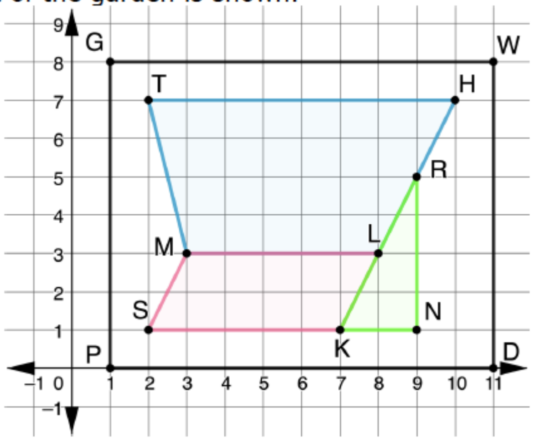
Justin created three sections in the garden: $THLM$, $LKSM$, and ΔKNR.

b. $THLM$ has vertices at $T(2,7)$, $H(10,7)$, $L(8,3)$, and $M(3,3)$. Classify $THLM$. Justify your answer.
Quadrilateral $THLM$ is a Trapezoid because both $\overline{TH}$ and $\overline{ML}$ have a slope of 0 so $\overline{ML} \parallel \overline{TH}$.

c. $LKSM$ has vertices at $L(8,3)$, $K(7,1)$, $S(2,1)$, and $M(3,3)$. Classify $LKSM$. Justify your answer.
Quadrilateral $LKSM$ is a parallelogram because both $\overline{MS}$ and $\overline{LK}$ have a slope of 2 so $\overline{MS} \parallel \overline{LK}$ and both $\overline{ML}$ and $\overline{SK}$ have a slope of 0 so $\overline{ML} \parallel \overline{SK}$.

d. ΔKNR has vertices at $K(7,1)$, $N(9,1)$, and $R(9,5)$. Classify ΔKNR. Justify your answer.
Triangle KNR is a right triangle. The slope of $\overline{KN}$ is 0 and the slope of $\overline{RN}$ is undefined, therefore the two lines are perpendicular and form a right angle.

The layout of the garden is shown.



e. If one square on the grid represents one square foot, how many feet of fencing will Justin need to go around $GWDP$?
36 feet

Students may struggle with understanding "one square on the grid represents one square foot." Listen for students who are confused by this, and take the time to show them that each square on the grid is 1 unit by 1 unit, so the area is 1 square unit.

f. Complete the table. If necessary, round to the nearest thousandth.

Section	Perimeter	Area
$THLM$	21.595 *feet*	26 *square feet*
$LKSM$	14.472 *feet*	10 *square feet*
ΔKNR	10.472 *feet*	4 *square feet*

g. What is the perimeter of the composite figure made by the three sections of the garden? If necessary, round to the nearest thousandth.
27.595 *feet*

h. What is the area of the three sections of the garden? If necessary, round to the nearest thousandth.
40 *square feet*

i. If a 40 pound (lb) bag of soil covers 12 square feet, how many bags of soil will Justin need to cover the three sections of the garden?
He would need 4 bags because he needs exactly $3\frac{1}{3}$ bags so he would round up since the bags must be bought in whole numbers.

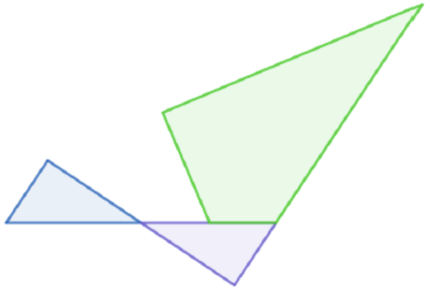
Consider having students use the instructional routine, "Take Turns," on question 2f. Each person in the group will take a turn at finding the perimeter and/or the area. Each group member can use this time to review each other's work and ask clarifying questions, if needed.

Consider prompting questions to help students make sense of the problem and as a check for understanding:

- *How is finding the perimeter of a composite figure different than finding the area of a composite figure?*
- *When finding the number of bags of soil, would you use the area or perimeter? Why?*
- *Why would you round up for the number of bags?*

13.14.2 Collaboration: Real-World Problems (continued)

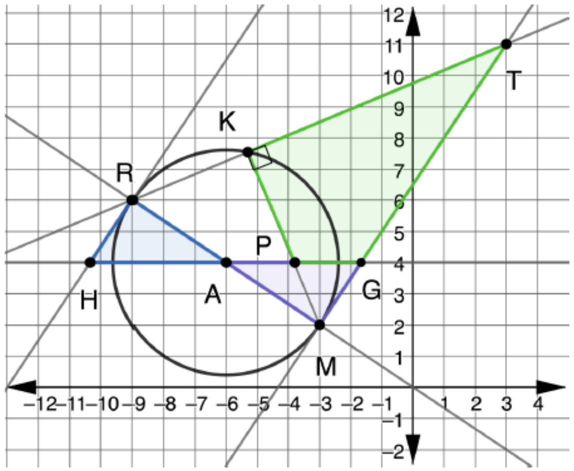
3. Edwin started creating a new logo design for the company office using the images shown.



To determine how to place the logo on the wall in the company's office, he uploaded his design to a computer program and got the following coordinates.

$M(-3,2)$	$R(-9,6)$	$P(-3.8,4)$	$T(3,11)$
$G(-\frac{5}{3},4)$	$H(-\frac{31}{3},4)$	$K(-5.3,7.5)$	$A(-6,4)$

Circle A has points R, M, and K on the circle. Lines TM and HR are tangent to circle A. Lines RM and HG are secant to circle A. Line segment KM is perpendicular to line TK.



Students may struggle with this question because some of the points are not whole numbers and cannot be easily identified on the coordinate plane. If necessary, support these students by incorporating the Desmos scientific calculator, which can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.

Consider asking guided questions to help students make sense of the information given:

- *What relationship does a tangent line have with the radius/diameter of a circle?*
- *What relationship does $\angle RKM$ have with the diameter of the circle?*
- *How can the diameter be used to find the $m\angle MKT$?*

a. Complete the table by finding the area of each triangle.

Triangle	Work	Area
ΔAHR	*Answers will vary.* $Area\ \Delta AHR = \frac{1}{2} \cdot HA \cdot 2$	≈ 4.333 *square units*
ΔAMG	*Answers will vary.* $Area\ \Delta AMG = \frac{1}{2} \cdot AG \cdot 2$	≈ 4.333 *square units*
ΔPGM	*Answers will vary.* $Area\ \Delta PGM = \frac{1}{2} \cdot PG \cdot 2$	≈ 2.133 *square units*
ΔTKM	*Answers will vary, depending on rounding.* $Area\ \Delta TKM = \frac{1}{2} \cdot TK \cdot MK$ $TK \approx 9.008$ $MK \approx 5.961$	≈ 26.85 *square units*

b. Explain how to use ΔPGM and ΔTKM to find the area of $TKPG$.

Take the area of triangle TKM and subtract the area of triangle PGM to find the area of polygon $TKPG$

c. What is the area of $TKPG$? If necessary, round to the nearest thousandth.
$26.85 - 2.133 \approx 24.717$ *square units*

d. What is the total area of Edwin's design? If necessary, round to the nearest thousandth.
$24.717 + 4.333 + 4.333 \approx 33.383$ *square units*

Consider having students use the instructional routine, "Take Turns," on question 3a. Each person in the group will take a turn at finding the area of each triangle. Each group member can use this time to review each other's work and ask clarifying questions, if needed.

13.14.3 Wrap-Up: Reflection

Component Overview: *Students complete a self-assessment based on their level of understanding of each key concept from Unit 13. It is designed to be done independently to gather specific, student-level data.*

13.14.3 Wrap-Up: Reflection

Circle the icon that best describes your understanding of each.

I need help. I got it. I could help someone.

Using coordinate geometry to determine if lines are parallel or perpendicular.	
Using coordinate geometry to partition a line segment or find the weighted average.	
Using coordinate geometry to classify triangles or quadrilaterals.	
Using coordinate geometry to justify definitions, properties, and theorems involving circles, triangles or quadrilaterals.	
Using coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	
Using coordinate geometry to solve mathematical and real-world involving perimeter or area of polygons.	

Answers will vary.

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.